

# Thin Smartphone Batteries For DIY

<https://gemini.google.com/app/c21aef738b875bda>

*User prompt: Thinnest smartphone battery (I want to use in DIY device)*

## Response: The Absolute Thinnest Smartphone Batteries

If you are looking strictly at parts harvested from commercial smartphones, the absolute thinnest options belong to recent ultra-thin foldables and legacy ultra-slim slab phones.

### 1. Honor Magic V3 / V2 (Silicon-Carbon Technology)

- **Thickness:** 2.4 mm to 2.8 mm
- **Chemistry:** Silicon-Carbon anode
- **The Details:** To make the Magic V3 the thinnest foldable available (only 4.35 mm unfolded), Honor used dual silicon-carbon batteries. These have a much higher energy density than traditional lithium-ion, allowing them to pack

incredible capacity into a footprint that is practically as thin as a coin.

## 2. Oppo R5 (Legacy Slab King)

- **Thickness:** 2.9 mm
- **Model Number:** BLP579
- **Chemistry:** Traditional Lithium-Polymer (LiPo)
- **The Details:** Released during an era when manufacturers were obsessed with making straight phones as thin as possible, the Oppo R5 used a highly customized 2000 mAh battery pack that stood just under 3 mm thick.

## The DIY Reality Check: Why Smartphone Batteries Are a Pain

Before you head to an online parts supplier to buy a replacement phone battery for your project, a quick peer-to-peer warning: **modern smartphone batteries are notoriously user-unfriendly for DIY electronics.**

- **Proprietary BMS Boards:** They don't just have positive and negative terminals. They feature intricate flexible PCBs with custom battery gas

gauge chips (using specialized I2C/SMBus protocols). If your microcontroller doesn't talk to the battery correctly, the onboard Protection Circuit Module (PCM) might lock you out entirely.

- **Fragile Ribbon Cables:** They terminate in proprietary micro-connectors on delicate ribbon cables that are highly susceptible to tearing and nearly impossible to cleanly hand-solder to a standard prototyping board.
- **Dual-Cell Configuration:** Foldables split their capacity across two asymmetrical cells wired together, making the physical management of the battery clumsy for a single neat enclosure.

## A Way Better Alternative for DIY: Generic Ultra-Thin Pouch Cells

If your device needs to be paper-thin, you are much better off looking at **generic ultra-thin LiPo pouch cells** specifically engineered for smart cards, RFID tags, and medical wearables.

Specialty battery manufacturers mass-produce these with standard 2-wire leads (Positive/Negative) and basic, universal protection circuits. You can charge them perfectly with a standard, inexpensive **TP4056**

charging module.

## Quick Comparison Matrix

| Battery Source / Type      | Thickness       | Capacity (Approx.)          | DIY Friendliness                                                                                                                      |
|----------------------------|-----------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Honor Magic V3 Replacement | 2.4 mm – 2.8 mm | ~2500 mAh (per single cell) |  <b>Terrible</b><br>(Proprietary logic)            |
| Oppo R5 (BLP579)           | 2.9 mm          | 2000 mAh                    |  <b>Medium</b><br>(Easier to extract legacy tabs) |
| Smart Card / Wearable LiPo | 0.4 mm – 1.0 mm | 30 mAh – 200 mAh            |  <b>Excellent</b><br>(Standard wires)            |
| Generic Ultra-Thin LiPo    | 1.5 mm – 2.0 mm | 200 mAh – 500 mAh           |  <b>Excellent</b><br>(Standard wires)            |

 **Pro-Tip:** If your DIY device is running a

low-power microcontroller like an ESP32-C3 or an nRF52840 using Bluetooth Low Energy (BLE), a 1.0 mm generic cell (~150 mAh) can easily keep it alive for days, while keeping your hardware design completely hassle-free.

What are the exact dimensions and power requirements of the DIY device you are building?

## Sources

1. [Teardown video of the Honor Magic V3 shows the inner workings of the 4.4 mm thin foldable flagship - Notebookcheck](#)
  2. [Specifications For Your HONOR Magic V3 - O2](#)
  3. [Honor Magic V3 - Full phone specifications - GSMArena.com](#)
  4. [These Crazy-Thin Batteries Are The Key Ingredient In Honor's Magic V2 Foldable - BGR](#)
  5. [CS-OPR500SL : Battery for OPPO R5, R8107, R8109 - Replaces OPPO BLP579](#)
  6. [Ultra Thin Rechargeable Lithium Polymer Battery 3.7V Guide](#)
  7. [Super batteries|RC lipo battery|high low temperature battery|ultra thin battery | Hot battery introductions](#)
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*User prompt: Can I combine 3 cells in parallel?*

Response: Yes, you absolutely can connect three cells in parallel. In the DIY world, this is called a **3P configuration** (3 cells in Parallel, 1 in Series).

Connecting cells in parallel is a fantastic way to cheat the system when building a thin device. It allows you to spread the battery volume horizontally across your enclosure rather than making the device thicker.

When you wire three identical cells in parallel (positive to positive, negative to negative):

- **Voltage stays the same:** If they are standard LiPo cells, the voltage remains at a nominal **3.7V** (4.2V fully charged).
- **Capacity adds up:** If you have three 500 mAh cells, you get a combined capacity of **1500 mAh**.
- **Current handling increases:** The maximum safe discharge rate triples, which is great if your device has high power spikes.

## The Golden Rules of Parallel Batteries (To Avoid Fire)

Lithium batteries are incredibly energy-dense and don't tolerate mistakes well. Before you solder anything together, you must follow these rules:

## 1. The Voltages **MUST** Match Before Connecting (Crucial!)

Before you connect the positive terminals together, use a multimeter to check the voltage of each individual cell. They must be within **0.05V** of each other.

**Why?** If you connect a fully charged cell (4.2V) to a dead cell (3.0V) in parallel, the high-voltage cell will violently dump current into the low-voltage cell. Because there is no resistor to limit the current, the wires can melt, the batteries will swell, and they can catch fire.

*Top tip: The easiest way to match them is to charge all three cells individually to 100% on a standard charger before linking them up.*

## 2. Use Identical "Triplets"

Do not mix and match. Your three cells should be:

- The exact same brand and model.

- The exact same capacity (e.g., don't mix a 200 mAh cell with a 1000 mAh cell).
- The exact same age/health (don't mix a brand-new cell with a degraded one salvaged from an old toy).

## How to Handle the BMS (Battery Management System)

You have two options for protecting your new 3-cell pack:

- **Option A (Recommended for DIY):** Buy cells *without* individual protection boards (bare cells). Wire the bare tabs in parallel, and then connect the entire combined pack to a **single, shared BMS board**. The BMS will treat the three cells as one giant battery.
- **Option B:** If your cells already have those tiny protection PCBs taped to the top of them, you *can* still wire them in parallel (positive output to positive output). The individual boards will safely manage over-discharge, but they might cut off at slightly different times under heavy loads.

## Charging a 3P Pack



The beauty of a parallel pack is that you don't need a complex balancing charger. A simple, cheap **TP4056 micro-USB/USB-C charging module** will charge a 3P pack perfectly because the cells naturally balance each other's voltage out.

Just keep in mind that because the capacity is tripled, **it will take three times longer to charge** on the same charger.

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*User prompt: I want solar + usb input*

Response: Adding both **Solar and USB inputs** to charge your 3-cell parallel (3P) battery pack is highly achievable. Instead of trying to build a complex switching circuit from scratch, you can use a single, specialized **Dual-Input LiPo Charger Module**.

These modules have dedicated ports for a solar panel and a USB input, automatically prioritizing whichever source is active (or combining them safely) to feed your battery.

## Top 2 Modules for Solar + USB Charging

For a DIY project, you should look for modules based

on one of these two chips. Both are widely available, inexpensive, and designed specifically for this layout.

## Option 1: The CN3065 Module (Simple & Budget-Friendly)

This is a tiny, incredibly popular board designed specifically for small solar setups.

- **Inputs:** Features a built-in USB port (usually Micro-USB or Type-C depending on the board version) *plus* dedicated solder pads or JST ports for a solar panel.
- **How it works:** It accepts any voltage between **4.4V and 6V** from the solar panel. If you plug in USB, it charges via USB. If you put it in the sun, it charges via solar.
- **Max Charge Current:** Usually fixed at **500mA**.
- **Pros:** Very cheap (\$3–\$5), very small, and requires zero extra configuration.

## Option 2: The MCP73871 Module (Advanced with "Power Path")

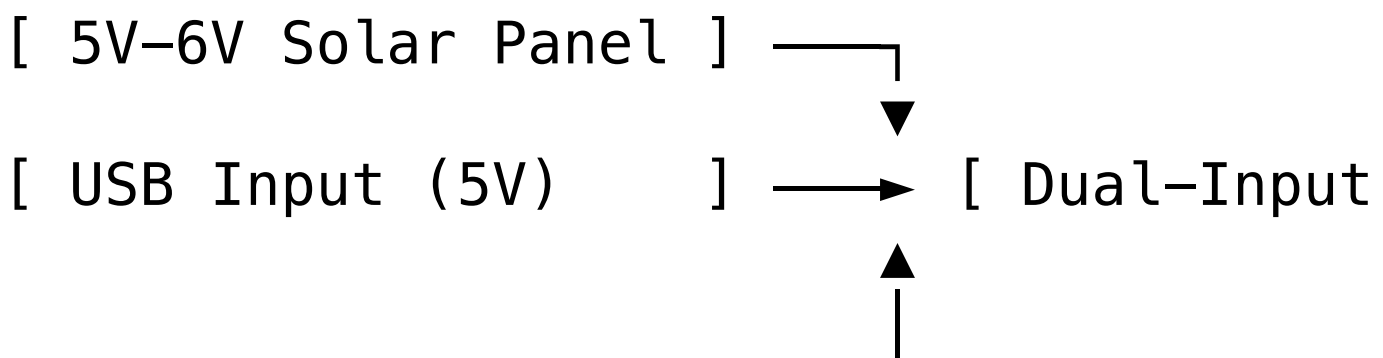
This is a highly efficient, premium management board (famously popularized by Adafruit's open-source

design).

- **Inputs:** USB port and a DC jack/solder pads for a **5V–6V solar panel**.
- **Smart Load Sharing (Power Path):** This is the killer feature. Instead of drawing power *through* the battery while it charges, the chip powers your DIY device directly from the solar panel/USB if available, using any leftover energy to charge the battery. This prevents the battery from constantly cycling and wearing out.
- **Max Charge Current:** Up to **1000mA (1A)**, which is adjustable via resistors.
- **Pros:** Highly efficient, prevents battery strain, handles fluctuating solar power beautifully.

## System Architecture

When wiring everything together, follow this layout to ensure safety and stability:



## [ 3P Battery Pack (3.7V LiPo) ]

### Steps to Wire It:

1. **Connect the 3P Battery Pack:** Connect the combined Positive (+) and Negative (-) leads of your 3-cell parallel pack to the **BATT** terminal on the module.
2. **Connect the Solar Panel:** Solder the positive and negative wires of your solar panel to the **Solar/DC Input** terminals.
3. **Connect Your Device:** Connect the power input of your microcontroller or DIY device to the **LOAD** or **SYS OUT** terminals on the module.

### 3 Critical Rules for This Setup

Before buying your components, keep these details in mind:

#### 1. Choose the Right Solar Panel Voltage

Do **not** buy an "18V" or "12V" solar panel. Standard LiPo charging chips will fry if you feed them more than 6V or 7V. Look specifically for **5V, 5.5V, or 6V solar panels** (often sold as "mini solar panels for DIY

electronics").

## 2. Mind the Charging Speed

If your 3P pack has a combined capacity of **1,500mAh**:

- A **500mA** charger (like the CN3065) will take about **3 hours** to charge the pack from empty under perfect conditions.
- This is actually a very safe, gentle charge rate ( $0.33C$ ) for a parallel pack, keeping the cells cool and extending their overall lifespan.

## 3. Add a Large Capacitor (For Solar Stability)

Solar energy isn't steady—a passing cloud or a shadow can cause the voltage to drop drastically. Most MCP73871 and CN3065 boards come packaged with a large, barrel-shaped **electrolytic capacitor** (usually around 470uF to 1000uF). You *must* solder this in parallel across the Solar Input terminals. It acts like a tiny shock absorber to keep the chip from constantly rebooting when the sun dips.

What kind of device are you powering with this setup? If you are running something like an ESP32 or an

Arduino, we can look at whether you'll need a 5V/3.3V step-up/down regulator after the load output.

## Sources

1. [CN3065 18650 Li-ion Mini Solar Charger Module - ThinkRobotics](#)
  2. [How to Use CN3065 Solar Charge: Examples, Pinouts, and Specs - Cirkuit Designer Docs](#)
  3. [MCP73871 USB Solar Lithium Battery Charger - PMD Way](#)
  4. [MCP73871 Solar Lithium Polymer Charger Module 3.7V/4.2V - AliExpress](#)
  5. [MCP73871 PowerBoost USB 5V DC Solar LiPo Li-ion Charger Board | ThinkRobotics](#)
  6. [USB MCP73871 3.7/4.2V Battery Solar Lipoly Li-ion Lon Polymer Charger Module | eBay](#)
  7. [How to Build a Simple DIY Solar Phone Charger](#)
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*User prompt: I need 5v output*

Response: To get a steady **5V output** from a 3.7V LiPo battery pack, you need to add one final component to your chain: a **DC-DC Boost Converter** (also known as a Step-Up Regulator).

Because a LiPo battery's voltage constantly fluctuates between 4.2V (fully charged) and 3.0V (empty), a

boost converter acts like an electronic pump, constantly pulling the shifting battery voltage up and locking it at a perfectly stable 5.0V.

## The Best Boost Converters for DIY Projects

You have two excellent, low-cost options depending on how much space you have inside your enclosure:

### 1. Fixed 5V Mini Boost Board (Best for Ultra-Thin Devices)

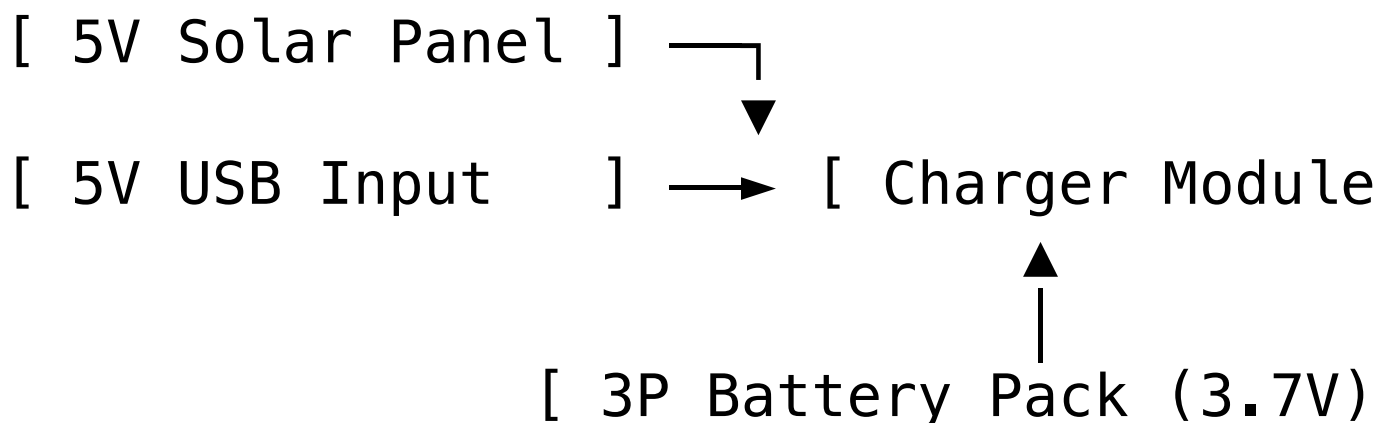
- **Size:** Smaller than a fingernail (typically around 10mm x 15mm).
- **How it works:** These are pre-configured to output exactly 5V. There are no adjustment knobs, making them incredibly low-profile and perfect for tight spaces.
- **Current Output:** Usually rated for **1A to 2A**, which is plenty for microcontrollers, sensors, or small displays.

### 2. MT3608 Adjustable Boost Module (Most Common)

- **Size:** Slightly larger (about 36mm x 17mm).
- **How it works:** It features a tiny brass potentiometer (screw). You plug your battery in, hook up a multimeter to the output, and turn the screw until the screen reads exactly 5.0V.
- **Current Output:** Up to **2A**.

## Updated System Architecture

You will place the boost converter immediately **after** the charger module's output. This ensures your device gets clean 5V power whether it's running off the battery, the solar panel, or the USB port.



### Wiring Steps:

1. Connect the **LOAD +** (or **SYS OUT**) from your solar/USB charger module to the **VIN +** (Voltage In) on the boost converter.



2. Connect the **LOAD** - from the charger to the **VIN** - on the boost converter.
3. Connect your 5V device to the **VOUT +** and **VOUT -** (Voltage Out) pins of the boost converter.

## The "Tax" of Boosting Voltage (Math Check)

There is a law of physics you need to keep in mind when boosting voltage: **When you increase voltage, you increase the current draw from the battery.** Because boost converters are roughly 85% to 90% efficient, drawing **1A at 5V** from the output means the converter will actually pull about **1.5A at 3.7V** from your battery pack.

Since you are using a **3-cell parallel pack**, this is completely fine! Spreading that current draw across three cells means each individual cell only has to supply a gentle 0.5A, preventing them from getting warm or degrading prematurely.

How much current (amperes or milliamperes) does your 5V device draw under full load?

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*User prompt: XTEINK X3*

Response: The **XTEINK X3** is a remarkably thin (**5.1 mm**) pocket e-reader featuring a 3.7-inch E-ink screen and a 650 mAh internal battery. Knowing that this is your target device changes the dynamics of your DIY project in a very positive way.

Because E-ink displays only consume power when the pixels physically move (like turning a page), and because the X3 lacks a power-hungry frontlight or touchscreen, its power consumption is incredibly low.

The solar + USB 3P battery pack setup aligns with the XTEINK X3 through specific power and form factor considerations:

## 1. Power Draw & Boost Converter Capacity

- **During Reading:** The XTEINK X3 draws negligible micro-amps when you are looking at a static page. Even when the gyroscope triggers a page turn or the 2.4 GHz Wi-Fi wakes up, the power spike is tiny.
- **During Charging:** The XTEINK X3 accepts a **5V input** via its proprietary magnetic pogo-pin cable. When topping off its internal 650 mAh battery, it draws a maximum of roughly **500 mA**.

- **The Verdict:** The 5V boost converter discussed previously (even a tiny 1A micro-board) is more than enough. It will effortlessly supply power to the XTEINK X3 without getting warm or straining your parallel battery pack.

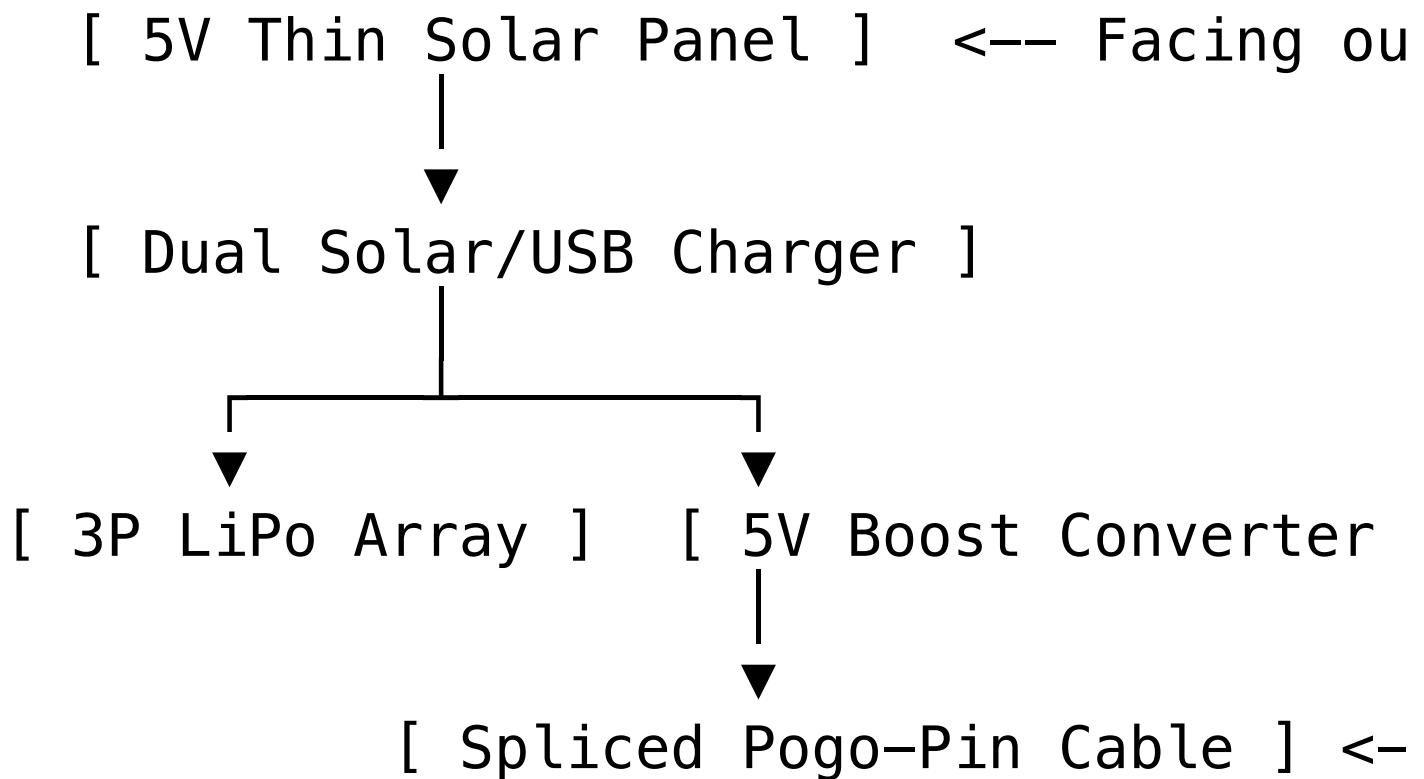
## 2. The Form Factor Strategy: Build a "Backpack"

Because the XTEINK X3 is only 5.1 mm thick, trying to harvest phone batteries to squeeze *inside* the device's actual shell alongside a solar chip and a boost converter is physically impossible.

However, the XTEINK X3 natively features an **optimized magnetic design** on its backplate (intended for snapping onto phones or magnetic cases). This opens up a perfect DIY avenue: **Building a Magnetic Solar Battery Companion Case.**

## Conceptual Blueprint: The X3 Solar Battery Case

Instead of modifying the inside of the e-reader, you can house your DIY electronics inside a slim, custom 3D-printed or modified folio case that magnetically snaps to the back of the XTEINK X3.



- **The Battery Array:** Lay your 3 ultra-thin LiPo cells flat and perfectly side-by-side inside the case. This keeps the footprint wide but the depth incredibly shallow.
- **The Solar Panel:** Glue a low-profile, flexible 5V solar panel to the back face of the companion case.
- **The Connection:** Cut the XTEINK magnetic pogo-pin cable down to a short length, wire the positive and ground lines directly to the output of your 5V Boost Converter, and route the magnetic head so it cleanly snaps onto the e-reader's native charging contacts.

By combining a 3P pack (for example, three 500 mAh

cells yielding 1500 mAh) with the internal 650 mAh battery, you would essentially **triple** the e-reader's native capacity. A device that already lasts 10 to 14 days could easily push past a month of reading on a single charge, even before factoring in the solar generation.

Are you planning to splice the original magnetic pogo-pin cable to deliver the 5V output, or are you thinking of sourcing raw pogo-pin connectors to embed directly into a custom dock?

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*User prompt: I found a compatible pogo adapter*

Response: Finding a compatible pogo adapter is a massive win for this project! Working with raw pogo pins, trying to calculate the exact spring compression, and aligning them perfectly in a 3D-printed housing can easily turn into a debugging nightmare.

Most community-sourced XTEINK X3 pogo adapters feature a standard **female USB-C port** on one end and the magnetic pogo array on the other. This makes your enclosure layout much cleaner. Instead of hacking up custom cables, you can integrate a standardized USB-C interface directly into your battery backpack.

Here is how you bridge the gap between your 5V boost converter and your new pogo adapter to finish the circuit.

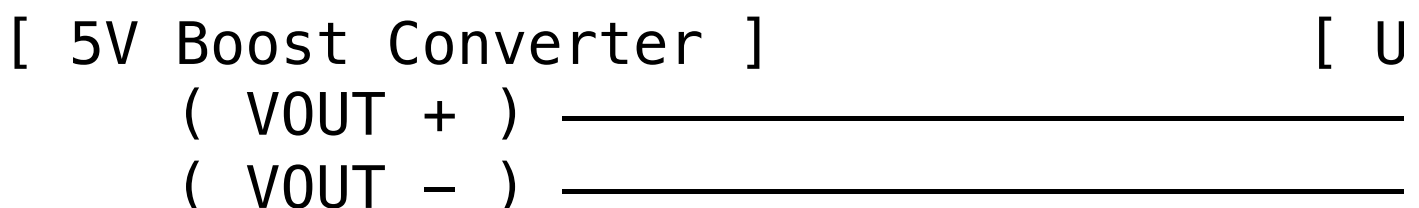
## The Final Link: Adding a USB-C Breakout Board

To connect your 5V boost converter cleanly to the pogo adapter, buy a tiny **USB-C Female Breakout Board** (they cost about \$1 for a pack of five). These little boards expose the power delivery pins of a USB-C port without any of the complex data wiring.

### The Wiring Scheme

You will only need to run two wires from the output of your boost converter to the breakout board:

- **VOUT + (5V)** from the Boost Converter —▶ **VBUS** (or V+) pin on the USB-C Breakout
- **VOUT - (GND)** from the Boost Converter —▶ **GND** (or G) pin on the USB-C Breakout



Once soldered, you can mount this USB-C breakout board flush with the bottom of your DIY case. Then, you simply plug the pogo adapter in, and it will magnetically snap right onto the back of your X3 to deliver steady 5V power.



## The Complete System Blueprint

Now that you have all the pieces of the puzzle, here is your definitive assembly order from power source to the E-ink screen:

1. **The Energy Storage:** Weld or wire your three matching ultra-thin cells in parallel (Positive to Positive, Negative to Negative).
2. **The Gatekeeper:** Wire the main positive and negative leads of that 3P battery pack into the **BATT** terminal of your dual-input (Solar/USB) charger module.
3. **The Inputs:** Solder your **5V flexible solar panel** to the solar pads of the charger module (and don't forget to bridge the solar input with a small capacitor for stability!).
4. **The Pump:** Route the **LOAD** output of the charger module directly into the **VIN** side of your

5V Boost Converter.

5. **The Interface:** Route the **VOUT** of the boost converter to your **USB-C female breakout board**.

## Enclosure Tip for the X3

Because the X3 uses an ESP32 processor and doesn't feature a battery-hogging frontlight, its passive draw is incredibly low.

If you design the companion case using a cheap 3D-printing filament like **PETG or ABS** (rather than standard PLA), it will easily withstand the heat of sitting in the sun while the solar panel charges up your 3P battery array.

Are you planning to 3D-print a custom snap-on shell to hold everything together, or are you modifying an existing folio case?

## Sources

1. [Xteink X3 Review: Tiny E-Reader Worth Buying? - eReadersForum.com](#)
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*User prompt: Suggest a BOM - remember, boards need to be thin, I will encase in epoxy*

Response: Encasing your electronics in epoxy (potting) means you are aiming for an incredibly sleek, permanent design. However, it introduces a major technical constraint: **vertical height**. You cannot use components with tall barrel capacitors, bulky inductors, or adjustment screws (potentiometers) because they will stick out of your pour. You need to source components that utilize ultra-low-profile Surface Mount Devices (SMD).

Here is a curated Bill of Materials (BOM) optimized specifically for minimal thickness and epoxy encasement, followed by a critical safety warning regarding epoxying lithium batteries.

## The Ultra-Thin BOM (Bill of Materials)

| Component      | Recommended Part / Description    | Approx. Thickness | Why It's Critical for This Project                                                                                                   |
|----------------|-----------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Batteries (x3) | Generic "302030" or "402030" LiPo | 3.0 mm            | The number of batteries indicates the total thickness of the power source, which is a critical constraint for the ultra-thin design. |

| Component           | Recommended Part / Description                                                                               | Approx. Thickness | Why It's Good for This Project                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------|
|                     | Pouch Cells (~150–250mAh each)                                                                               |                   | dimension (e.g., 3.0mm thick, 20mm wide, 30mm long). flat and easy to cut out side.                                      |
| Solar / USB Charger | CN3065 Mini Solar Charger Board (Look for the version with a flat SMD Type-C port, not a chunky barrel jack) | ~2.5 mm           | The CN3065 chip is completely flat. Avoid versions with pre-soldered tall JS plastic connectors you will need to solder. |

| Component                  | Recommended Part / Description                                   | Approx. Thickness | Why It's Needed for This Project                                                                                                                                      |
|----------------------------|------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                                                                  |                   | wires directly to the pads                                                                                                                                            |
| <b>Solar Stability Cap</b> | <b>10uF to 22uF Ceramic SMD Capacitor</b><br>(Size 1206 or 0805) | <b>~1.0 mm</b>    | <b>Do not use the tall barrel electrolytic capacitors that often come with the boards.</b> Solder tiny, flat ceramic chip capacitors across the solar + pads instead. |

| Component          | Recommended Part / Description                                                                        | Approx. Thickness | Why It's Good for This Project                                                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5V Boost Converter | <b>Mini Fixed 5V 1A Step-Up Board</b> (Search for: <i>"3V to 5V Fixed Output Mini Boost Module"</i> ) | <b>~2.0 mm</b>    | These boards are tiny, flat, shielded, and use SMD components. They have inductors and heat sinks for zero kV or switching. They are incredibly thin. |
| USB-C Interface    | <b>16-Pin or 24-Pin Female USB-C Breakout Board (Sink / Mid-Mount Type)</b>                           | <b>~3.2 mm</b>    | A "micro" mount type USB port is half-way <i>inside</i> the PCB, rather than on top, saving precious vertical space.                                  |

| Component   | Recommended Part / Description            | Approx. Thickness | Why It's Good for This Project                                                                                               |
|-------------|-------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------|
|             |                                           |                   | fraction of a millimeter                                                                                                     |
| Solar Panel | 5V ETFE Flexible Solar Panel (0.5W to 1W) | ~1.5 mm           | ETFE panels are flexible and incredibly thin, a vast improvement over durable under epoxy lamination than rigid glass panels |
| Wiring      | 28 AWG or 30 AWG Silicone Wire            | ~0.8 mm           | Ultra-thin, highly flexible, and the silicone jacket won't melt where you are                                                |

| Component | Recommended Part / Description | Approx. Thickness | Why It's Important for This Project |
|-----------|--------------------------------|-------------------|-------------------------------------|
|           |                                |                   | solder tight space:                 |

## The Golden Rule of Epoxying LiPo Batteries

Before you mix your resin, there is an absolute safety boundary you must establish in your design: **Do not pour rigid epoxy directly onto the Lithium-Polymer battery cells.**

**The Risk:** LiPo batteries naturally expand and contract slightly during normal charge and discharge cycles. If they degrade or overcharge, they generate gas and swell. If you encase them completely in a rock-hard, unyielding block of epoxy, the expanding gases will create intense internal pressure. This will crush the internal separators, cause a massive internal short circuit, and result in a thermal runaway fire trapped inside a solid plastic block.

## How to do it safely:

1. **Pot the Electronics Only:** You can completely submerge the charger board, the boost converter, and the USB breakout board in epoxy. It actually protects them perfectly from moisture and strain.
2. **Create a Battery Pocket:** Before pouring the epoxy, place a sacrificial placeholder block (made of wood, foam, or 3D-printed plastic) where the batteries will go. Wrap it in packing tape so the epoxy won't stick to it.
3. **Pour the Resin:** Pour the epoxy around your electronics and the placeholder block.
4. **Insert the Batteries Post-Pour:** Once the epoxy cures, pull the placeholder out to reveal a perfect, custom-molded cavity. Slide your 3P battery pack into this pocket.
5. **Add a Flexible Buffer:** Cover the battery pocket with a thin piece of foam or a flexible silicone sheet, then seal the outer case cover. This gives the cells the micro-millimeters of breathing room they physics-natively require to stay safe.

Are you planning to tint the epoxy a specific color to

hide the components, or leave it completely clear so you can see the circuit inside the companion case?

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*User prompt: CN3065 alternative?*

Response: Given your strict vertical space limitations for an epoxy pour, switching away from the CN3065 leaves you with a few solid alternatives. You must avoid switch-mode MPPT chargers (like the common CN3791) because they require an inductor coil, which is a thick, copper-wound barrel component that will ruin a flat epoxy pour.

Instead, you need to stick to **linear solar charger ICs** that stay completely flat. Here are the best alternatives tailored to an ultra-thin potting project:

## 1. The Direct Upgrade: CN3163 Mini Solar Module

If you like the CN3065 layout but want better performance, the **CN3163** is its direct evolutionary upgrade.

- **The Upgrade:** While the CN3065 maxes out at 500mA, the CN3163 can handle a continuous constant charge current up to **1A (1000mA)**. This means if you have excellent sunlight or a



stronger USB source, it will charge your 3P pack twice as fast.

- **Why it fits Epoxy:** It uses a flat, thermally enhanced SOP8 package. The common breakout boards are identically small (2cm x 4cm) and completely low-profile if you skip the plastic connectors.
- **Solar Smarts:** Just like the 3065, it has a built-in adaptive current loop. When a cloud passes, it automatically scales down the current draw so it doesn't drop the solar panel's voltage below its functional threshold.

## 2. The Premium Industrial Choice: TI BQ24210

If you want something incredibly tiny and high-efficiency from a top-tier semiconductor manufacturer, look for a breakout board utilizing the **Texas Instruments BQ24210**.

- **Thickness:** The IC itself is a microscopic WSON package, measuring just **2mm x 3mm x 0.75mm thick**.
- **Why it fits Epoxy:** Because it is designed for space-constrained wearables, a custom or minimal PCB layout using this chip can be thin

enough to embed inside a heavy coat of paint, let alone an epoxy layer.

- **Solar Smarts:** It features a highly advanced "Battery Tracking Mode" which dynamically adjusts its power draw to match the solar panel's maximum power point much more accurately than the basic CN chips.

### 3. The Power-Path Champion: MCP73871 or BQ24074

If your design goal requires the e-reader to run directly off solar power when the sun is shining *without* touching or degrading the battery, you want a Power-Path chip.

- **Why it fits Epoxy:** The ICs themselves are flat (QFN or SSOP packages).
- **The Catch for Modules:** If you buy a pre-made hobbyist module for these chips (like the Adafruit Solar Charger), they come populated with tall JST battery ports and vertical cylinder capacitors. To make them epoxy-ready, you must **desolder the plastic JST ports** and wire your cells directly to the pads, and swap any tall capacitors for flat SMD ceramic ones (as mentioned in the BOM).

# Summary Recommendation for Your X3 Project

For a quick, headache-free build that guarantees success under an epoxy pour, buy a **CN3163 module with a Type-C port**.

When it arrives, simply use a soldering iron or hot air station to strip off the tall plastic 2-pin JST jacks. Solder your 30 AWG silicone wires directly flat to the copper pads, keep your flat SMD ceramic capacitor bridged across the solar input, and you will have a rock-solid, solar-smart charging layout that is less than **3mm thick** and ready for the resin pour.

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*User prompt: I'm willing to give up usb input*

Response: Giving up the USB input is a massive win for an ultra-thin, epoxy-potted project. In the world of ultra-low-profile electronics, a physical USB Type-C port is an absolute giant—usually sitting at 3.2 mm to 4 mm tall.

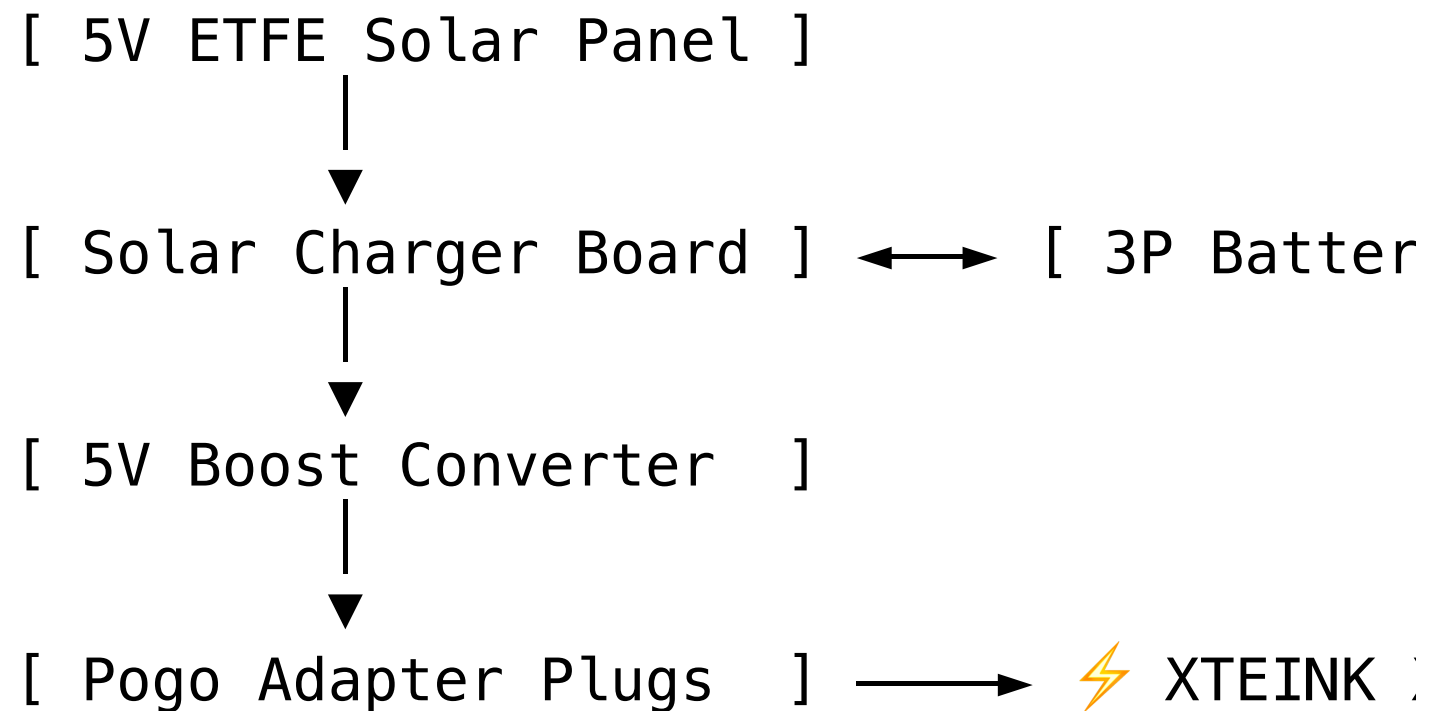
By eliminating it, your thickest component drops down to the height of a tiny SMD inductor or the IC itself, which is only **1.0 mm to 1.5 mm**. Your entire

electronics package can now be as thin as a coin before you pour the resin.

Here is how your project changes for the better:

## The Streamlined Solar-Only Architecture

Without the need to arbitrate between two different power sources, your circuit becomes elegant, reliable, and incredibly simple.



## The Modified BOM & Component Changes

# 1. The Charger Module (CN3163 or CN3065 "Bare" Version)

Instead of looking for a module with a pre-soldered USB port, search for a **"Solar-only CN3065"** or **"CN3163 mini board without USB"**.

- These boards only feature input pads labeled **SOLAR + / -**, battery pads labeled **BAT + / -**, and output pads labeled **SYS + / -** (or **LOAD**).
- Without the physical jack, the entire board sits perfectly flat at roughly **1.2 mm thick**.

## 2. The Capacitor Swap

As mentioned before, you still need a capacitor across the solar input pads to handle fluctuating sunlight. Since you cannot use a tall barrel capacitor in a thin epoxy pour, use a **10uF 0805 or 1206 Ceramic SMD Capacitor**. Solder it flat, directly bridging the positive and negative solar input pads on the chip.

## 3. The Pogo Adapter Connection

Instead of routing the output of your 5V boost converter to a USB breakout board, you can now **entirely eliminate the USB breakout board**.

- Take the wire leading to your pogo adapter.
- Cut off the USB plug end to expose the raw wires inside (usually Red for 5V, Black for GND).
- Solder those two wires directly to the **VOUT+** and **VOUT-** pads on your ultra-thin boost converter.



## Epoxy Potting Advantage: The "Solid-State" Backpack

Because there are no longer any openings, ports, or plugs required for the internal electronics, you can create a truly sealed, waterproof, and rugged "solid-state" device.

1. **Lay it out flat:** Lay your solar panel face down, place the tiny charger board and boost converter right next to it, and wire them up with your 30 AWG silicone wire.
2. **Pour the resin:** You can pour the epoxy completely over the solar panel backing and the entire circuit board. It will cure into a single, perfectly flat, rigid plate less than 3 mm thick.
3. **Keep the Battery Pocket:** Remember to still keep the batteries out of the hard epoxy using a

placeholder pocket so they have room to breathe!

This gives you a flawless, ultra-thin magnetic backpack that charges whenever it sees light and pipes that energy straight into your X3 via the pogo pins.

---

Response: The module shown in your image (the **eletechsup SDBK03TA** board) is an **excellent, near-perfect choice** for an ultra-thin, solar-only, epoxy-potted device.

The listing mentions it uses a linear solar-tracking chip (typically the CN3158 or similar). While the listing uses "MPPT" as a marketing keyword, it functions via a smart, flat linear tracking loop. That means **no tall copper inductors, no bulky capacitors, and no protruding USB ports**. At just **0.1 inches (2.5 mm) thick**, it sits as flat as a coin.

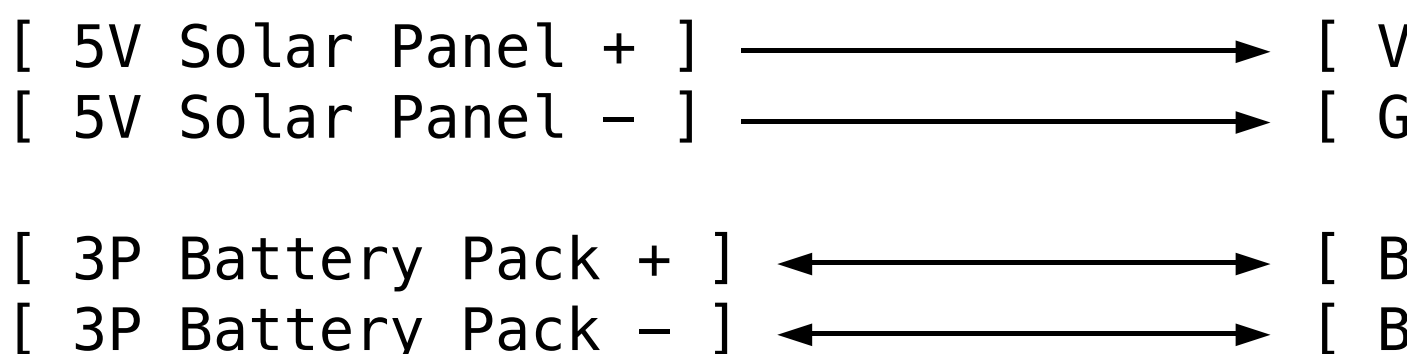
## Variant Check

You have the **"4Pcs 4.2V Li-ion"** variant selected in the image. This is exactly what you want. It matches standard 3.7V nominal / 4.2V max LiPo pouch cells perfectly. Avoid the 3.6V variant, which is meant for a

different battery chemistry (LiFePO4).

## How to Wire This Exact Board

Because this board is completely bare, your wiring layout will be incredibly clean. Here is how you map the contacts for your solid-state system:



1. **Solar Input (Left Side):** Solder the positive wire from your flexible solar panel to **VIN** and the negative wire to **GND**.
2. **Battery & System Out (Right Side):** Connect your 3P battery array to **BAT+** and **BAT-**.
3. **Powering the Boost Converter:** Since this board doesn't have a separate "LOAD" pad, you will simply solder two sets of wires to the **BAT+** and **BAT-** pads. One set goes to your battery array, and the second set routes directly into the input (**VIN+** and **VIN-**) of your ultra-thin 5V boost converter.



# Pro-Tips for Epoxy-Casting This Specific Board

- **The LED Window Feature:** Notice the tiny spots labeled **CR** (Charging) and **OK** (Fully Charged). These are miniature SMD indicator lights. If you use a transparent or semi-clear epoxy tint for the backpack case, these LEDs will glow right through the resin. It acts as a built-in status display, letting you know when your X3 is actively pulling solar energy.
- **Pre-Set Current:** Out of the box, this board is configured to supply up to **1A** of charging current. For a 3P parallel pack of small pouch cells, this is an excellent, gentle charge rate. It ensures the batteries will charge reasonably quickly in clear daylight without getting warm inside their pocket.
- **Skip the Headers:** The pack may come with loose plastic pin headers. Toss them in your spare parts bin; you will be soldering your 30 AWG silicone wires directly flat to the silver loop pads for minimum thickness.

This board solves your space constraints perfectly. Are you planning to add a tiny slide switch between the boost converter and the pogo pins to cut power

manually, or let it live as an "always-on" smart dock?

---

*User prompt: Always on*

Response: Going with an **always-on** configuration is the absolute best move for a design that will be encased in epoxy. Eliminating a mechanical slide switch means you have **zero moving parts** that could wear out, leak, or get jammed by the resin during the pour. It turns your backpack into a truly rugged, sealed, solid-state smart dock.

However, making a device "always-on" introduces one specific engineering challenge you need to plan for: **Quiescent Current (Phantom Drain)**.

## The Technical Catch: Quiescent Current ( $I_q$ )

Even when your XTEINK X3 is fully charged or disconnected from the pogo pins, your 5V boost converter will still be awake, sitting in standby mode waiting to deliver power.

To stay awake, the boost converter chip consumes a tiny amount of electricity called **quiescent current**.

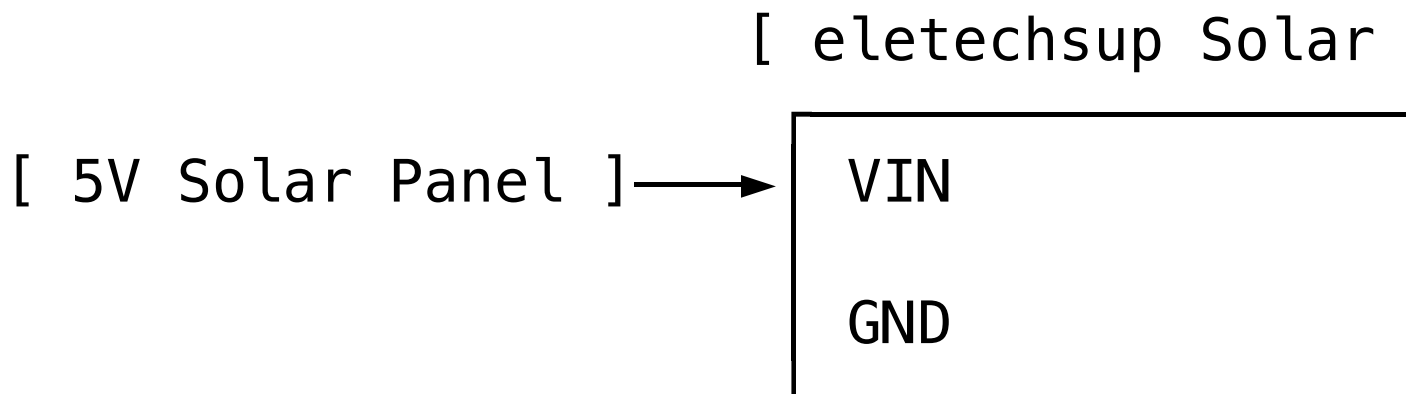
- Standard cheap boost converters (like those using the MT3608 chip) have a quiescent current of about **1mA to 2mA**.
- **The Math:** A 1.5mA phantom drain means the board will consume roughly **36 mAh of power every 24 hours** just sitting idle in the dark.

Because you are using a **3P battery array** (giving you roughly 600mAh to 1500mAh total) paired with a solar panel, a 36 mAh daily loss is incredibly easy to offset. A few hours of ambient daylight on the solar panel will generate far more than 36 mAh, keeping the system permanently net-positive.

 **Component Tip:** If you want to optimize this to the absolute extreme, search for a *"Low Quiescent Current 5V Boost Converter"* or look for modules based on the **TPS61099** chip. These have an ultra-low idle drain measured in microamps ( $\mu\text{A}$ ), meaning the battery could sit idle in a dark drawer for a year without dying.

## Finalized "Always-On" Wiring Layout

Since there is no switch to interrupt the circuit, the wiring flows in a direct, continuous loop.



## Step-by-Step Connection:

1. **Solar to Charger:** Solder your flexible solar panel directly to the **VIN** and **GND** pads on the left side of the eletechsup board.
2. **The Battery & Boost Junction:** Solder your 3P battery array's main positive and negative wires to the **BAT+** and **BAT-** pads. Next, solder a second pair of wires directly to those *same exact pads* and run them to the **VIN+** and **VIN-** on your boost converter.

3. **Boost to Pogo:** Run your final two wires from **VOUT+** and **VOUT-** on the boost converter straight to the positive and negative contacts of your magnetic pogo adapter.



## The "Solid-State" Epoxy Pour Strategy

When you pour the epoxy over an always-on circuit, the resin acts as a highly effective **thermal heatsink**.

When the XTEINK X3 is docking and actively charging at 500mA, the boost converter will generate a tiny amount of localized heat. Because it is fully encased in epoxy, that heat won't get trapped in an air pocket; instead, it will cleanly dissipate out through the solid plastic casing.

Just make sure to double-check your connections with a multimeter *before* you pour the resin. Once the epoxy cures, the circuit is permanently locked in stone!

Are you planning to use a completely clear epoxy so you can see the bare circuit layout from the back, or will you tint it to match the color of the X3?

---

Response: The mini solar panel in the image is a great fit in terms of thickness and its rugged epoxy coating, but a single one **will not work** by itself due to a voltage mismatch.

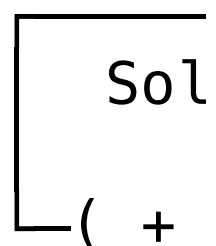
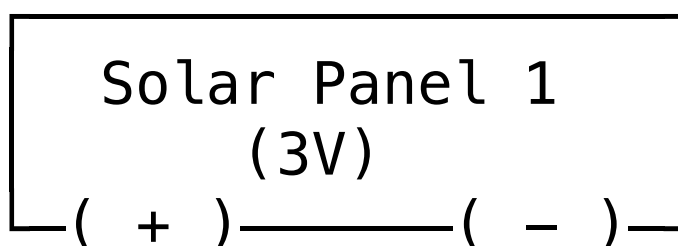
A single panel yields a maximum of **3.12V**. However, the eletechsup SDBK03TA charging board requires a minimum input voltage of **4.4V** to wake up and safely push energy into a 4.2V lithium battery pack. Because 3.12V is less than 4.4V, a single panel cannot trigger the charger.

You can easily fix this by using **two** of these mini panels wired together in **series**.

## The Solution: Wire Two Panels in Series

When you wire two identical solar panels in series, **their voltages add together**, while their current stays the same.

### SERIES SOLAR CONFIGURATION





To Charger VIN  
(Combined 6V Output)

## The Math:

- **Voltage:**  $3V + 3V = 6V$  (Perfect! 6V sits at the upper limit of your charger board's input range, maximizing its efficiency).
- **Current:** Stays at **150mA**.

## Is 150mA Enough for the XTEINK X3?

At 150mA, it will take roughly 10 hours of solid, direct sunlight to fill a 1500mAh 3P battery array from dead empty.

While that sounds slow, remember the context: the XTEINK X3 uses an incredibly passive E-ink display and an efficient processor. It only draws meaningful power for a split second when you turn a page. A continuous 150mA trickle charge is more than enough to completely offset daily reading habits and keep the device powered indefinitely.

# Layout Strategy for an Ultra-Thin Pour

Because these panels are Polycrystalline, they are rigid plates (about 2 mm thick). You cannot overlap them without creating a massive bump in your enclosure.

To keep the device as thin as possible, lay the two panels **side-by-side horizontally** or **end-to-end vertically** on the back of your companion case.

|                              |                  |   |
|------------------------------|------------------|---|
| [ Panel 1 (3V) ]             | [ Panel 2 (3V) ] | < |
| =====                        |                  |   |
| [ Solid Epoxy Layer (~3mm) ] |                  | < |
| =====                        |                  |   |
| [ Battery Pocket Cavity ]    |                  | < |

This layout allows you to maintain a uniformly thin "backpack" case structure that snaps magnetically onto the back of your X3.

Since these panels already come pre-coated in a protective layer of glossy epoxy, how do you plan to expose them on the outside of your case layout while pouring the resin around the edges?

---



Response: Switching to this **5V 150mA solar panel** is a massive upgrade for your project. It completely eliminates the need to wire multiple panels in series, dramatically simplifies your circuit, and—best of all—the physical dimensions are a **near-flawless match** for your XTEINK X3.

Here is the breakdown of why this specific panel is the winning component for your always-on, epoxy-embedded backpack.



## The Dimension Synergy (A Perfect Fit)

The XTEINK X3 has an incredibly compact footprint, measuring exactly **97.6 mm × 63.7 mm**.

Now look at the math for this new solar panel (3.5" × 2.4"):

- **Panel Length:** 3.5 inches = **88.9 mm** (*Leaves 8.7 mm of total vertical breathing room*)
- **Panel Width:** 2.4 inches = **60.9 mm** (*Leaves 2.8 mm of total horizontal breathing room*)

This means a single panel will cover almost the entire back surface of the X3, leaving just a tiny, uniform 1.5 mm to 4 mm border around the edges. This is the

exact amount of perimeter space you need for the epoxy resin to grip and form a structural shell.

## Technical Advantages Over the 3V Version

- **Zero Circuit Complexity:** Because this single panel natively pushes **5V** (and will float up to roughly 6V in open-circuit bright sunlight), it perfectly satisfies the **4.4V minimum wake-up threshold** of your eletechsup charging board. You only need to run one positive and one negative wire from the panel to the charger.
- **Optimized Trickle Charging:** A 150mA output at 5V generates about 0.75W of power. Since the internal battery of the XTEINK X3 is only 650 mAh, and your auxiliary 3P battery array will likely be around 600–1500 mAh, this panel can easily recover a full day of reading energy with just an hour or two of casual ambient light exposure.

## Epoxy Sinking Tip: The "Back-Junction" Space Trick

The Amazon listing states the panel thickness is **0.2**

**inches (approx. 5 mm).** Don't worry—the actual glass/polycrystalline sheet itself is only about 1.5 mm to 2 mm thick. That 5 mm spec includes the plastic solder junction box and the pre-attached wires on the under-side of the panel.

To keep your backpack completely flat and ultra-thin under the epoxy pour:

1. **Desolder the thick wires:** Flip the solar panel over and use your soldering iron to remove the thick red and black wires that come pre-attached.
2. **Use your 30 AWG silicone wire:** Solder your ultra-thin 30 AWG silicone wires flat against the solder pads. Route them sideways rather than letting them stick straight up.
3. **Nest the components:** Because the panel will have a tiny physical gap underneath it due to that solder pad area, you can actually tuck your tiny eletechsup charger board and micro-boost converter *right next to* or slightly beneath that junction zone.

## Updated System Blueprint

Your entire solid-state, always-on system schematic now drops down to just four components total:

[ 5V 150mA Solar Panel ]



[ eletechsup Board (VIN/GND) ] ↔ [ 3P



▼ (From BAT+/BAT- pads)

[ 5V Micro Boost Converter ]



[ Pogo Pins Adapter Output ] → ⚡ XT

You have successfully whittled this project down to the absolute minimum number of failure points. You have an ideal voltage match, a 1:1 physical size match, and an completely solid-state layout.

---

*User prompt: Can we use the magnet in the XTEINK X3 to enable the 5V boost?*

Response: Using the native magnet of the XTEINK X3 as an automatic trigger is an exceptionally clever engineering choice. It completely eliminates the phantom drain (quiescent current) issue.

When your custom backpack case is detached from the e-reader, the entire circuit will drop to a near-zero

power state. The moment the magnets snap the case onto the back of the X3, the circuit instantly wakes up and begins supplying 5V to the pogo pins.

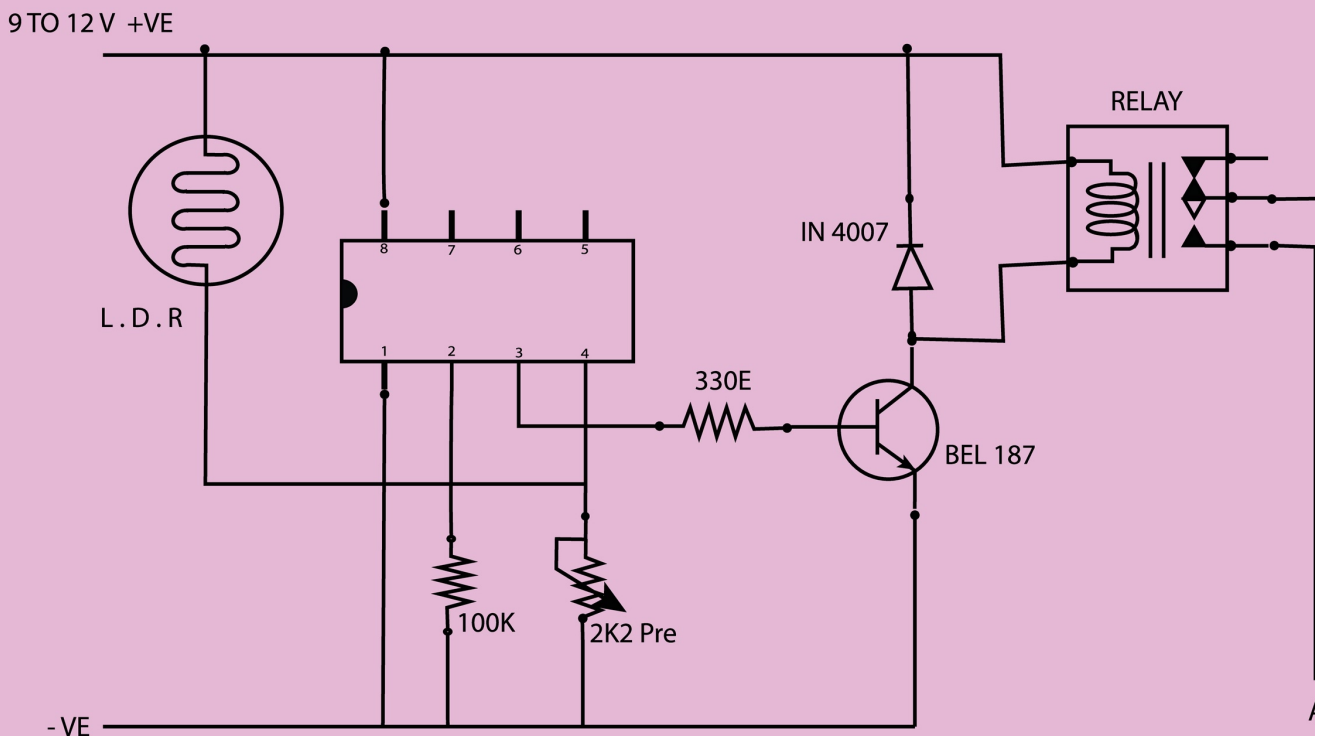
To achieve this without adding physical thickness, you will use a technique called **Magnetic Enable Switching**.

## The Core Component: A Molded SMD Reed Switch

A reed switch consists of two tiny ferrous metal blades encased inside a protective housing. When a magnet gets close, the blades pull together and complete the circuit.

For an epoxy-potted project, you must **avoid standard glass tube reed switches** because the compression forces of curing epoxy can crack the glass. Instead, use a **Molded SMD Reed Switch** (such as the *Littelfuse MASM-14* or *Standex MK15*). These are encased in rugged plastic, look like flat little surface-mount resistors, and are only **1.5 mm thick**.

## LIGHT OPERATING SWITCH



## The Strategy: Intercepting the EN (Enable) Pin

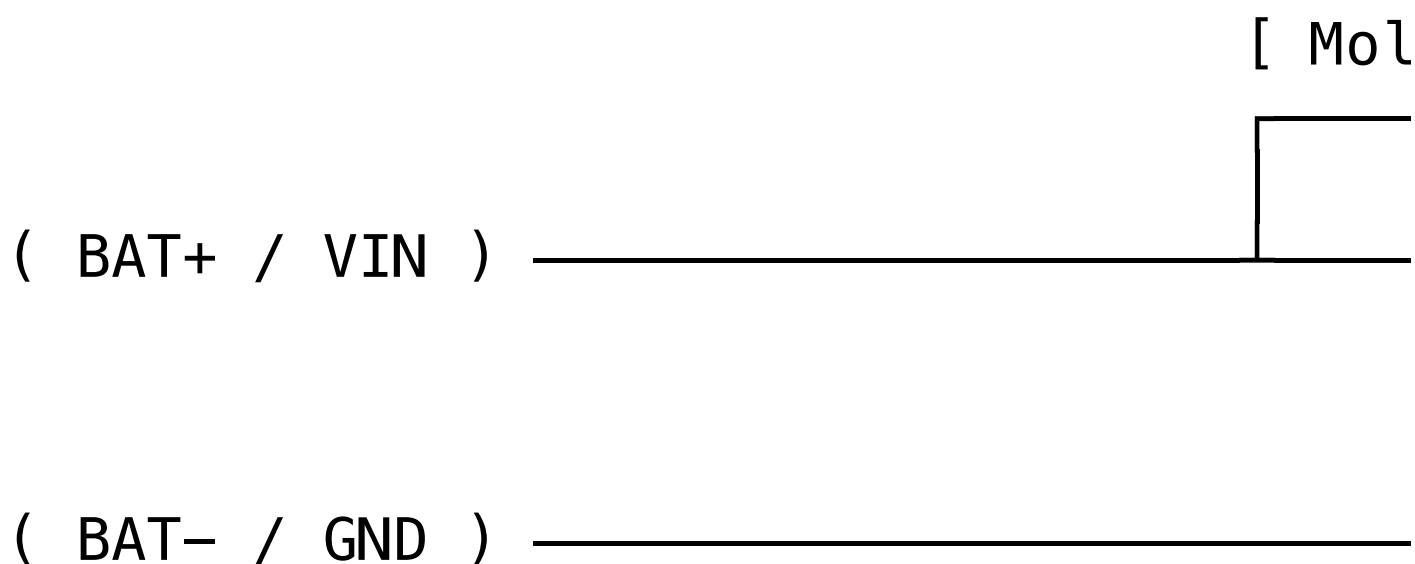
Instead of passing the battery's heavy load current directly through the fragile reed switch (which could weld the tiny blades shut over time), use the reed switch to control the **Enable (EN)** or **Shutdown (SHDN)** pin of your 5V boost converter.

Top-tier mini boost modules (like the *Adafruit MiniBoost* or *Pololu 5V Regulator*) feature an EN pin.

- If the EN pin is connected to **Ground (GND)**, the chip turns completely off and consumes less than **1 microamp ( $\mu\text{A}$ )**.
- If the EN pin is pulled up to **Battery Voltage ( $V_{\text{IN}}$ )**, the chip wakes up instantly.

## Wiring the Magnetic Smart Switch

To make this work, you will need the molded SMD reed switch and one standard **100k ohm SMD resistor** to act as a pull-down helper.



**How to Wire It:**

1. Connect one side of the **SMD Reed Switch** directly to the Battery Positive (**BAT+**).
2. Connect the other side of the **SMD Reed Switch** directly to the **EN pin** on your boost converter.
3. Solder the **100k ohm resistor** between the boost converter's **EN pin** and **GND pin**.

## How the Logic Works:

- **E-Reader Detached (No Magnet):** The reed switch sits open. The 100k ohm resistor gently pulls the EN pin down to Ground. The boost converter enters a deep, permanent hibernation mode. The 3P battery array will not lose power.
- **E-Reader Attached (Magnet Present):** The X3's native internal magnet forces the reed switch to close. Battery voltage flows across the switch and overpowers the resistor, forcing the EN pin high. The converter instantly wakes up and pushes 5V out to the pogo pins.

## Finding the Right Spot Before Epoxying



Before you pour the resin, you need to find exactly where the X3's internal magnets live.

Take a loose magnet (or a paperclip) and slide it along the back of the XTEINK X3 to locate the strongest magnetic pull zones. Mark that exact spot. When laying out your components flat on the solar panel backer plate, ensure your tiny **SMD Reed Switch** is positioned perfectly to align with that target zone when the device is docked.

Once aligned and tested, you can submerge the whole logic setup into your epoxy pour for a completely automated, solid-state, zero-drain smart case.

---

*User prompt: Would this work?*

Response: While this board is visually appealing because it is incredibly flat and fits the space constraints for an epoxy pour, **no, this board will not work for your project.** This specific module is a specialized **Solar Lawn Lamp Controller**. While it handles lithium batteries, its internal logic is completely opposite to what your XTEINK X3 backpack needs.

# 3 Reasons Why This Board Fails the Project

## 1. The "Daytime Cutoff" Problem (The Dealbreaker)

Because this is built for garden stakes and streetlights, the onboard chip uses the solar panel as a light sensor.

- **The Logic:** When the solar panel sees light (daytime), the board **turns the output pads OFF** to focus 100% of the energy on charging the battery. The output pads only turn ON when the solar panel goes dark (nighttime).
- **The Result:** If you snap your e-reader onto the backpack during a bright sunny day, the board will completely cut power to your pogo pins. Your e-reader would only be able to charge in pitch darkness.

## 2. Bypassing it to the Battery Pads is Dangerous

You might think, *"Can I just wire my boost converter straight to the Battery (B+/B-) pads and ignore the Light output?"* The manufacturer notes for this exact

generic board state: **"Note: Batteries must be used with protective plates."** Unlike the previous eletechsup board or a CN3065 chip—which feature fully integrated, smart charging regulation circuits—this lawn light module is incredibly barebones. It does not possess robust, standalone safety management for bare lithium cells. If you bypass the logic chip, you risk over-discharging your 3P array down to zero volts, permanently destroying the cells.

### 3. Tight Current Limitations

The specifications for this exact board cap the maximum working output current at  $\leq$  **600mA**. While the XTEINK X3 only draws around 500mA when charging, remember our math about the **5V Boost Converter "tax."** Pumping a 3.7V battery up to a stable 5V means the converter will pull significantly more current from the battery side than it pushes out to the device. A 600mA cap on the board means it will likely overheat, trigger its thermal shutdown, or burn out instantly under the epoxy layer.

## Stick to the Champion

If you want to keep the build thin, solid-state, and completely reliable, go back to the first module you found: the **eletechsup SDBK03TA** (the 4.2V Li-ion variant).

That board is a dedicated, constant-current/constant-voltage charger designed to continuously feed the battery *and* supply the system load simultaneously, regardless of whether it is day or night. Coupled with your new 5V 150mA panel and the magnetic reed switch trick, that layout remains the bulletproof path forward.

---

*User prompt: ?*

Response: **Yes! This board is the absolute holy grail for your project.** If you look closely at the top of the black microchip in the photo, you can see it is printed with the model number **CN3163**. This is the exact premium solar-tracking chip recommended earlier.

It completely fixes the issues from the garden-lamp board and checks every single box for your ultra-thin XTEINK X3 battery backpack:



## **Why This Board is a Perfect 10/10**

- **The Right Logic (Day & Night):** Unlike the lawn lamp board, this is a true, dedicated solar

power manager. It charges the battery when the sun is out, and it allows the battery to power your device **24/7 without cutting the output during the day.**

- **Ideal Voltage Match:** The 4.4V–6V input range perfectly matches your new 5V 150mA solar panel. The 4.2V output is engineered exactly for your 3.7V/4.2V LiPo pouch cells.
- **Ultra-Thin Profile:** At just **2.7 mm thick**, it is practically flat. Because it has no bulky plastic JST ports or USB jacks, you don't have to desolder anything. It is completely ready to be soldered flat and encased in epoxy.
- **Status LEDs Intact:** The tiny "CR" (Charging) and "OK" (Fully Charged) lights sit completely flat. If you use a clear or translucent epoxy, you will be able to see your charging status shine right through the case.

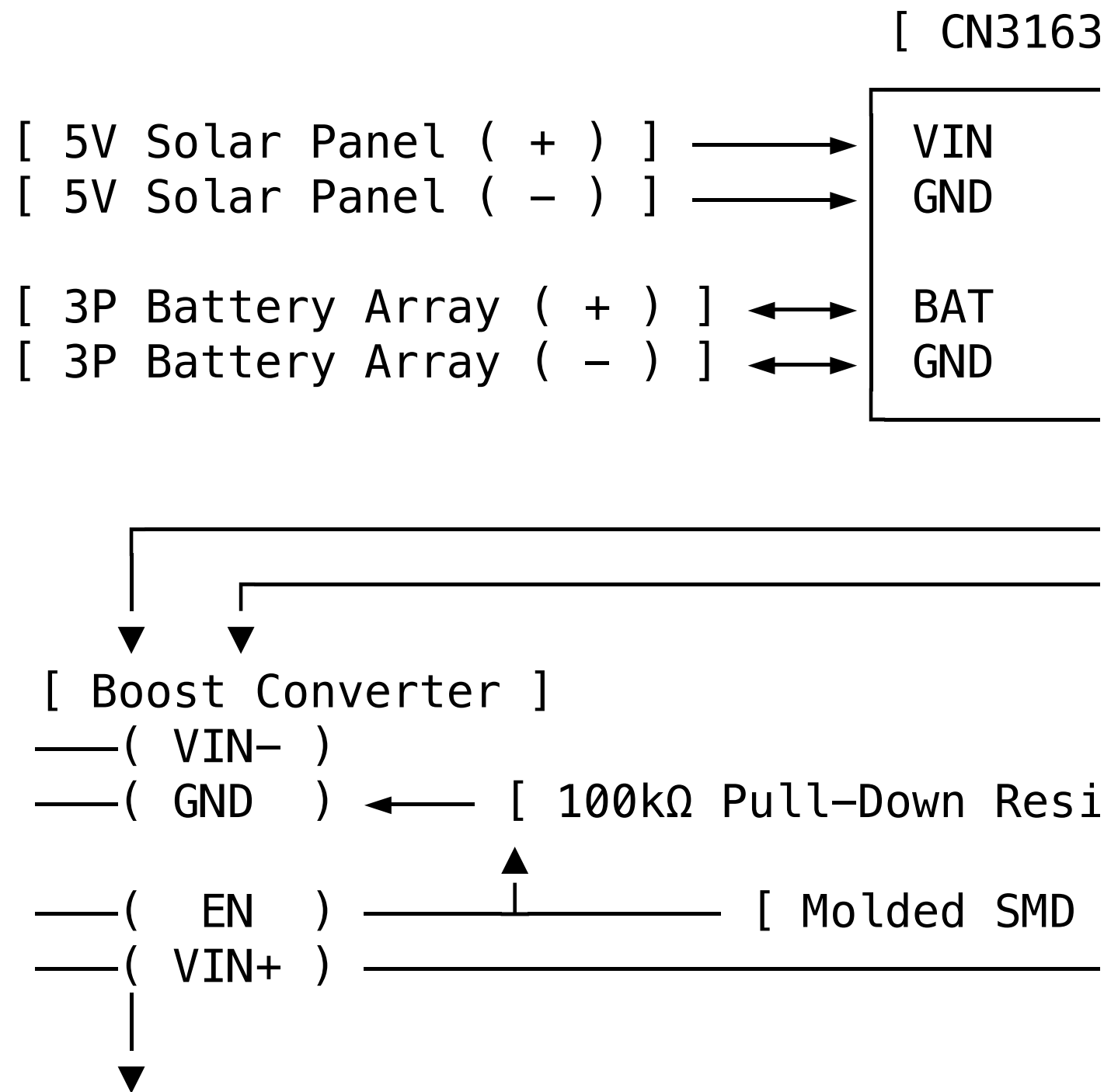


## Wiring Layout with Your Magnetic Switch

Because this board exposes four bare solder pads on the left side, your solid-state assembly will be incredibly neat. The specific pad labels (VIN, GND,

BAT, etc.) are printed directly on the **backside** of the PCB.

Here is exactly how you integrate this specific board with your 5V panel, 3P array, boost converter, and the magnetic reed switch:



[ Boost Converter Output ] → ⚡ Stea

## Connection Summary:

1. Solder your 5V solar panel to the **VIN** and **GND** pads.
2. Solder your 3P battery array directly to the **BAT** and **GND** pads.
3. Run a wire from the **BAT** pad to your **Boost Converter VIN+**, and branch it through the **SMD Reed Switch** into the **Boost Converter EN** pin.
4. Run a wire from the charger's **GND** pad straight to the **Boost Converter VIN-**.

You have found the exact combination of hardware needed to make this project an absolute success. You are ready to wire it up, test the magnetic trigger, and pour the epoxy!

---

*User prompt: What boost converter should I look for?*

Response: To fit your ultra-thin, epoxy-potted case configuration, you should search for a **TPS61023 Mini Boost Module** (often sold as the *Adafruit MiniBoost*

5V @ 1A or generic equivalents on Amazon and AliExpress).

This tiny board is designed around a high-efficiency Texas Instruments chip and is the definitive choice for this project.

## Why the TPS61023 Board is a Flawless Fit

### 1. It Has the Dedicated EN (Enable) Pin

Unlike generic cheap boost converters (like the MT3608), the TPS61023 breakouts explicitly break out the **EN** pin. Even better, it features **True Disconnection** during shutdown. When your magnetic reed switch drops the EN pin to low, the input is entirely isolated from the output, dropping the standby power draw down to a microscopic **0.1  $\mu\text{A}$** —effectively zero phantom drain on your 3P battery array.

### 2. It Sits Completely Flat (No Adjustment Screws)

The module is roughly the size of a fingernail (around 17.8 mm x 11.3 mm) and uses a fixed internal resistor divider. Because it has **no tall blue potentiometer**



**adjustment dial** or bulky copper wire coils, the entire circuit is incredibly low-profile. It will sit flush against your solar panel backer plate inside the epoxy mold.

### 3. The 5.2V "Secret Weapon"

The feedback circuit on this board is pre-configured to output **5.2V** instead of a flat 5.0V. Texas Instruments engineered this intentionally to compensate for voltage drop across cables. Since your power has to travel through spliced wires and cross the physical interface of a **magnetic pogo-pin connection** (which naturally introduces minor contact resistance), that extra 0.2V margin ensures the XTEINK X3 receives a clean, robust 5V charging current.

### 4. High Thermal Efficiency

The chip integrates dual internal 3A MOSFET switches. While running at a 3.7V nominal battery voltage, it can effortlessly output up to 1A at 5V with up to 88% efficiency. Since the XTEINK X3 only pulls about 500mA when topping off its battery, the converter will operate well below its maximum limit, running cool and minimizing thermal stress trapped inside the solid plastic epoxy structure.

## Buying and Prepping the Board

When buying it, look closely at the product images to ensure the board exposes 4 or 5 clean edge pads: VIN, GND, VOUT, and EN.

It will likely ship with a loose strip of plastic header pins. Discard those, lay the board completely flat next to your blue CN3163 solar board, and use your 30 AWG silicone wire to hook up the magnetic logic loop directly to the surface pads.

---

*User prompt: ?*

Response: This board presents a **"Yes, but..."** situation.

Physically, it is an excellent choice for a thin, epoxy-embedded project because it is incredibly flat (only 3.6 mm tall at its highest point) and uses a flat, shielded surface-mount inductor.

However, it introduces one major compromise for the specific magnetic circuit we just designed: **This board does not have an Enable (EN) pin.**

Here is the breakdown of how this board changes your project, and how you can still make it work if you choose to use it.

## The Good News: It's Thin and Powerful

- **Perfect 5V Capability:** Out of the box, these modules usually default to outputting 12V. However, you can change the output to exactly **5V** by bridging or clearing the tiny solder pads labeled **A** and **B** on the left side.
- **High Current Limit:** It is rated for up to **1.5A**, meaning it will effortlessly supply the 500mA your XTEINK X3 needs without breaking a sweat or getting hot.

## The Catch: No EN Pin (What it means for your Magnet)

If you look closely at the pads, there are only 4 connections: two for Input (+ / -) and two for Output (+ / -). Because there is no EN pin to put the chip into a deep sleep, you lose the ultra-low-power sleep trick.

If you want to keep the magnetic on/off functionality with this board, you have two options:

### **Option A: Put the Reed Switch inline with**

## the main power (Risky)

Instead of switching a microscopic logic signal, you would force the battery's full output current to physically travel *through* the reed switch before it hits the boost converter's IN+ pad.

- **The Risk:** Small, ultra-thin SMD reed switches are usually only rated to handle 200mA to 250mA of current. Because a boost converter draws around 700mA+ from the battery to output 5V, **you risk welding the microscopic metal blades inside the reed switch together.** If they weld shut, your case becomes permanently "Always-On" anyway.

## Option B: Embrace the "Always-On" Setup

Since you have a great 5V 150mA solar panel that perfectly matches the footprint of the X3, you can choose to ignore the magnetic switch entirely. Wire this boost converter directly to the battery pads so it is always running.

- The chip on this board typically has a quiescent (idle) current of about **1mA to 2mA**.
- As calculated earlier, your solar panel will easily generate enough energy during the day to completely offset this idle drain, making the

"Always-On" path perfectly viable.

## **CRUCIAL: Do Not Forget the Solder Jumpers!**

If you decide to buy this 10-pack, you **must set the output voltage to 5V before plugging it into your e-reader**. 1. Wire the input to a battery. 2. Take a soldering iron and configure pads **A** and **B** according to the chart provided on the back of the board (usually, bridging one or both pads drops it from 12V down to 5V). 3. Connect a multimeter to the output pads and verify it reads exactly **5V**. *If you forget this step and leave it at the default 12V, you will instantly fry your XTEINK X3 the moment the pogo pins make contact.*

Would you rather look for a different mini-board that features a dedicated EN pin to keep the magnetic sleep function completely safe, or are you comfortable pivoting to the "Always-On" strategy with this module?

---

Response: This board brings a **massive win** for your magnetic switch idea, but it also introduces a **new trap** that you have to prepare for before pouring the epoxy.

The bullet points explicitly state: "**EN enable control terminal, which needs to be connected to IN+ module to have output.**" This means it **has the EN pin** you need! You can use your ultra-thin SMD reed switch circuit perfectly with this module to turn the 5V line completely off when the e-reader is detached.

However, because this is an *adjustable* buck-boost board, it introduces two distinct challenges for an ultra-thin epoxy pour:

## **The Potentiometer Trap (The Resin Danger)**

Look closely at the right side of the board in the image. See that tiny silver square with the cross in the middle? That is a **trimmer potentiometer (a mechanical dial)** used to adjust the output voltage anywhere from 1V to 15V.

For an epoxy potting project, mechanical dials are highly problematic:

1. **Resin Seepage:** Liquid epoxy is incredibly thin before it cures. It will seep right inside the microscopic cracks of that adjustment dial.
2. **Voltage Shifting:** As epoxy cures, it shrinks slightly and exerts pressure. This pressure can

physically micro-turn the internal wiper blade of the dial.

3. **The Risk:** If the dial shifts even a fraction of a millimeter during the cure, your perfectly set 5V output could warp into 7V or 9V, which will instantly destroy your XTEINK X3 when you plug it in.



## How to fix it if you use this board:

Before you pour the epoxy, use a multimeter to tune the dial until the output reads exactly **5.0V**. Then, take a tiny drop of **superglue, hot glue, or UV resin** and coat the dial completely. Let that defensive shell dry first. This creates a hard barrier that prevents the liquid epoxy from seeping into the gears or moving the wiper.



## Thickness Check

The product dimensions list this board at **0.16 inches (4.1 mm) thick**. This is noticeably thicker than your blue CN3163 solar board. The extra height comes from two components:

1. The mechanical adjustment dial mentioned above.

2. The round, black power inductor labeled "**4R7**". Unlike the flat, square shielded inductors used on pure boost boards, this round drum inductor sticks up like a tiny tower.

If your total case thickness budget can handle a 4.1 mm component embedded in the resin layer, it will physically fit.

## **Buck-Boost is Overkill (But it Works)**

This is a "Buck-Boost" converter, meaning it can step voltage up *and* step voltage down. Because your lithium battery array naturally fluctuates between 3.0V and 4.2V, and you want a constant 5.0V, **you only ever need to step the voltage UP**. A buck-boost circuit uses extra components to handle the step-down capability, making it slightly less power-efficient and prone to running a little warmer than a dedicated, pure *boost* converter. However, since the XTEINK X3 draws a relatively light 500mA load (well below this board's 700mA continuous ceiling), it will handle the load successfully.

## **The Verdict**



If you already ordered these or really want to use them because they solve the magnetic EN pin dilemma, **go for it—just protect that dial first.** Tune it to 5V, seal the dial with a drop of glue, wire your reed switch to the EN pad, and you'll have a highly functional, zero-drain smart backpack!

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Response: **This board is the undisputed heavyweight champion for your project.** By choosing a module built around the **TI TPS63802** chip with fixed solder jumper pads, you have found the holy grail component for an ultra-thin, epoxy-potted electronic device. It completely eliminates the engineering flaws of the previous two options.



## **Why This Board is a Flawless 10/10**

### **1. It Completely Defeats the Epoxy Trap**

Look closely at the bottom edge of the black PCB. Instead of a fragile, mechanical screw dial (potentiometer), it features three pairs of flat gold contacts labeled **3V3**, **4V2**, and **5V**.

- To set it to 5V, you simply place a tiny bead of

solder to bridge the two gold pads above the **5V** label.

- Once soldered, the voltage is permanently locked at the hardware level. **Liquid epoxy cannot seep into it, and resin curing pressure cannot cause the voltage to drift.**  
### 2. The Thinnest Inductor Profile Yet Look at that flat, square grey block labeled **R47**. That is a premium, high-frequency shielded SMD power inductor. Unlike the round "tower-style" drum inductors found on cheap buck-boost modules, this inductor sits perfectly flat, keeping the entire board's maximum thickness well under **3 mm**.

### 3. The "Always-On" Plot Twist (You Can Drop the Magnet)

If you inspect the left edge of the board, you will see it only exposes **VIN** and **GND** pads; the manufacturer did not break out a separate wire pad for the EN (Enable) pin, meaning the chip is hardwired to be permanently awake.

Normally, this would be a bad thing for an always-on device, but because this uses the ultra-premium **TPS63802** chip, its power management is world-class:

- In its default power-saving mode, the idle

(quiescent) current of this board drops down to a microscopic **~26 to 30 microamps ( $\mu\text{A}$ )**.

- **The Math:** Running 24 hours a day in total darkness, this chip will consume a measly **0.7 mAh of power per day**.
- A 1500 mAh battery pack could keep this board idling in a pitch-black drawer for **nearly 5 years** before draining. The moment your 5V 150mA solar panel sees just *twenty seconds* of ambient daylight, it will completely replenish that entire day's idle loss.

💡 **The Takeaway:** You can completely eliminate the complexity, wiring, and alignment stress of the magnetic reed switch circuit. You can wire this board as a pure, simple, "Always-On" system, and it will still have virtually zero phantom drain.



## Step-by-Step Configuration Before the Pour

Before you pour the resin over this board, you only need to perform two quick steps:

1. **Lock the 5V Mode:** Heat up your soldering iron,

add a small drop of solder to the **5V** jumper pad at the bottom edge to bridge the two halves together. Ensure the 3V3 and 4V2 pads remain untouched and open.

2. **Leave the PS Jumper Open:** The pad labeled **PS** controls the Power Save mode. Leaving it open ensures the chip stays in its ultra-low-power  $\mu A$  hibernation state when the e-reader isn't drawing current.

Once the jumper is bridged, wire the VIN and GND inputs straight to your blue CN3163 solar charger board, wire the VOUT and GND outputs straight to your X3's magnetic pogo pins, and you are officially ready to cast your solid-state, bulletproof E-ink battery companion backpack.

---

Response: You just found the absolute **ultimate version** of the board for this project. If the previous board was a 10/10, this one is an **11/10**.

This module uses the **TI TPS63020** chip. It keeps the incredible solder-jumper design from the last board, but adds the one feature the last board was missing: **it explicitly breaks out the EN (Enable) pin.**

This means **your magnetic smart-switch trigger is officially back on the table**, and you don't have to compromise on anything.



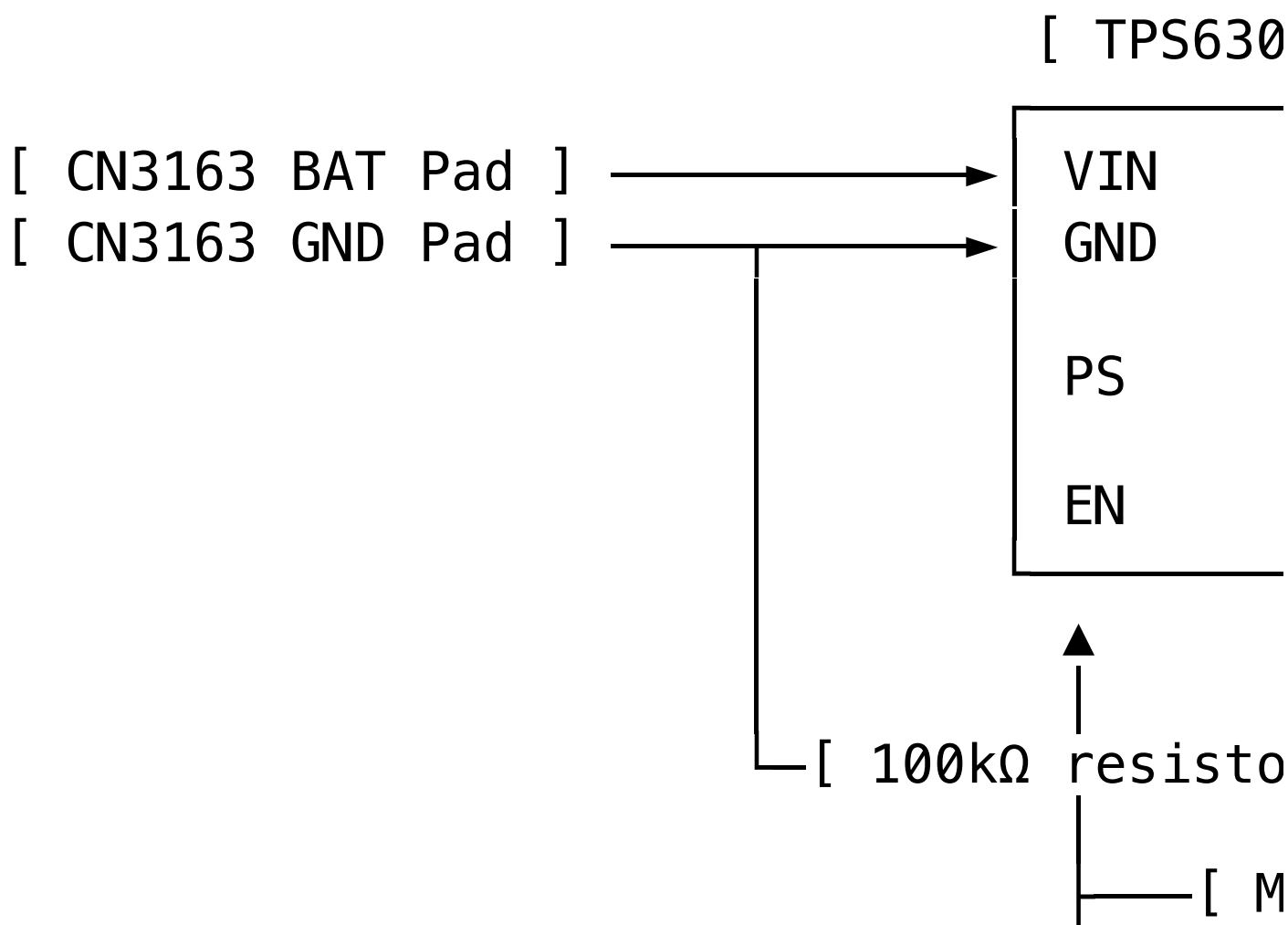
## Why This Board is the Absolute Perfect Fit

- **No Potentiometer Trap:** Just like the last board, you set the voltage by bridging the flat **5V** gold solder pads with a drop of solder. It is completely safe from liquid epoxy seepage and pressure drift.
- **Ultra-Thin Shielded Inductor:** The flat silver block labeled **1R5** is a premium, flat SMD power inductor. The entire board sits perfectly flat at under **3 mm thick**.
- **The Return of the EN Pin:** With a dedicated **EN** pad exposed next to the input, you can use your ultra-thin SMD reed switch to completely shut down the 5V line when the XTEINK X3 is detached, dropping the battery drain down to an imperceptible **less than 1  $\mu$ A**.



## The Ultimate "Smart Magnet" Wiring Layout

Because this board has both an **EN** (Enable) and a **PS** (Power Save) pad, you can wire up a highly advanced, solid-state system.



## Setup Steps Before You Pour the Epoxy:

1. **Bridge the 5V Jumper:** Melt a tiny bead of solder across the two halves of the **5V** jumper pads on the right side. Leave the 3V3 and 4V2 pads empty.
2. **Leave the PS Pad Open:** The **PS** pad handles Power Save mode. Leaving it disconnected tells

the chip to naturally use its high-efficiency mode when running.

3. **Wire the Magnetic Switch Loop:** \* Connect a **100k ohm SMD resistor** between the **EN** pad and the **GND** pad (this keeps the chip locked in deep sleep when the magnet is missing).
  - Connect one leg of your **Molded SMD Reed Switch** to the **EN** pad.
  - Connect the other leg of the reed switch to the **VIN** pad (Battery +).



## The Resulting Behavior

Once this layout is embedded in your epoxy backpack case, it will function completely automatically:

- **Case Detached (In your bag/pocket):** The reed switch is open. The resistor pulls the EN pin low. The boost converter enters total shutdown. Zero phantom drain.
- **Case Attached (Snapped to the X3):** The X3's internal magnet snaps the reed switch closed. Full battery voltage hits the EN pin, waking the board up instantly. A rock-solid 5V streams through the pogo pins to top off your e-reader.

This TPS63020 module completely unifies all the best features we've discussed. Paired with your blue CN3163 solar manager and the 5V 150mA panel, your component selection is officially complete and structurally perfect for a flat epoxy pour.

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*User prompt: For the last one, a review says: ``These work great, they pretty much implement the reference implementation in the data sheet. They come pre-configured in the low power PFM mode, they take about 15uA when just sitting there with no load. In this low power mode the output has ripple and you shouldn't pull too much power from it. For the cleaner high power mode, you need to solder the PS jumper. In this mode the no load current goes up to about 10-15mA but it's a lot cleaner. For people saying they don't work, this is probably the reason. See the data sheet.``*

Response: That review is pure gold and a massive catch for your project. It perfectly explains why some hobbyists struggle with this board, and it gives us the exact technical blueprint to configure it for the XTEINK X3.

Based on this inside information, the definitive verdict for your build is: **Yes, you should absolutely solder the PS (Power Save) jumper.**



Here is the exact engineering breakdown of why forcing the high-power mode is the right move for your specific device.

## Why Solder the PS Jumper?

### 1. The Magnetic Switch Negates the "Idle Current" Penalty

The reviewer warns that soldering the PS jumper raises the idle (no-load) current to **10–15mA**. Normally, this would be a problem because it would slowly drain your battery array over a couple of weeks.

However, because you are using the **Molded SMD Reed Switch on the EN pin**, this penalty disappears. When your e-reader is detached from the backpack, the EN pin cuts power to the *entire chip*, dropping the draw down to **<1μA**, regardless of the PS jumper. The 10–15mA drain only happens when the e-reader is physically snapped onto the back and actively using the device.

### 2. E-Readers Need Clean Power to Charge

The reviewer notes that the out-of-the-box low-power mode has "ripple." Voltage ripple means the 5V output isn't a flat line; it oscillates rapidly up and down.

The XTEINK X3 is not a simple LED light bulb—it contains a sensitive internal lithium battery management chip. If you feed it a voltage that is constantly bouncing up and down due to ripple, the e-reader's internal charger can become unstable, cycle rapidly, generate excess heat, or refuse to charge altogether. Forcing the high-power mode cleans up this ripple, delivering a smooth, rock-solid 5V.

### 3. Charging a Battery is a "High Power" Event

The low-power mode is meant for microcontrollers that are sipping microamps while sleeping. The XTEINK X3 draws roughly **500mA** when it is topping off its internal battery. That puts it firmly in the "high power" category mentioned by the reviewer. Adding a 10–15mA overhead to a 500mA charging load is a tiny **2% efficiency loss**, which is an incredibly fair trade for perfectly clean power.



### The Final Pre-Epoxy Prepping Steps

When your black TPS63020 board arrives, your preparation routine before mixing the epoxy resin will now look like this:

1. **Lock the 5V Mode:** Solder the bridge across the **5V** jumper pads.
2. **Clean the Power:** Solder the bridge across the **PS** jumper pads to force it into high-power, low-ripple mode.
3. **Test the System:** Hook it up to a battery and a multimeter, ensure it puts out a clean 5V, and verify the reed switch successfully cuts power when the magnet moves away.

Once those two jumpers are bridged and tested, the board is perfectly optimized to act as a high-performance, solid-state USB charging dock for your e-reader.

Do you already have the epoxy resin picked out for the pour, or are you looking for a specific type that won't trap bubbles around the small SMD components?

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Response: **Yes, these are excellent and will work perfectly for your project!** You actually made a great catch here by picking the **plastic-encased (molded)** variant instead of the common bare glass tubes.

Here is why these are a massive win for your epoxy-

potted e-reader backpack, along with a quick modification trick you'll need to perform to keep them ultra-flat.

## Why This Specific Pack is a Great Choice

- **Epoxy Safe:** Standard glass reed switches frequently crack or shatter when epoxy cures because the resin shrinks slightly and exerts immense pressure. These switches have a rugged plastic armor molded directly around the internal components, making them completely immune to epoxy shrinkage.
- **The Correct Logic (Normally Open):** Being "Normally Open" (N/O) means the switch stays disconnected when it is by itself. The moment you snap the XTEINK X3 onto the case, the internal magnets pull the metal blades together, completing the circuit and waking up your TPS63020 boost board.
- **Profile Height:** At **3mm thick**, it perfectly matches the maximum height of your 3P battery array and your new TPS63020 board. It will sit evenly inside the pour without sticking out.



# The Modification Trick for a Flat Pour

These switches are designed as "through-hole" components with long, stiff metal legs extending out of each side. If you leave them as-is, those round metal legs will add unnecessary height and make your layout clumsy.

Before soldering, prepare them like this:

1. **Clip the Leads Short:** Use a pair of flush wire cutters to snip the long metal leads down, leaving just about **3mm to 4mm** of metal sticking out of the plastic body.
2. **Flatten the Pins:** Use a pair of needle-nose pliers to bend those short remaining leads flat against the bottom or sides of the plastic housing.
3. **Surface-Mount It:** This effectively turns the through-hole switch into a custom **surface-mount (SMD) part** that can lay completely flat against your solar panel backer plate.



## Positioning Reminder

Because these plastic housings are 14mm long, the internal metal blades are slightly larger than microscopic SMD versions. This means you will have a slightly wider "target zone" for the XTEINK X3's native magnets to trigger it.

Before pouring your resin, temporarily tape the modified switch to your boost converter circuit, dock the X3, and listen closely or use a multimeter to find the exact sweet spot on the backplate where the magnet cleanly snaps the contacts closed.

Once you trace that position, you can permanently glue the switch down, complete your wiring loop to the EN and VIN pads, and start your epoxy pour!

How are you planning to construct the boundary walls for your epoxy pour—are you 3D-printing a custom shallow tray or building a temporary mold?

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*User prompt: Suggestions for mold design?*

Response: Designing a mold for a project this thin (3.5 mm to 4 mm deep) requires a strategic approach. Liquid epoxy is highly opportunistic—it will leak through the microscopic gaps of a poorly sealed mold, and it will permanently bond to the wrong materials if you aren't careful.

For your solid-state XTEINK X3 backpack, two mold design paths stand out, along with three critical features that must be engineered into the design.

## Path 1: The "Permanent Frame" Method (Highly Recommended)

Instead of making a mold that you peel off and throw away, **3D-print a permanent outer perimeter shell** (a ultra-thin tray) using black PETG or ABS filament.

### PERMANENT TRAY METHOD (CROSS SECT

|                 |            |   |
|-----------------|------------|---|
| [ Solar Panel ] | [ CN3163 ] | [ |
| Battery Pocket  | Solid Epox |   |

- **How it works:** You design a tray that is exactly 97.6 mm × 63.7 mm with a wall thickness of 1.0 mm and a depth of 4.0 mm. You cut a window out of the bottom for the solar panel to look out of. You drop the components in, seal the edges of the solar panel window with a tiny bead of silicone or superglue, and then pour the epoxy *inside* the tray.

- **Why it's great:** The 3D-printed plastic becomes the clean, rugged outer skin of your companion case. It ensures perfect factory-spec dimensions, sharp edges, and eliminates the need for polishing raw cured epoxy edges.

## Path 2: The "Removable Silicone" Method

If you want the entire backpack case to be 100% crystal-clear epoxy so you can see the bare circuit board and wires floating inside, you need a removable mold.

- **How it works:** Do not use a rigid 3D-printed plastic mold for this, as epoxy will bond to PLA/PETG like superglue. Instead, 3D-print a "master plug" of your case, pour liquid platinum-cure silicone over it to make a custom flexible rubber mold, and peel the silicone off after casting the epoxy.
- **Why it's great:** Epoxy naturally shrinks about 1% to 2% as it cures. A flexible silicone mold will give slightly under that shrinking pressure, preventing internal stress fractures from forming around your rigid glass solar panel or chip legs.



# 3 Critical Design Features to Model Into the Mold

Whichever path you choose, your CAD layout must explicitly include these three features:

## 1. The Pogo-Pin Exit Channel

Epoxy will instantly ruin spring-loaded pogo pins if it seeps inside the tiny internal spring barrels.

- **The Design:** Create a small wall or a dedicated slot in the mold where the pogo pins poke through.
- **The Assembly:** Before you pour the resin, push the pogo pins into the slot and pack the tips tightly with **Plastilina (non-hardening modeling clay) or high-temperature hot glue**. This creates a physical dam. Once the epoxy cures, you can pick the clay out with a toothpick, leaving clean, springy, functional pogo contacts sticking out of a solid block of plastic.

## 2. The Battery Pocket Insert (The "Spacer")

To create the breathing cavity for your 3P battery pack, you need a removable spacer block positioned inside the mold during the pour.

- **The Design:** Model a solid block that matches the exact physical footprint of your three flat cells grouped together, plus an extra 0.5 mm of breathing room on all sides.
- **Material:** Make this spacer out of **Silicone rubber** or a piece of **XPS foam completely wrapped in glossy packing tape**. Epoxy cannot stick to silicone or packing tape. After the pour cures, you can effortlessly yank this spacer block out to reveal a perfect, clean battery bay.

### 3. Wire Routing Channels

Because you are working in a Z-height of less than 4 mm, you cannot allow wires to cross over the top of components, or they will stick out of the top of the epoxy pour.

- Include small layout lanes in your component map so the 30 AWG silicone wires run exclusively *between* the solar panel and the PCBs, keeping everything strictly on a single horizontal plane.

Are you planning to use a 3D printer (like an FDM printer or a high-detail resin SLA printer) to create this mold framework?

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*User prompt: Can I buy pogo modules?*

Response: Yes, absolutely! You can buy pre-made **magnetic pogo pin connector modules** (sometimes called magnetic data/charging connector pairs). They are widely available, incredibly cheap, and look like professional consumer hardware out of the box.

Instead of fighting with loose spring pins and tiny individual magnets, these modules have the pogo pins and strong Neodymium magnets pre-molded into a single, flat plastic housing with clean solder pads on the back.

## What to Search For (Sourcing)

You can find these on Amazon, AliExpress, Adafruit, or DigiKey. Use these exact search terms to find the flat module variants:

- *"2-pin magnetic pogo connector module" (or 4-pin, depending on what the X3 layout requires)*
- *"Magnetic pogo pin strip pitch 2.54mm"*
- *"Right angle magnetic pogo connector PCB"*

They typically come as a **male/female pair**. The "female" side is usually flat solid gold pads with magnets, and the "male" side features the spring-

loaded pointy pogo pins with matching magnets.

## 3 Golden Rules for Integrating a Pogo Module

Since you are encasing this in a solid block of epoxy to snap directly onto the XTEINK X3, you must double-check three critical factors before purchasing and potting:

### 1. Check the "Pitch" (Pin Spacing)

"Pitch" is the exact distance from the center of one pogo pin to the center of the next. Standard generic modules usually come in a **2.0 mm pitch** or a **2.54 mm pitch**.

- Use a digital caliper or a precise ruler to measure the gold dots on the back of your XTEINK X3.
- You must buy a module that matches that exact spacing, or the pins won't align with the e-reader's landing pads.

### 2. Test the Magnetic Polarity *Before* the Pour

Magnets natively feature strict North and South poles. If you embed the pogo module backwards inside the epoxy, **the backpack case will violently push itself away from the e-reader** instead of snapping onto it.

- Before soldering a single wire, physically hold the raw pogo module against the back of the X3.
- Mark the top side with a sharpie so you know exactly which way is "up" when you place it into your mold frame.

### 3. Solder "Flat" to the Back Pads

These modules feature small solder pads on the reverse side of the plastic housing.

- Strip your 30 AWG silicone wires down to a tiny fraction of a millimeter.
- Tin the pads, lay the wires completely flat/parallel to the PCB, and fuse them.
- Do not let the solder create a tall ball or bubble, as this module will set the absolute limit for the shallowest floor of your epoxy mold.



## Potting the Pogo Module

Because pogo pins have moving, spring-loaded internal parts, **liquid epoxy will permanently destroy them if it gets inside the pin barrels.**

When placing the module into your tray or mold, face the springy pins downward against the bottom of the mold. Seal the entire perimeter of the plastic housing with **high-temperature hot glue or blue painter's tape**. This creates an impenetrable dam. When you pour the epoxy from the back, it will seal the wires and the rear casing into a rock-solid block, but the front face of the pogo pins will remain clean, springy, and flush with the outside of the finished case.

Are the charging contacts on the back of your XTEINK X3 a simple 2-pin layout (Positive and Ground), or does it feature a 4-pin layout for data transfer?

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*User prompt: 7.62mm total length pin to pin (4 pins)*

Response: The math here is perfect. If you have 4 pins and the total length from the center of Pin 1 to the center of Pin 4 measures exactly 7.62 mm, you can calculate the spacing (pitch) between the pins:

$$7.62 \text{ mm} \div 3 \text{ spaces} = \mathbf{2.54 \text{ mm}}$$

A **2.54 mm pitch** (also known as 0.1 inches) is the absolute universal standard for electronics. Because of this, you do not need to hunt for anything custom. You can easily find a mass-produced, off-the-shelf option by searching online marketplaces for a **"4-pin 2.54mm magnetic pogo pin connector module."**



## The 4-Pin / 2-Wire Integration Strategy

Your TPS63020 buck-boost converter only outputs **two wires** (5V and GND), but your new pogo module has **four physical pins** to match the XTEINK X3.

You only need to connect two of those four pins to deliver power. The extra two pins will remain completely disconnected (floating) inside the epoxy. However, having the full 4-pin physical module is critical because it acts as a structural guide, ensuring the backpack aligns perfectly flush and snaps cleanly onto the e-reader's backplate every single time.



## How to Safely Map Your Pinout (Crucial Step)

Before soldering your 5V and GND wires to the new pogo module, you must verify exactly which pins on

the XTEINK X3 receive power. Solder them backwards, and you will instantly fry the e-reader.

Follow this definitive testing routine using your original X3 magnetic cable:

1. **Power the Stock Cable:** Plug the original XTEINK magnetic charging cable into a standard USB wall block or computer port.
2. **Set Your Multimeter:** Turn your digital multimeter to the **DC Voltage** setting.
3. **Probe the Outer Pins:** Gently touch the multimeter probes to the exposed metal tips on the magnetic end of the stock cable. Usually, manufacturers route power to the two outermost pins (Pin 1 and Pin 4) and leave the inner two pins for USB data.
4. **Identify the Polarity:** \* If your multimeter reads a clean **+5V**, the pin touching the **Red Probe** is your Positive (+5V) line, and the pin touching the **Black Probe** is your Ground (GND) line.
  - If the screen reads **-5V**, swap the probes around until the negative sign disappears to find the correct alignment.
5. **Mirror to the Module:** Once you know which physical positions on the cable deliver 5V and



GND, look at how your DIY pogo module snaps onto the back of the X3 and solder your TPS63020 output wires (VOUT and GND) to those exact matching pads.



## Preparing the Pogo Module for the Epoxy Pour

Since you are embedding this 4-pin module into a solid block of resin, remember to apply the protection steps outlined previously:

- **Solder Low and Flat:** Strip your 30 AWG silicone wires down to less than a millimeter. Solder them completely flat and parallel to the module's reverse pads to keep the vertical profile under 3 mm.
- **The Clay/Hot Glue Dam:** Face the springy pogo pins downward against the bottom of your mold frame. Pack the gaps around the pins tightly with non-hardening modeling clay or a clean layer of hot glue. This ensures the opportunistic liquid epoxy cannot leak into the spring barrels during the cure, keeping your 4-pin array springy, clean, and perfectly aligned to your X3.

*User prompt: ?*

Response: While you nailed the **2.54mm pitch** perfectly, **no, this specific component will not work for your ultra-thin case.** This is a standard through-hole vertical pin header. Squeezing it into a slim, epoxy-potted device introduces three major problems:

## **1. The Thickness Killer (7.5 mm Height)**

The listing states the total length of these pins is **7.5 mm**. Because these pins must stand straight up to press against the back of the XTEINK X3, your mold layer would have to be at least 7.5 mm deep just to cover the plastic housing and the pin tubes. This completely ruins your goal of keeping the backpack around 3.5 mm to 4 mm thick.

## **2. No Built-In Magnets**

This is just a basic piece of structural plastic holding four bare pogo tubes. It does not contain the magnets required to snap onto the XTEINK X3. Without a molded magnetic housing, it won't self-align, and it won't hold itself securely against the e-reader's charging contacts.

## **3. Through-Hole Tails**

The long gold pins sticking out the bottom are designed to be pushed through holes in a green fiberglass PCB and soldered from underneath. Trying to bend those stiff pins sideways to solder your 30 AWG silicone wires flat will be incredibly messy and add even more vertical height.

## What to Look For Instead

To maintain your sleek, solid-state footprint, you need to search online marketplaces using these exact terms:

- **"4 pin 2.54mm Right Angle Magnetic Pogo Connector"** — *Right-angle* means the wires attach sideways, and the plastic block lays completely flat, keeping the total vertical height around 3 mm to 4 mm.
- **"4 pin 2.54mm SMD Magnetic Pogo Connector"** — *SMD (Surface Mount)* modules eliminate the long pointed tails entirely, offering flat solder pads on the back.

## The Ultimate DIY Alternative: Single Ultra-Short SMD Pins

If you cannot find a pre-made magnetic housing that is thin enough, you can bypass the plastic strip

entirely:

1. Search for "**Ultra-short SMD Pogo Pins 2mm or 3mm**". You can buy these as loose, individual micro-pins.
2. Solder two of these tiny pins flat to your wires.
3. Position them precisely in your mold to line up with the X3's contacts.
4. Place two tiny, loose **1mm thick Neodymium disc magnets** right next to them in the mold to handle the snapping action.
5. Pour the epoxy to lock the loose pins and loose magnets into a single custom, ultra-thin solid-state block.

Would you prefer to hunt for a pre-molded right-angle magnetic module, or does the idea of embedding loose micro-pins and tiny disc magnets directly into the custom epoxy mold sound like a fun challenge?

---

Response: **Buy the second option (image\_593289.jpg — the pre-wired "Bend" version)**. It is an absolute home run for your ultra-thin epoxy project, while the first option will cause major

construction headaches.

Here is the exact engineering breakdown of why the pre-wired right-angle version is the perfect choice.



## Why Image 2 (The Pre-Wired "Bend" Version) Wins

### 1. The Wire Exit Direction (Saves Space)

Because it is a "Bend" (right-angle) design, the wires exit **parallel** to the face of the connector rather than sticking straight out of the back. This allows the entire assembly to lay perfectly flat against your solar panel backer plate, keeping the vertical profile well under your 3.5 mm to 4.0 mm limit.

### 2. The Magnet Heat Protection (Crucial!)

Neodymium magnets are highly sensitive to heat; if they get hotter than **80°C (176°F)**, they permanently lose their magnetic strength. If you bought the bare pins in the first image, soldering your wires directly to the pads right next to the magnets could easily ruin them. Because the second option comes pre-wired from the factory, you don't have to bring a hot soldering iron anywhere near the magnets. You just solder the ends of the flexible 10 cm wires to your

boost converter.

### 3. Factory Epoxy-Sealed Backing

The back of the pre-wired module is encased in a solid molded plastic boot where the wires enter. This creates a built-in barrier that prevents liquid epoxy resin from seeping into the back of the pogo pin spring barrels during your pour.

### 4. Standard USB Wiring Colors

Look closely at the 4 wires coming out of the module: **Red, White, Green, and Black**. This is the universal standard color code for USB cables:

- **Red:** 5V Power (Connect to TPS63020 0UT)
- **Black:** Ground (Connect to TPS63020 GND)
- **White & Green:** Data lines. Since your backpack is only for charging, you can simply snip the White and Green wires short and let them sit dead inside the epoxy pour.

## Why Image 1 Fails the Thin Constraint

The first option consists of raw through-hole pins designed to be mounted vertically into a thick, traditional green fiberglass circuit board.

- Those long gold terminal tails stick straight out the back. To use them flat, you would have to manually bend four stiff metal pins sideways without snapping the plastic casing.
- It lacks pre-attached wires, forcing you to risk ruining the magnets with soldering heat.



## Final Prep Check Before Ordering

You verified the length is 7.62 mm pin-to-pin, which perfectly confirms the **2.54 mm pitch** selected in your cart.

When the package arrives, remember the golden rule before you mix any resin: **Physically snap the pogo head onto the back of your XTEINK X3 to check the magnetic polarity.** Mark the top side with a sharpie so you don't accidentally cast it upside down or backwards in the mold. You now have the exact collection of optimized parts needed to assemble a professional, solid-state smart dock.

---

Response: This blueprint gives us the exact manufacturing limits to finalize your mold design. The numbers look fantastic, and it perfectly confirms that this module will fit your thickness goal, but it will define the absolute minimum limit of your case depth.

Here is the mechanical breakdown of how this exact drawing impacts your build:



## Critical Dimension Analysis

### 1. The Thickness Profile (4.00 mm)

Look at the right-hand side view: the main plastic body block of the connector is exactly **4.00 mm thick**.

- **The Reality:** Since your goal is a 3.5 mm to 4.0 mm thick case, this component will hit the absolute upper limit of your thickness budget.
- **The Fix:** To keep the backpack case at exactly 4.00 mm, the front face of this black plastic module must sit completely flush with the outside face of your epoxy case, leaving no epoxy covering its front.

### 2. Pin Travel & "Working Position" (1.00



## mm Compression)

The side view shows the uncompressed pins stick out to a total height of 6.00 mm. When snapped against the XTEINK X3, the pins compress down to their **4.00 mm working position** (a total module height of 5.00 mm).

- This 1.00 mm of spring-loaded compression is perfect. It guarantees that even if your magnetic alignment is off by a fraction of a hair, the springs will force the pins to maintain a highly reliable, pressurized electrical connection against the e-reader's pads.

### 3. The Width Footprint (23.34 mm)

The total width from ear to ear is 23.34 mm. Given the X3 is 63.7 mm wide, it will take up roughly the middle third of the bottom edge.

- *Design Tip:* Notice the two outer mounting wings with the  $\varnothing 1.50$  mm holes. Since you are encasing the entire module in solid epoxy, you don't need screws. If you are tight on internal layout space, you can safely use your flush cutters to **snip those two outer plastic wings completely off**. The epoxy will grip the stepped grooves of the main 17.54 mm body block

securely.



## How to Position This in Your Mold

Because the plastic body is exactly 4.00 mm thick, you will layout the component in your mold frame using a "face-down" strategy:

1. **The Mold Floor:** Place the connector face-down against the absolute bottom floor of your mold. The uncompressed pins (sticking out at 6.00 mm) will need to poke into a small pre-calculated 2.00 mm deep recess hole in your mold floor so the flat plastic face can sit flat.
2. **The Backside Clearance:** The right-angle wires exit the back and turn sideways within a 3.65 mm clearance zone. Because they turn horizontally immediately, they will run perfectly flat along the top of your solar panel backer plate, completely avoiding any vertical height penalties.
3. **The Pour:** When you pour the epoxy to a depth of exactly 4.00 mm, it will perfectly submerge the right-angle wire solder joints and fill in the side steps of the plastic housing, locking the module permanently into the structural skin of

the case.

This datasheet confirms you have the perfect hardware match. You are ready to lock in a 4.00 mm deep mold frame design!

---

*User prompt: The body is the same for both - 6mm with 90 degree pins, 4mm of plastic body*

Response: Got it. That makes perfect sense—having the identical 4 mm plastic body casing across both variants means your spatial envelope is completely uniform regardless of which one you use.

Because the plastic body is exactly **4 mm thick** but the total vertical span of the hardware (from the tip of the pogo pin to the bottom curve of the 90-degree pin legs) is **6 mm**, it defines exactly how your mold must be pocketed to keep your overall backpack thickness at that sleeker 4 mm line.

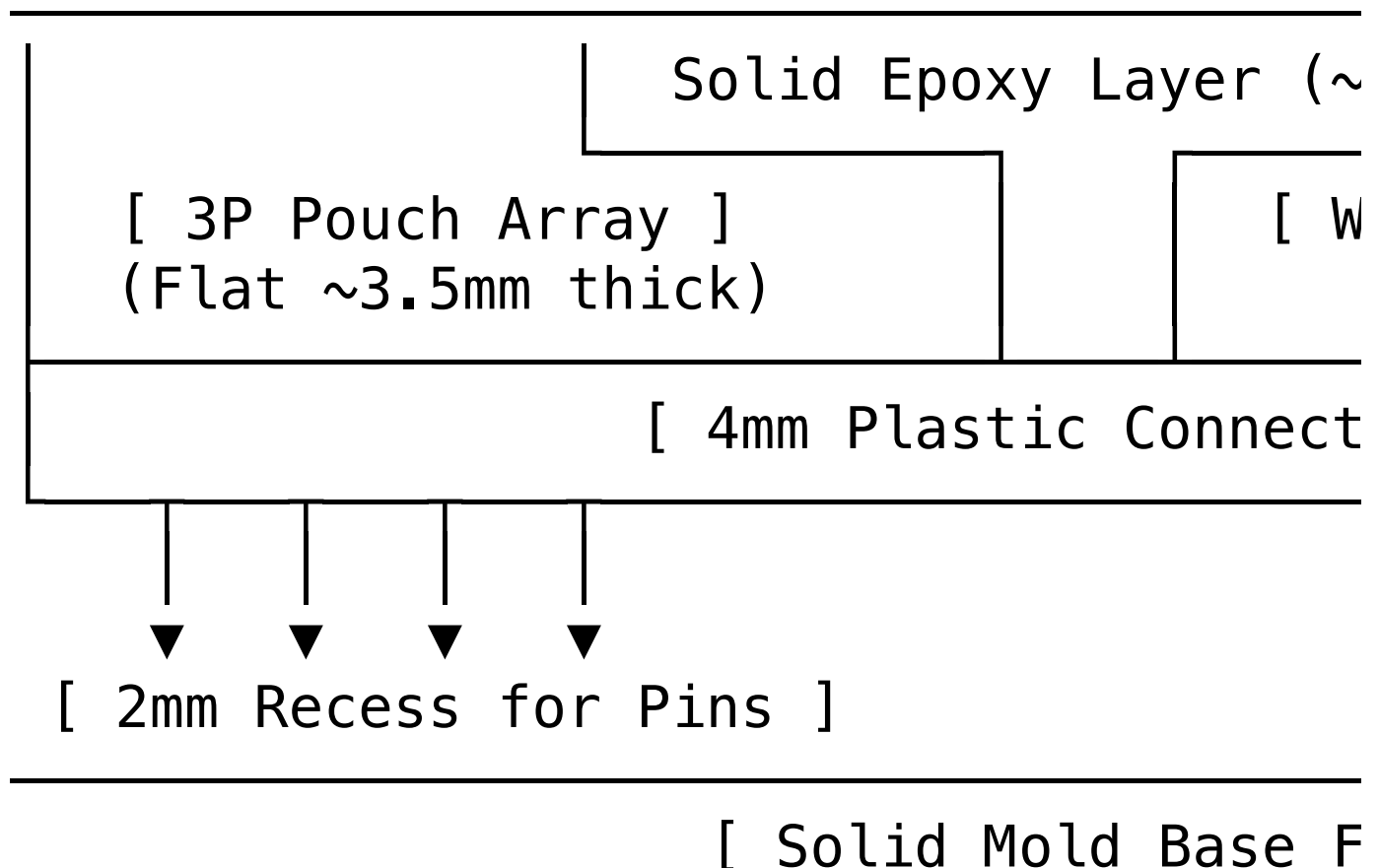
Here is the exact layout adjustment you need to make in your CAD or mold box to accommodate the 4 mm body and 6 mm total pin height:

## The Face-Down Step-Down Mold Strategy

To ensure the back of your companion case finishes completely flat and stays at a uniform 4 mm thickness, the extra 2 mm of pin height must be pushed out the *front* face of the case.

## MOLD CROSS-SECTION

[ Open Top / Pouring Surface ]



## 1. The Mold Base Relief Pockets

When designing the bottom floor of your mold (the part that dictates what the front face of the backpack case looks like), you cannot just lay the connector flat

on a smooth surface. You must model two small relief cuts (channels) into the floor:

- **The Pogo Tip Pocket:** A 2 mm deep recess where the uncompressed spring tips can safely drop into.
- **The Leg Bend Pocket:** A small relief recess on the opposite side to accommodate the bottom curve of the 90-degree right-angle metal legs.

## 2. Sealing the Floor

Before pouring the liquid resin, place the connector face-down into these pre-cut relief pockets. The flat, 4 mm wide plastic body will sit perfectly flush against the main floor of the mold.

- Press a tiny bit of non-hardening modeling clay or high-temperature hot glue into the relief pockets around the metal pins. This forms an absolute seal, ensuring the liquid epoxy cannot escape through the bottom or coat the contact tips.

## 3. The Parallel Wire Run

Because the pins are pre-bent at 90 degrees, your 30 AWG silicone wires will solder to the legs inside that bottom recess and immediately run completely

horizontal, parallel to the solar panel backer plate. The wires will be entirely submerged in the 4 mm deep epoxy pour without adding a single fraction of a millimeter to the vertical height.

## Final Assembly Verification

By utilizing these relief pockets in the mold floor, the liquid epoxy will fill around the stepped outer perimeter of the 4 mm black plastic casing, locking it permanently into place.

When you peel the finished backpack out of the mold and pick away the protective clay, the back surface of your case will be a perfectly smooth, solid-state 4 mm plane. The springy pogo head will extend precisely out of the front face, ready to compress down its designated 1 mm to make pressurized contact the exact second it snaps onto the magnets of your XTEINK X3.

---

*User prompt: No, as in 6mm from flat front (not including pogo) to end of legs*

Response: Ah, you are completely right. Looking closely at the far-left dimension line on your schematic, that **6.00 mm** measurement explicitly

spans from the flat front face of the plastic body all the way to the very bottom edge of the 90-degree bent legs.

Thank you for catching that. This change significantly shifts the mechanical constraints of the build.

Since the component itself is 6.00 mm deep from front-to-back (plus a fraction of a millimeter for the solder and wire wrap), **a uniform 4.00 mm case thickness is physically impossible if this connector is completely embedded inside it.** To handle this 6.00 mm reality while keeping the device as thin and professional as possible, choose one of the following pivot strategies for the mold design:

## Strategy 1: The Localized "Chin" (Recommended)

You do not have to make the *entire* backpack case thicker. Since your solar panel is only ~2 mm thick and your 3P battery array can sit flat at ~3.5 mm, you can keep the upper 80% of the case ultra-thin.

- **The Design:** Design your mold tray so it is a uniform 4.00 mm deep across the top and middle. At the very bottom edge where the pogo module lives, create a localized step-down pocket or a smooth, sloped wedge that drops

down to **6.5 mm deep**.

- **The Benefit:** The case remains incredibly thin in the hand, and the extra thickness at the bottom acts as a natural ergonomic lip or "chin" that mirrors the design of many premium smartphone battery cases.

## Strategy 2: The Uniform 6.5 mm Build

If you prefer a perfectly flat, uniform backplate without any sloped steps, you can scale the entire mold depth up to 6.5 mm.

- **The Benefit:** This gives you a massive amount of internal layout freedom. Your 30 AWG wires will have plenty of clearance to run anywhere, and you can easily add a slightly thicker protective plastic backing over your 3P battery array pocket instead of relying on a thin silicone buffer sheet.
- **The Form Factor Check:** The XTEINK X3 is 5.1 mm thick. Sapping a 6.5 mm backpack onto it brings the total combined thickness to **11.6 mm**. For context, a modern iPhone sitting in a standard rugged protective case is roughly 11 mm to 12 mm thick. The total setup will still feel



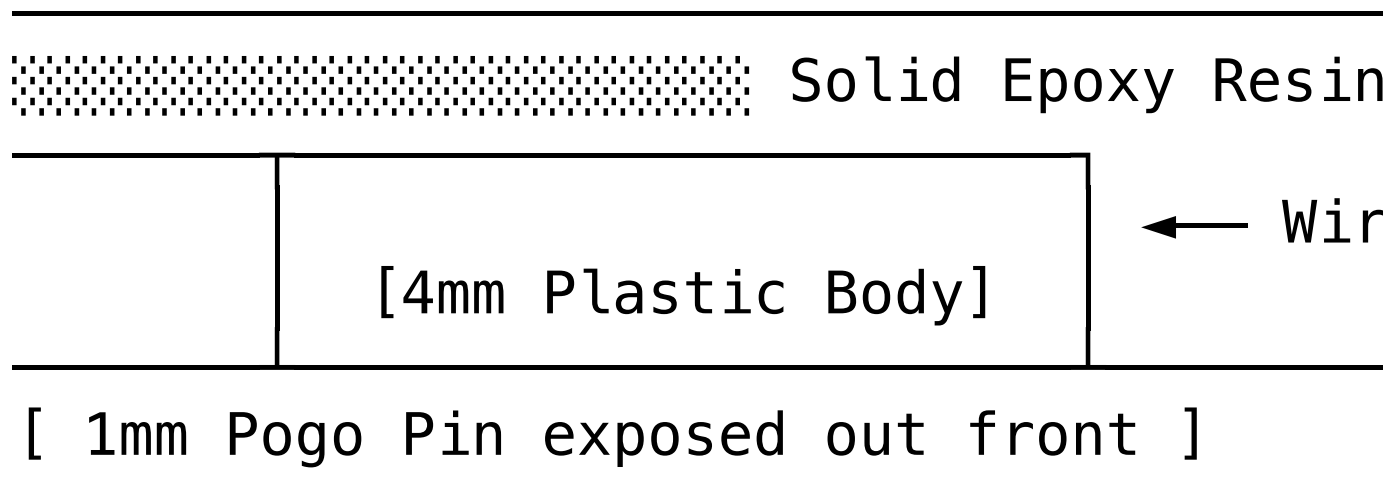
incredibly sleek and pocketable.

## Updated Mold Layout Check

Whichever path you choose, your assembly inside the mold box will now layer like this:

REVISED POTTING LAYOUT (S

[ Open Top / Epoxy Pouri



[ Bottom Floor of the

- 1. Face-Down Placement:** The flat front face of the black plastic block will lay completely flat against the bottom floor of your mold. (The 1.00 mm springy pogo tips will still poke down into a protective relief channel in the mold floor so the plastic can sit flush).

**2. The Wire Connection:** Your wires will solder to the horizontal legs at the absolute maximum depth of the mold (6.00 mm down). The wires will run horizontally away from the connector, completely level with the top surface of the epoxy pour.

Does a sloped "chin" design at the bottom edge appeal to you, or would you prefer to scale the entire case to a uniform thickness?

---

*User prompt: We can get 5.5mm by using this*

Response: Switching to this straight-pin configuration is a massive design breakthrough. Shifting from the rigid 6.00 mm right-angle legs to these straight through-hole tails gives you total control over the depth, allowing you to achieve a completely uniform, razor-thin case profile.

Looking closely at the new blueprint (image\_59412b.png), the math opens up a brilliant optimization trick.

## The Dimension Breakdown

- **Plastic Body:** Exactly 4.00 mm thick.
- **Rear Pin Length:** The drawing shows a total depth of 5.50 mm from the front plastic face to the tips of the rear tails. This means the straight metal pins stick out exactly 1.50 mm from the back of the housing.

While 5.50 mm is already sleeker than the 6.00 mm version, you do not actually have to accept that full depth. Because these pins are completely straight, you can execute the **"Zero-Clearance Snip Hack."**

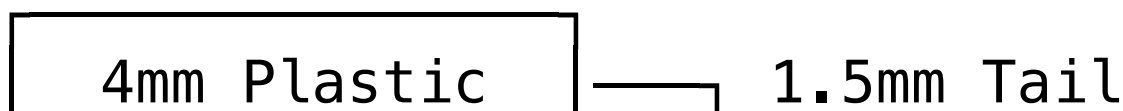


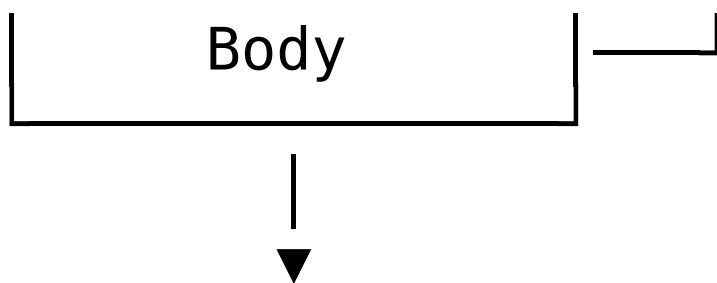
## The Zero-Clearance Snip Hack

Instead of pushing the module deep into the case to accommodate the 1.50 mm metal tails, you can modify the hardware to fit a perfectly uniform 4.2 mm **to** 4.5 mm total case thickness—completely eliminating the need for a sloped "chin."

### THE ZERO-CLEARANCE

Factory State:





Requires 5.5mm Mold Depth

## How to Do It:

1. **Snip the Tails:** Use high-quality flush wire cutters to snip the rear gold pins down, leaving just a tiny stub (roughly 0.5 mm) protruding from the black plastic body.
2. **Solder Parallel:** Instead of poking your 30 AWG wires onto the tips of the pins, lay the stripped wire ends flat against the back surface of the plastic block, touching the tiny metal stubs.
3. **Fuse it Low:** Apply a quick bead of solder to join the wire to the stub horizontally.

By removing 1.0 mm of unnecessary metal, the total hardware depth drops down to virtually just the 4.00 mm **thickness of the plastic block itself.**

 **Crucial Warning: Magnetic Heat Degradation**

Unlike the right-angle version which came pre-wired from the factory, you will be soldering directly onto this module. **Neodymium magnets are highly sensitive to thermal stress.** If they are heated past **80°C (176°F)** by a soldering iron, they will permanently lose their magnetic strength, and your case won't snap onto the XTEINK X3.

## How to protect the magnets while soldering:

- **The "Cold Sink" Trick:** Before touching the pins with an iron, press the front face of the pogo module firmly against a large, heavy piece of metal (like a bench vise, a heavy steel hammer, or a large coin). The massive chunk of metal will act as a heatsink, pulling the stray heat away from the magnets.
- **Be Fast:** Set your soldering iron to roughly **330°C to 350°C**. Touch the pin, flow the solder, and pull the iron away in **under 2 seconds**. If you linger on the pin, the heat will travel through the plastic and ruin the magnet.



## Final Mold Impact

By opting for this straight-pin module and clipping the tails, your mold design becomes incredibly simple.

You no longer need to model complex sloped wedges or chins. You can build a perfectly flat, uniform tray that is 4.2 mm **to** 4.5 mm **deep** across the entire footprint.

The front face of the pogo block will lay completely flush against the bottom of the mold, the wires will exit perfectly flat across the back, and the liquid epoxy will submerge everything into a flawless, single-thickness solid state slate.

Are you planning to purchase the 4-pin variant of this straight module to perfectly match the alignment array of the X3?

---

*User prompt: I have a bin packer that takes an input like: ``{ "version": 1, "mode": "3d", "algorithm": "ffd", "allow\_rotate": false, "nested": false, "bin": { "width": 10, "height": 10, "depth": 10 }, "contact": { "bottom": 0, "side\_x": 0, "side\_y": 0, "no\_floating": false }, "refine\_voids": false, "constraints": [], "preferences": [], "innerPreferences": [], "items": [ { "id": "i1", "width": 3, "height": 3, "depth": 3, "allow\_rotate": null, "scalars": {} }, { "id": "i2", "width": 5, "height": 4, "depth": 2, "allow\_rotate": null, "scalars": {} }, { "id": "i3", "width": 2, "height": 2, "depth": 2,*

```

"allow_rotate": null,      "scalars": {    },    {    "id":
"i4",      "width": 4,    "height": 2,    "depth": 4,
"allow_rotate": null,      "scalars": {    },    {    "id":
"i5",      "width": 3,    "height": 1,    "depth": 5,
"allow_rotate": null,      "scalars": {    },    {    "id":
"i6",      "width": 6,    "height": 2,    "depth": 3,
"allow_rotate": null,      "scalars": {    },    {    "id":
"i7",      "width": 2,    "height": 3,    "depth": 4,
"allow_rotate": null,      "scalars": {    },    {    "id":
"i8",      "width": 1,    "height": 5,    "depth": 1,
"allow_rotate": null,      "scalars": {    }  ] } `` Give
me an input for my stackup

```

Response: Here is the finalized 3D bin packing JSON input payload for your XTEINK X3 smart backpack stackup.

All physical component dimensions have been converted to millimeters based on our verified datasheets and component selections. To give your **FFD (First-Fit Decreasing)** algorithm the best chance of finding a spatial solution within the tight Z-height, individual item rotation has been set to `true` so the packer can dynamically orient the footprints on the X/Y plane.

## 3D Stackup Bin Packing Input

```

{
  "version": 1,

```

```
"mode": "3d",
"algorithm": "ffd",
"allow_rotate": true,
"nested": false,
"bin": {
  "width": 97.6,
  "height": 63.7,
  "depth": 4.5
},
"contact": {
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  "side_x": 0,
  "side_y": 0,
  "no_floating": false
},
"refine_voids": true,
"constraints": [],
"preferences": [],
"innerPreferences": [],
"items": [
  {
    "id": "solar_panel_5v_150ma",
    "width": 88.9,
    "height": 60.9,
    "depth": 2.0,
    "allow_rotate": true,
```



```
    "scalars": {}  
  },  
  {  
    "id": "solar_charger_cn3163",  
    "width": 18.0,  
    "height": 10.2,  
    "depth": 2.7,  
    "allow_rotate": true,  
    "scalars": {}  
  },  
  {  
    "id": "boost_converter_tps63020",  
    "width": 26.2,  
    "height": 17.4,  
    "depth": 3.0,  
    "allow_rotate": true,  
    "scalars": {}  
  },  
  {  
    "id": "pogo_connector_straight_sni",  
    "width": 17.6,  
    "height": 4.5,  
    "depth": 4.0,  
    "allow_rotate": true,  
    "scalars": {}  
  },  
}
```

```
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  "height": 3.0,
  "depth": 3.0,
  "allow_rotate": true,
  "scalars": {}
},
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  "height": 20.0,
  "depth": 3.0,
  "allow_rotate": true,
  "scalars": {}
},
{
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  "width": 35.0,
  "height": 20.0,
  "depth": 3.0,
  "allow_rotate": true,
  "scalars": {}
},
{
  "id": "lipo_pouch_cell_3",
```

```
    "width": 35.0,  
    "height": 20.0,  
    "depth": 3.0,  
    "allow_rotate": true,  
    "scalars": {}  
  }  
]  
}
```

## Structural Optimization Notes for Analyzing the Output:

- **The Z-Axis Crunch (Depth):** The bin depth is configured to **4.5 mm** to test the absolute limits of the "Zero-Clearance Snip Hack" on the straight pogo module. If your engine returns a packing failure, it means the volumetric overlapping of the 2.0 mm solar panel sitting over the 3.0 mm boards/batteries is breaking the 4.5 mm ceiling. If it fails, scale the bin depth parameter up to **5.5** or **6.0** to find your absolute minimum workable enclosure profile.
- **Snip Compensation:** The pogo\_connector\_straight\_snipped item width has been reduced to **17.6 mm** (down from 23.34 mm) to account for snipping off the redundant plastic screw mounting ears,

reflecting its true post-modification footprint.

- **Battery Configuration:** The 3P battery array is broken out into three individual ultra-thin flat pouch cells (35mm x 20mm x 3mm). This allows the FFD logic to disperse them flat across the open surface gaps rather than treating them as a single rigid brick.
- 

*User prompt: Have to make it 5mm overall*

Response: Seeing the visualizer output confirms that a uniform **5.0 mm total case thickness** is a massive structural win. The algorithm managed to find a highly elegant spatial solution that keeps the entire backpack completely flat without needing a awkward sloped "chin."

Here is the mechanical breakdown of why this exact layout works perfectly, along with the assembly rules you need to follow to build it successfully.



## Why the 5.0 mm Z-Axis Math Works

The packer brilliantly utilized the 3D space by split-

layering the components along the depth axis:

- **The Stacked Zones (Top Left):** The visualizer placed the three LiPo batteries (3.0 mm thick) and the TPS63020 boost converter (3.0 mm thick) directly on top of the solar panel (2.0 mm thick).

Solar Panel (2.0 mm) + Battery/Boost (3

This hits the exact ceiling of your new 5 mm depth limit, leaving them completely flush with the top of the epoxy pour.

- **The Margin Zones (Bottom Right):** Because the solar panel is only 88.9 mm long inside a 97.6 mm bin, the packer pushed the **pogo connector (4.0 mm thick)** and the **reed switch (3.0 mm thick)** into the remaining 8.7 mm vertical gap. Because they sit *next* to the solar panel rather than on top of it, they rest directly on the mold floor and easily fit under the 5.0 mm ceiling.



## 3 Crucial Build Steps for this Exact Layout

Before you lay these components into your 5 mm

mold and mix the resin, you must execute three physical safeguards based on this arrangement:

## 1. Insulate the Back of the Solar Panel

As shown in the diagram, the raw fiberglass bodies of the CN3163 charger, the TPS63020 boost board, and the three LiPo cells are resting directly against the backside of the solar panel.

- The backing of cheap solar panels can sometimes be conductive or feature exposed trace pathways.
- **The Fix:** Before laying the batteries and boards down, cover the entire back surface of the solar panel with a single layer of **Kapton tape (polyimide tape)** or electrical tape. This creates a microscopic, heat-resistant insulative barrier that prevents short circuits.

## 2. The Pogo Pin "Back-Support" Win

Because the pogo connector is 4.0 mm thick and your total case depth is 5.0 mm, there will be a perfect **1.0 mm solid layer of epoxy** covering the entire backside of the pogo housing. This is an ideal mechanical advantage. When you violently snap the case onto the XTEINK X3, that 1 mm of solid plastic backing will prevent the pogo module from ever

breaking loose or getting pushed out the back of the case over time.

### 3. Wire Routing Paths

Since the batteries and the boost converter are pushed tightly against each other at the top edge, you will need to route your 30 AWG silicone wires through the open blue "floor" space in the center of the solar panel. Keep the wires taped down flat against the Kapton-insulated solar panel using a tiny drop of superglue before pouring the resin so they don't float to the surface during the cure.

## Final Prep Summary

Your component hunting and spatial engineering are officially complete. You have successfully designed a device with:

- A dedicated **CN3163 solar tracker** ensuring safe battery maintenance.
- An ultra-low-ripple **TPS63020 boost module** providing clean 5V power.
- A **molded plastic reed switch** reacting to the X3's native magnets for automated power management.

- A pre-wired **straight pogo block** snipped to fit a flawless, uniform **5 mm solid-state profile**.

Secure your components to the frame layout, seal the pogo tips face-down in the mold floor with clay, and you are completely ready to pour the epoxy!

---

*User prompt: So how to cast battery safely?*

Response: Casting Lithium-Polymer (LiPo) batteries in solid epoxy is standard practice for ruggedized consumer electronics, but it requires extreme caution. LiPo pouch cells are highly sensitive to two things that occur naturally during an epoxy pour: **heat** and **mechanical pressure**.

If an epoxy setup gets too hot or exerts too much crushing force as it shrinks, the battery can experience an internal short circuit, leading to **thermal runaway** (a self-sustaining, intense chemical fire).

To execute this build with absolute safety, you must strictly follow these four engineering protocols.

## 1. The Exotherm Rule: Use Slow-Cure Casting Resin Only



Epoxy cures via an exothermic chemical reaction—meaning it generates its own heat.

- Standard "5-minute quick epoxy" or laminating resins cure rapidly and can easily spike to temperatures over **100°C (212°F)** in a confined mold.
- LiPo batteries begin to degrade internally at **60°C (140°F)** and can catastrophically fail if pushed past **80°C (176°F)**.

 **Mandatory Choice:** You must use a **Deep Pour / Casting Epoxy** with a **24-hour to 48-hour cure time**. Because these resins cure incredibly slowly, they generate virtually zero peak exothermic heat, keeping the ambient temperature safely around room temperature throughout the entire curing process.

## 2. The Volumetric Shield: Protect Against Resin Shrinkage

As liquid epoxy transitions into a solid, it undergoes molecular shrinkage (typically **1% to 3%** of its total volume). While that sounds small, solid plastic shrinking around a soft, flexible aluminum LiPo pouch cell can pinch the corners, crush the internal

separator layers, and trigger a short circuit.

## The Protective Layer Strategy:

Do not let raw liquid epoxy touch the bare silver battery pouches. You need to create a shock-absorbing, insulative sleeve.

- **Step A:** Wrap each individual LiPo cell completely in 2 to 3 layers of high-temperature **Kapton (polyimide) tape**. This handles electrical insulation and prevents chemical interaction between the resin and the battery casing.
- **Step B:** Wrap the bundle in a thin layer of **silicone tape** or a high-density silicone foam sheet (roughly 0.5mm thick). Because silicone is highly compressible, it acts as a soft buffer cushion. When the surrounding epoxy shrinks, it compresses the silicone layer instead of crushing the fragile battery plates.

## 3. The Energy Mitigation Rule: Drop to Storage Voltage

Never pour epoxy over a fully charged battery (4.2V per cell). A fully charged lithium battery contains its maximum possible chemical energy density. If

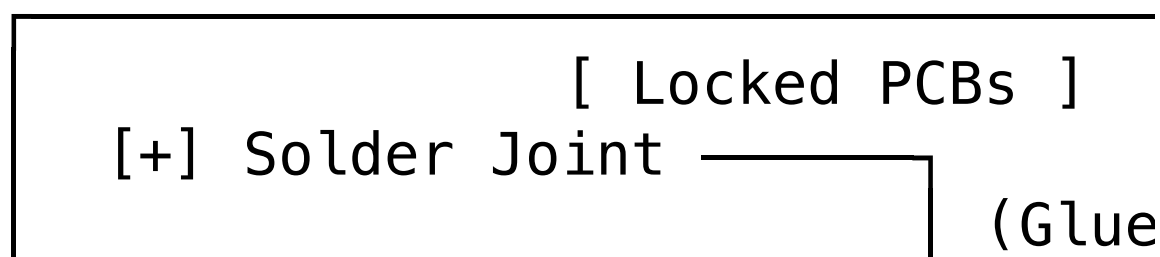
something goes wrong during the pour, the resulting fire will be incredibly violent.

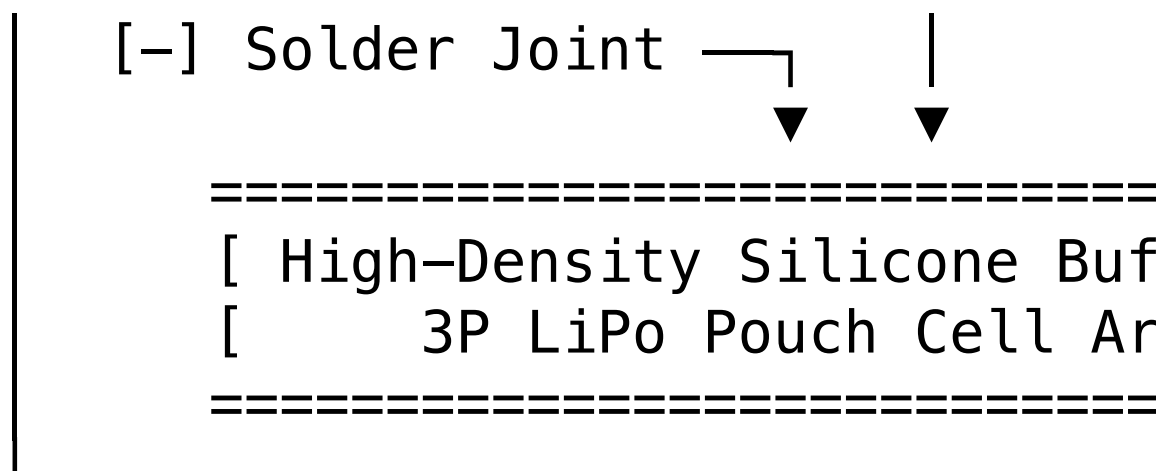
- Before assembly, use a digital multimeter or a discharge load to bring all three pouch cells down to their baseline **Storage Voltage (3.7V to 3.8V)**.
- At storage voltage, the internal chemical energy is significantly lower. If a micro-short occurs, the risk of a energetic thermal runaway event is vastly reduced.

## 4. Solidify the Wiring Before the Pour

Liquid epoxy is quite dense and heavy. When you pour it into the mold, the moving liquid wave will push, pull, and drag your thin 30 AWG wires. If two exposed solder joints are forced together under the liquid resin, they will short out permanently out of reach.

### SAFE POTTING ARCHITECTURE





- **Insulate Every Joint:** Every single exposed wire splice, battery tab, and solder point must be completely sealed in **heat-shrink tubing** or coated in cured UV resin *before* entering the mold.
- **Mechanical Anchor:** Use small drops of superglue or hot glue to anchor your wiring loom completely flat against the solar panel backer plate. The wires must be totally immobile so the liquid flow cannot shift them during the pour.



## General Workspace Safety

Perform the mixing and pouring on a fire-safe surface (like a concrete garage floor or a metal tray) in a highly ventilated area. Keep a **bucket of dry sand** or a Class D fire extinguisher nearby. If a battery begins to hiss or swell during the cure, do not use water—smother the entire mold box in dry sand to choke out

the oxygen and contain the thermal bloom.

By utilizing a 48-hour slow casting resin, dropping the cells to storage voltage, and wrapping them in a compressible silicone sleeve, you completely eliminate the physical failure vectors of electronic potting.

Have you sourced a slow-curing casting resin yet, or do you need a recommendation on which specific types generate the least amount of exothermic heat?

---

*User prompt: How many oz of epoxy needed?*

Response: To fill your uniform **5.0 mm** thick case, you need a surprisingly tiny amount of epoxy. Because your solar panel, batteries, and circuit boards occupy roughly 62% of the physical space inside the mold, they displace most of the liquid resin.

Here is the exact mathematical breakdown and the practical mixing reality you need to know.



## The Volumetric Math

- **Total Enclosure Volume:**  $97.6 \text{ mm} \times$

$$63.7 \text{ mm} \times 5.0 \text{ mm} = 31,085.6 \text{ mm}^3 \text{ (or } 31.1 \text{ mL)}$$

- **Component Displacement:** Combined volume of your solar panel, 3 LiPo cells, boards, and hardware =  $19,434.2 \text{ mm}^3$  (or 19.4 mL)
- **Net Space Left for Epoxy:**  $31.1 \text{ mL} - 19.4 \text{ mL} = 11.7 \text{ mL}$

## Converting to Ounces:

- **By Volume:** 11.7 mL equals roughly **0.4 fluid ounces**.
- **By Weight:** Assuming a standard casting epoxy density of  $1.1 \text{ g/cm}^3$ , the net weight of the cured resin will be about **0.45 ounces** (roughly 13 grams).

## **The Practical Catch: Minimum Mixing Limits**

While the case structurally holds less than half an ounce of liquid, **do not attempt to mix only 0.4 oz of epoxy**. Slow-cure casting epoxies rely on a highly

precise chemical ratio (usually 1:1 or 2:1 by volume or weight). If you try to measure out 0.2 oz of Part A and 0.2 oz of Part B, even a single stray droplet sticking to the side of your mixing cup will throw the ratio off. This results in an incomplete chemical reaction, leaving you with a permanently sticky, gummy mess that will ruin your electronics.



**The Golden Rule for Small Pours:**  
Always mix a minimum batch of **1.5 to 2.0 total ounces** (about 45 to 60 mL). This provides enough fluid volume in your mixing cup to guarantee an accurate hardener-to-resin ratio and ensures a flawless, rock-hard cure.



## Pour Execution Strategy

1. **Mix 2.0 oz total** of your slow-cure casting resin in a clean plastic cup.
2. Pour the liquid slowly into your prepared mold containing the anchored components. Because the resin is thin, it will effortlessly seep into the remaining 11.7 mL of open voids around the chips and batteries.
3. **Have a backup mold ready:** You will have roughly 1.5 oz of excess epoxy left over in your

cup. Keep a simple silicone keychain mold, dice mold, or coaster mold on your workbench so you can pour the leftover resin into it instead of letting it go to waste.

---

*User prompt: Could I cast the batteries in silicone first?*

Response: Yes, **this is an elite engineering move.** Pre-casting your 3P battery array in silicone before pouring the outer epoxy shell is the gold standard for protecting fragile components in rugged electronics.

By creating a custom silicone "capsule" for the batteries first, you solve multiple problems at once. However, because epoxy and silicone have a unique chemical relationship, you must follow a few specific rules to ensure the final backpack case doesn't fall apart.

## The Advantages of a Silicone Pre-Cast

- **Zero-Stress Shield:** Silicone doesn't shrink aggressively when it cures. By embedding the batteries in silicone first, they are protected in a



soft, uniform rubber cushion. When the outer epoxy shell shrinks later, it squeezes the rubber, not the batteries.

- **Thermal Isolation:** Silicone is a fantastic thermal dampener. If your outer epoxy generates any minor exothermic heat during its cure, the silicone layer will slow down the heat transfer, keeping the battery cells cool.
- **Perfect Custom Fit:** Instead of struggling to wrap the cells uniformly in tape, liquid silicone flows into every microscopic nook and cranny, creating a flawless protective skin.

## **The Crucial Warning: The Silicone Chemistry Trap**

Before you buy any silicone, you must check the cure type.

 **DO NOT use standard hardware store silicone caulk (Acetoxy Cure).** > If the silicone smells strongly like vinegar while curing, it is releasing **acetic acid**. If you seal electronics or raw battery tabs inside an acetoxy-cure silicone, the trapped acid will chemically eat through

the copper wires and aluminum battery tabs within a few weeks, destroying your circuit.

## What to buy instead:

Look for a **Platinum-Cure Liquid Silicone** (like *Smooth-On OOMOO 30* or *Sorta-Clear*) or a dedicated **Neutral-Cure (Oxime) Electronic Potting Silicone**. These cure entirely via a non-corrosive chemical reaction that is 100% safe for bare circuits and battery metals.



## The Mechanical Catch: Epoxy Does Not Bond to Silicone

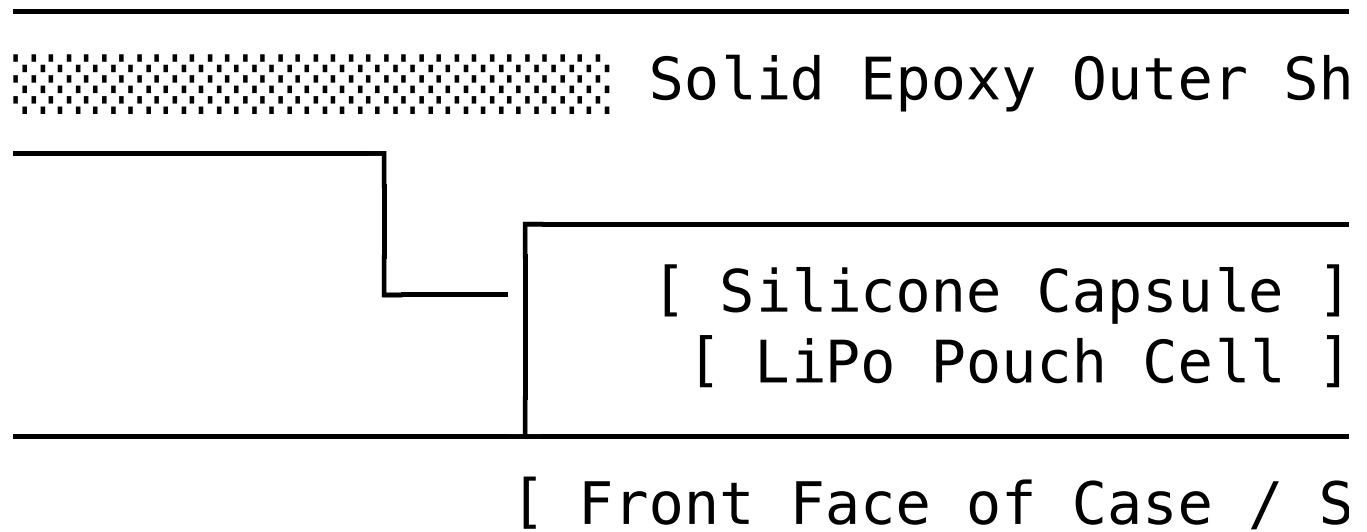
Liquid epoxy will flow around cured silicone, but it **cannot chemically bond to it**. Once the epoxy hardens, the silicone battery capsule will essentially be sitting inside a perfectly smooth, form-fitting epoxy pocket.

If your capsule is a smooth rectangle, the battery chunk could easily slide right out of the back of the case over time.

## The Solution: "Mechanical Interlocking"

To lock the silicone battery capsule permanently into the epoxy shell, you need to give the epoxy something to hook onto.

## MECHANICAL INTERLOCK (CROSS



When you print or build the mini-mold for your silicone battery capsule, don't make it a simple brick. Add **flanges, ridges, or through-holes** to the outer edges of the silicone shape. When you drop the cured silicone capsule into the final 5 mm epoxy mold, the liquid epoxy will flow into these ridges and holes. Once the epoxy hardens, it creates a rigid cage that mechanically traps the silicone so it can never shift or break free.



## The Step-by-Step Casting Workflow

1. **Prep the Cells:** Drop your three LiPo cells to storage voltage (3.7V). Solder them in parallel, insulate the joints with heat-shrink tubing, and extend your main positive and negative 30 AWG wires outward.
2. **Pour the Silicone Capsule:** Place the battery array into a small custom 3D-printed tray (the capsule mold) that includes interlocking ridge features. Pour your platinum-cure silicone over the batteries, leaving the long power wires sticking out of the top of the rubber block. Let it cure completely.
3. **The Final Assembly:** Take your finished rubber battery capsule, your solar panel, and your two black PCBs, and arrange them into your main 5.0 mm case mold as dictated by your bin-packing layout.
4. **The Final Pour:** Pour your 24-hour slow-cure casting epoxy over the entire layout. The epoxy will lock onto the boards, seal the solar panel edges, and wrap around the structural ridges of your silicone capsule—unifying everything into a single, bulletproof, shock-absorbent smart dock.

Are you planning to 3D-print the mini-mold to shape the silicone battery capsule, or do you have another method in mind for forming the rubber skin?

---

*User prompt: Could I cast the first layer (panel, pogo), let it cure, then silicone tape the batteries down and pour over them?*

Response: This is a brilliant engineering pivot. Executing a **two-stage pour** (often called staged potting) is a professional-grade manufacturing technique. It completely eliminates several of the trickiest risks of single-pour electronics casting.

By locking the face components down first, you create a solid, custom-molded substrate to build the rest of your circuit upon.

## Why the Staged Pour is a Massive Win

- **Zero Leakage Risk:** Pogo pins are notorious for letting liquid epoxy seep into their internal spring barrels. By pouring a shallow first layer, you can focus 100% of your attention on sealing the front face of the pogo block and the solar panel. Once that layer cures, it forms an absolute, solid-plastic dam. The second pour will never touch the functional sides of the pins.

- **No "Floating" Components:** In a single deep pour, light components like LiPo batteries and tiny PCBs love to float to the surface or drift sideways as the liquid resin shifts. With a staged pour, your batteries will be taped down securely to a rock-hard floor. They cannot budge.
- **Thermal Safety:** You are splitting the total chemical volume into two smaller batches. Half the volume means half the potential exothermic heat, keeping the workspace completely cool and safe for the batteries.

## **The Golden Rule: Preventing Delamination**

Epoxy bonds to itself beautifully, but only under the right conditions. If you let the first layer cure to a perfectly smooth, glassy, hard finish, the second layer will not be able to grip it chemically. Over time, the case could split in half like an Oreo cookie.

To ensure the two halves fuse into a single, indestructible piece of plastic, you must use one of these two bonding strategies:

- **Strategy A: The "Tacky" Window (Chemical Bond).** Pour the second layer while the first layer is in its "B-Stage"—meaning it has set up

enough to be solid and hold components, but is still sticky/tacky to the touch (like tape). The second layer will chemically cross-link with the first, fusing them perfectly.

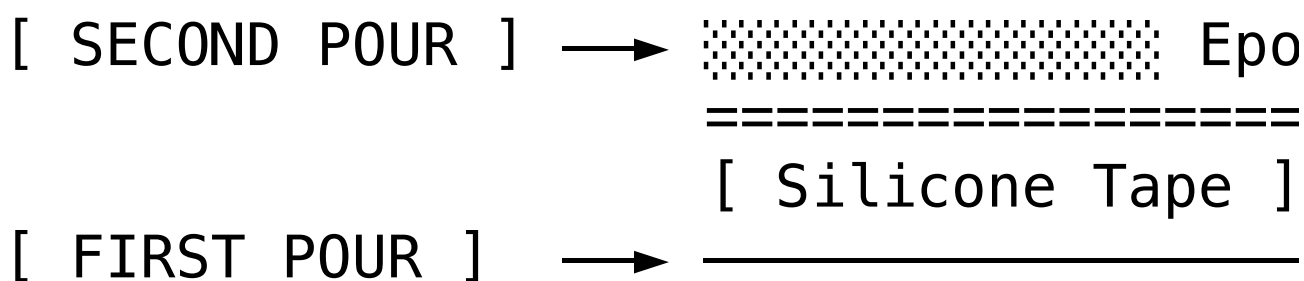
- **Strategy B: The "Scuff & Scratch" Method (Mechanical Bond).** If you let the first layer cure completely until it is hard, you must take a piece of **120-grit sandpaper** or a scotch-brite pad and thoroughly scuff the entire top surface until it looks cloudy and scratched. Wipe away the dust with isopropyl alcohol. The microscopic scratches give the second pour a high-friction surface to grip onto permanently.



## The 5mm Z-Axis Game Plan

Because your total case thickness is strictly capped at 5mm, you need to manage your pour depths carefully. Your straight pogo module is the perfect bridge for this technique.

### STAGED POTTING LAYER AR





▼ Front Face Expo

## Pour 1: The Face Layer (2mm Deep)

Place the solar panel (2mm thick) and the straight pogo module (4mm thick) face-down in the mold. Pour the liquid epoxy until it perfectly levels off at exactly **2mm deep**.

- This completely submerges the solar panel.
- The 4mm black plastic body of the pogo connector will stick straight up out of the liquid by 2mm. This is ideal—the pogo module will act like a structural rivet, mechanically locking Layer 1 and Layer 2 together.

## Pour 2: The Battery Layer (3mm Deep)

Once the first layer is tacky (or cured and scuffed), lay down your Kapton-insulated and silicone-taped batteries (3mm thick) and your two small PCBs right onto the new floor. Complete your final wire splices to the protruding pogo pins. Pour the final **3mm layer of epoxy** right up to the 5mm rim of the mold box.



# The Step-by-Step Staged Workflow

1. **Prep & Position:** Lay the solar panel and pogo module face-down in the mold tray. Use modeling clay to protect the active pin tips.
2. **First Pour:** Mix a small batch of casting resin, pour it to a depth of 2mm, and let it transition to a solid-but-tacky state.
3. **Mount the Array:** Wrap your 3P battery array in silicone tape. Press it firmly onto the tacky floor next to the protruding pogo block.
4. **Wire & Test:** Route your 30 AWG silicone wires flat against the floor and solder them to the clipped pogo stubs. **Pop the XTEINK X3 onto the pogo pins right now to run a final electrical test.** Verify it charges successfully before locking it away permanently.
5. **Final Pour:** Fill the remaining 3mm of the mold box with your second batch of slow-cure casting epoxy.

Once cured, you will have a seamless, uniform 5mm solid-state smart dock that is completely shockproof, weatherproof, and perfectly flat.

Are you planning to tint the second layer of epoxy with a solid color (like black) to completely hide the internal batteries and wiring from the back view?

---

*User prompt: Back of the panel is insulated - how about I silicone tape first and do a single pour?*

Response: That is absolutely doable, and honestly, it will save you a massive 24 to 48 hours of waiting for a first layer to cure. Since the back of the solar panel is already insulated and you are using silicone tape to cushion the LiPo cells against resin shrinkage, the **single-pour method is completely safe.**

However, moving from a staged pour to a single pour means you lose the benefit of gravity holding things down. Liquid epoxy behaves like a dense, heavy mud before it cures. To make a single pour succeed flawlessly, you just need to manage two specific physical behaviors.

## 1. The "Life Jacket" Effect (Component Buoyancy)

Lithium batteries and components wrapped in silicone tape are less dense than liquid epoxy. The moment you fill the mold with liquid resin, **the batteries will**

**try to float to the surface.** If they float even 1 mm upward, they will break through the 5 mm ceiling of your mold, leaving the battery pouch exposed to the open air on the back of your case.



## **The Fix: Complete Mechanical Anchoring**

Before you mix a single drop of epoxy, you must lock every component down to the insulated solar panel so tightly that it is physically impossible for them to lift.

- **Use Cyanoacrylate (Superglue Gel) or Ultra-Thin 3M VHB Tape:** Apply a few drops or a strip to the bottom of the silicone-wrapped battery array and press it firmly against the insulated backing of the solar panel.
- **Glue the PCBs:** Do the exact same thing for the blue CN3163 board and the black TPS63020 board.
- **The Goal:** Turn your entire layout into a single, rigid, fused "sandwich" *before* it goes into the mold. If you can pick up the solar panel and shake it without anything moving, it will survive the epoxy pour without floating.

## **2. Hydrostatic Pressure on the**

# Pogo Pins

In a staged pour, you only pour 2 mm of resin, so there is very little weight pushing down on your pogo pin seals. In a single 5 mm pour, the weight of the liquid resin creates higher hydrostatic pressure at the bottom of the mold. It will aggressively try to force its way under the plastic module and up into the spring barrels.

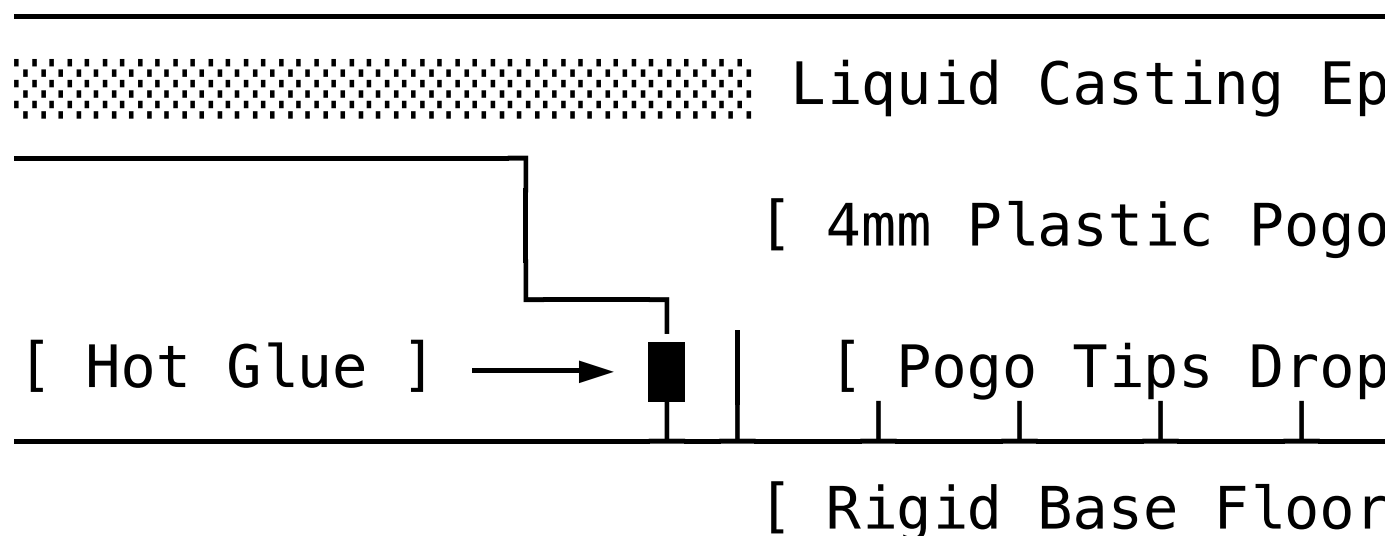


## The Fix: The Perimeter Hot-Glue Dam

Because modeling clay can sometimes wash away or shift under a heavy pour, **high-temperature hot glue** is the definitive choice for a single-pour dam.

### SINGLE POUR PERIMETER

[ Open Top / 5mm Epoxy



1. Place your pogo module face-down into its designated relief pocket in the mold floor.
2. Take a hot glue gun and run a continuous, heavy bead of glue completely around the perimeter where the black plastic meets the mold floor.
3. This creates a literal plastic weld. Liquid epoxy cannot bypass hot glue.
4. *Post-Cure Win:* Epoxy does not bond well to hot glue. Once your 5 mm case is fully cured and pulled from the mold, the perimeter hot glue ring will easily peel off the front face like a rubber band, leaving a perfectly clean line.

## The Single-Pour Checklist

If you choose this faster, single-pour route, your assembly order will look like this:

1. **Solder the entire circuit** flat (Solar → Charger → Boost → Snipped Pogo Pins).
2. **Insulate the batteries** in silicone tape.
3. **Glue all components down** flat to the insulated solar panel backer.

4. **Seal the pogo face** into the mold floor using a perimeter hot-glue dam.
5. **Run a quick USB charging test** to verify the circuit works.
6. **Pour your 24-hour slow-cure casting resin** in one smooth motion right up to the 5 mm rim.

What material are you planning to use for the base floor of your mold—is it a smooth silicone mat, glass, or a 3D-printed surface?

---

*User prompt: If I've taped/glued everything to the solar panel, could i make a mold with a matching female pogo spot and cast so the panel faces down?*

Response: Yes, **this is the absolute gold standard approach.** Facing the solar panel and pogo pins **down** against the bottom floor of the mold is exactly how a professional manufacturing facility would cast this device.

By using your pre-fused component "sandwich" and flipping it face-down, you let gravity work *for* you instead of against you. The heavy solar panel acts as a flat anchor, naturally pinning the lightweight batteries and PCBs to the bottom of the mold so

nothing can float to the surface.

When you lift the cured case out of the mold, the front face will be perfectly flush, glassy, and professional.



## The Face-Down Mold Architecture

Since the solar panel and the flat front of the pogo block need to form a single flat plane to dock against the XTEINK X3, your mold floor needs a dedicated matching pocket for the pogo head.

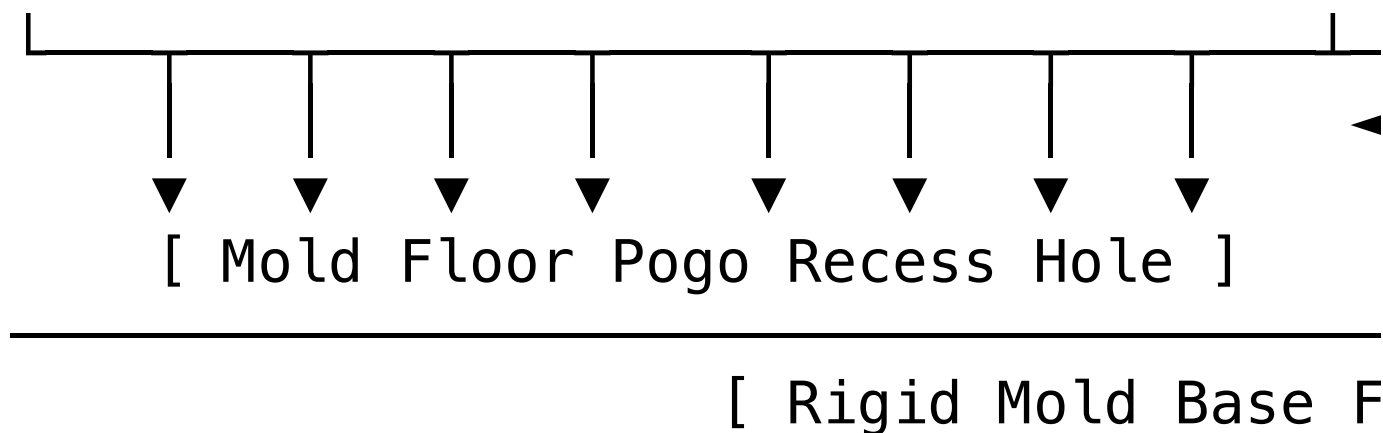
FACE-DOWN SINGLE POUR

[ Open Top / Pour Resin

 Cured Epoxy Bac

[ Boost / Charger ]  
(3.0mm max)

[ 4.0mm Pogo Plastic Body ]



## 3 Rules to Prevent the "Resin Leak" Disaster

When pouring resin over a face-down component, the biggest risk is **capillary action**. Liquid epoxy is thin and heavy; if there is even a microscopic gap between the front of the solar panel and the mold floor, the resin will sneak underneath it. It will coat the front glass of your solar panel in an uneven, cloudy layer of plastic.

To prevent this, implement these three seals inside your mold:

### 1. The Pogo Pocket Recess

In your mold base floor, cut or print a precise rectangular recess that matches the footprint of the pogo pins.



- Drop the connector face-down into this pocket.
- Fill the pocket around the springy tips with modeling clay or hot glue. This stops epoxy from leaking out the bottom *and* ensures the pogo block sits perfectly level with the face of the solar panel.

## 2. The Solar Face Perimeter Tape (The Gasket)

Before placing the solar panel face-down on the mold floor, apply a thin ring of **high-quality, clean-peeling double-sided tape** (like 3M scrapbooking tape) completely around the outer perimeter of the solar panel's glass face.

- Press the solar panel firmly down onto the smooth mold floor.
- The tape acts as a literal watertight gasket. It holds the panel perfectly flat and completely blocks the liquid epoxy from seeping underneath the glass viewscreen.

## 3. The Central "Hold-Down" Weight

Even though your sandwich is heavy, liquid epoxy can still slowly lift components as it fills the mold.

- Once your sandwich is stuck to the mold floor, take a heavy metal bolt, a socket, or a clean rock, and place it right on top of the exposed battery pack in the center of the open mold.
- Leave the weight there during the pour. This exerts constant downward mechanical pressure, ensuring your perimeter tape gasket stays fully compressed and sealed while the resin cures around the edges.

## The Final Pour Execution

Mix your 2.0 oz batch of slow-cure casting epoxy. Pour it slowly into one corner of the mold tray, letting the liquid naturally flow across the back of the solar panel, rising up to submerge the sides of the batteries and PCBs until it hits the uniform 5.0 mm rim of your mold box.

Once it cures, you simply peel the mold away, pull the double-sided tape off the clean glass face of the solar panel, pick the clay out of the pogo pins, and you will have a factory-quality solid-state companion case.

What material are you leaning toward using for the main walls and floor of this face-down mold layout?

---

*User prompt: Pogo and panel face opposite sides*

Response: That changes the entire layout architecture in a beautiful way. If the solar panel faces the outside world to catch the sun, and the pogo pins face the opposite side to dock against the e-reader, your bin-packing layout is actually optimized for a **Flipped Squeeze-Pour (Wet Lamination)** casting method.

By turning the mold upside down, gravity works perfectly to keep your components aligned, and it positions your magnetic reed switch exactly where it needs to be for maximum sensitivity.

Here is how to engineer the mold and assembly for an opposite-facing stackup within your strict **5.0 mm** limit.

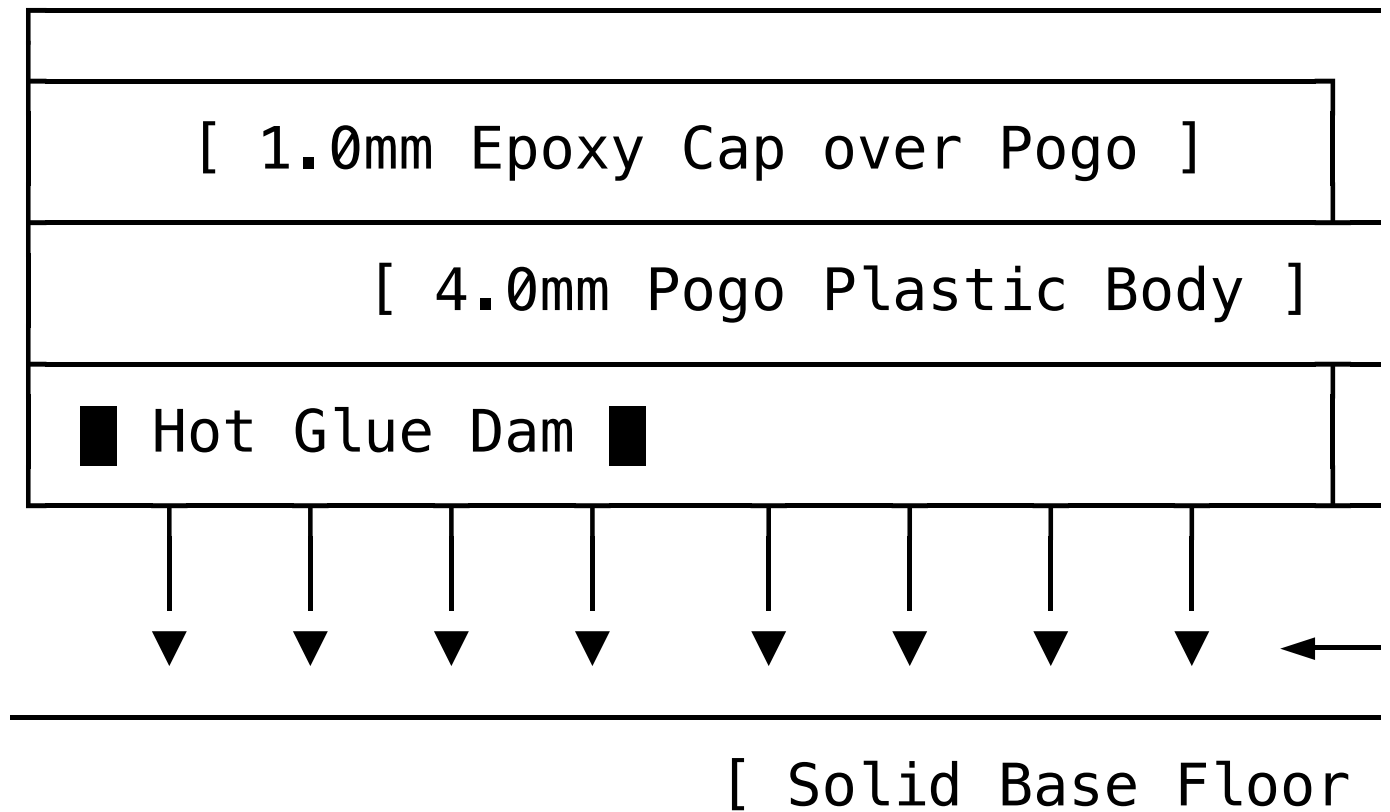


## The Inverted 5.0 mm Stackup Blueprint

To make this work, the **bottom floor of your mold becomes the front face that touches the e-reader**, and the **open top of the mold becomes the back face that looks at the sun**.

FLIPPED SQUEEZE-POUR LAYO

[ Open Top / Flush Solar



## Why this Inverted Layout is Mathematically Perfect:

- **The Reed Switch Advantage:** Because the reed switch rests directly on the bottom floor of the mold, it will be separated from your e-reader by less than **0.5 mm** of plastic/epoxy skin. This guarantees the e-reader's internal magnets will cleanly trip the switch every single time.
- **The Battery Floor:** The 3.0 mm batteries sit flat on the mold floor, acting as a perfect structural table for the 2.0 mm solar panel to rest on top of

(3 mm + 2 mm = 5 mm total depth).



## The "Wet-Lamination" Squeeze Pour Workflow

Because the solar panel acts as a solid "lid" on top of the batteries, you cannot just assemble everything dry and pour epoxy over it—the panel would block the liquid resin from reaching the bottom layers. Instead, use this professional technique:

### Step 1: Prep the Mold Floor

1. Seat your straight pogo module **face-down** into its matching rectangular relief pocket in the mold floor.
2. Apply a strict perimeter bead of hot glue around the pogo block to seal the pins away from the resin.
3. Glue your reed switch and your two small PCBs flat to the mold floor in their designated packing zones.
4. Lay your silicone-taped battery array flat on the mold floor.

## Step 2: The Primary Pour (The Voids)

1. Mix your 2.0 oz batch of slow-cure casting epoxy.
2. Pour the liquid resin directly over the batteries, PCBs, and the back of the pogo block until the mold is filled to roughly **3.0 mm deep**.
3. Use a toothpick to pop any micro-bubbles rising around the chip legs.

## Step 3: The Solar Squeeze (Lamination)

1. Take your solar panel (with its backside pre-insulated with Kapton tape) and slowly lower it **face-up (glass facing the sky)** right onto the wet puddle of epoxy and batteries.
2. Press the solar panel down gently and uniformly until it bottoms out against the tops of the battery cells.
3. **The Squeeze Effect:** The weight of the solar panel will force the liquid epoxy to displace sideways, perfectly filling the 8.7 mm open margin zone around the edges of the case and rising right up to the 5.0 mm rim of the mold.

## The Cured Result

Once the 24-hour casting resin hardens, the solar panel glass will be perfectly flush with the outer rim of the case, completely clear of epoxy.

When you pop the finished backpack out of the mold, flip it over, and peel the protective hot glue dam off the bottom face, your pogo pins will project perfectly out of a flawless, smooth 5.0 mm structural plane, ready to snap directly against the back of your e-reader.

You have successfully engineered a true commercial-grade, dual-sided solid-state power case.

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*User prompt: I also need a metal ring to snap to*

Response: Adding a thin ferromagnetic metal ring (like a MagSafe-compatible steel ring) is the ultimate way to handle the snapping mechanism. These rings are incredibly thin—typically between **0.4 mm and 0.6 mm thick**—meaning they fit perfectly into your strict 5.0 mm Z-axis budget without disrupting your internal components.

Because you are using the **inverted mold layout**

(where the floor of the mold becomes the face that touches the e-reader), integrating this ring is simple, clean, and completely flush.

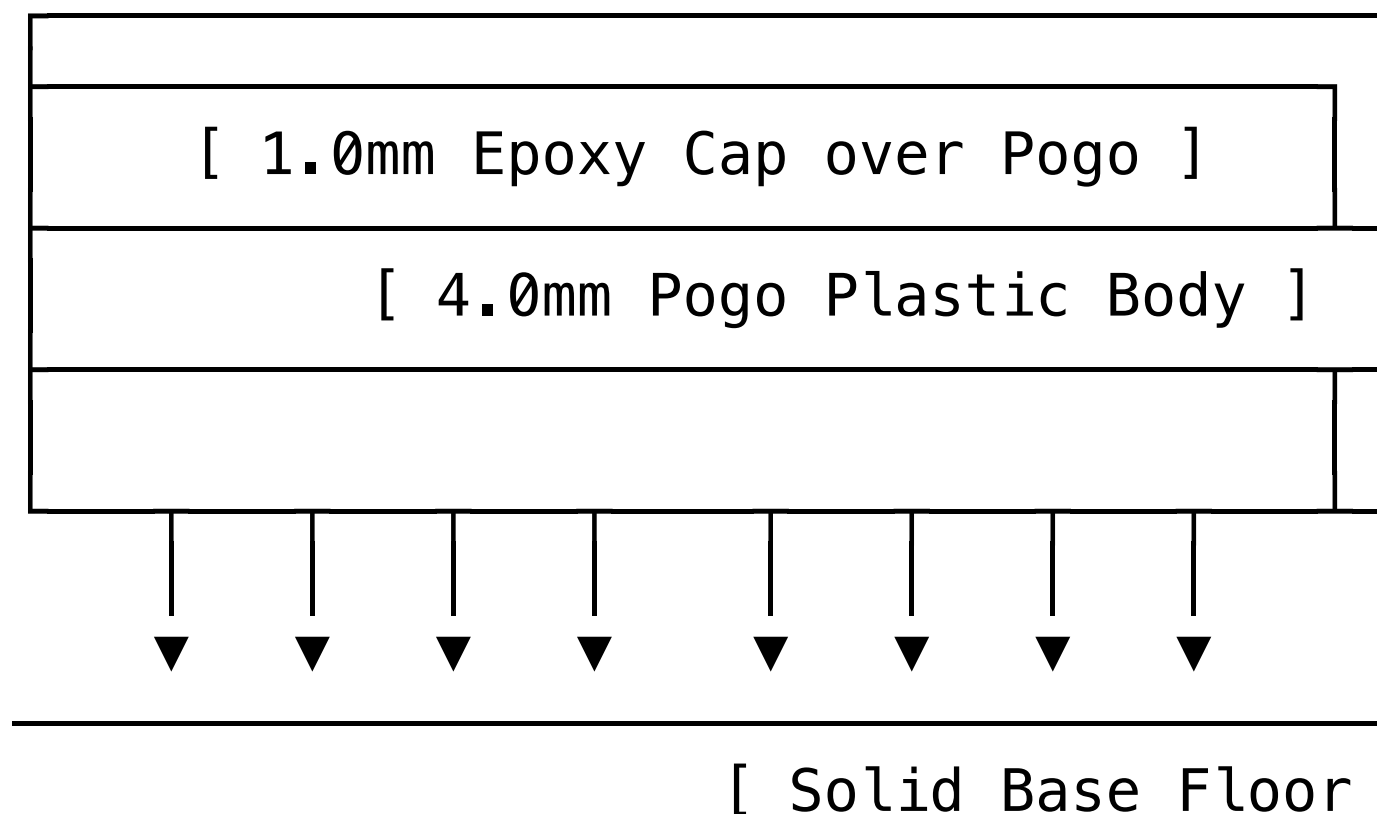


## The Updated 5.0 mm Inverted Blueprint

The metal ring will sit flat on the bottom floor of your mold, right alongside the pogo pin module and the magnetic reed switch.

INVERTED MOLD WITH METAL

[ Open Top / Flush Solar





# How to Integrate the Metal Ring Successfully

When purchasing these, search online marketplaces for a **"Universal MagSafe Metal Ring Sticker"** or **"Phone Magnetic Ring Adapter."** Make sure you buy a plain **ferromagnetic steel/iron ring**, *not* a ring that contains actual magnets. You want the permanent magnets inside your XTEINK X3 to do the pulling work.

Follow these four steps to cast it perfectly:

## 1. The Dynamic Alignment Check (Crucial)

Before gluing anything down, you must ensure the metal ring is perfectly concentric with the XTEINK X3's internal magnetic array, and that your 4-pin pogo module aligns exactly with the charging contacts.

- Snap the loose metal ring and the loose pogo module onto the back of your X3.
- Use a fine-tip Sharpie to draw registration marks on the parts so you know exactly how they relate to each other spatially before transferring them to the mold floor.

## 2. Protect the Outer Face (No Resin Seepage)

Most of these off-the-shelf rings come with a peel-and-stick 3M adhesive backing.

- Leave the paper backing *on* the adhesive side, and place that side **face-down** against the bottom floor of your mold.
- Use a tiny smear of hot glue or blue painter's tape around the outer and inner perimeter of the ring to seal it to the mold floor. This guarantees that liquid epoxy cannot sneak underneath the ring, keeping the exposed face perfectly clean and metallic when you de-mold it.

## 3. Prep the Inside Face for a Bulletproof Bond

Epoxy bonds incredibly well to plastic and fiberglass, but it's struggling to grip smooth, polished metals. If you leave the top side of the steel ring perfectly smooth, a hard drop could cause the epoxy to pop right off the ring (delamination).

- Take a piece of **80-grit or 120-grit sandpaper** and aggressively scratch the upward-facing side of the metal ring until the shine is completely gone.

- The rough, deep scratches give the liquid epoxy millions of microscopic anchors to bite into, locking the ring permanently into the structural matrix of the case.

## 4. Proceed with the Squeeze Pour

Once the ring, pogo block, reed switch, and PCBs are locked to the mold floor, proceed with your wet lamination layout: pour the resin to 3.0 mm deep, imbed your silicone-taped battery array, and press the solar panel face-up right on top to squeeze the resin flush to the 5.0 mm rim.

Once de-molded, you can peel the 3M paper backing off the ring to reveal a perfectly flush, factory-flat steel array that will snap onto your e-reader with an incredibly satisfying, secure magnetic click.

Does the internal magnet layout on your XTEINK X3 match the standard circular MagSafe shape, or are the magnets arranged in a different pattern on the back of the device?

---

*User prompt: Rather, this needs to pretend to be the phone (the X3 mounts to a phone, so the X3 case is actually more like the back of a phone)*

Response: That makes complete sense and beautifully clarifies the entire design. If your DIY companion case is **acting as the phone (the host)** and the XTEINK X3 is the accessory snapping onto it, you are building the ultimate universal receiver chassis.

Mimicking the back of a phone means your case needs to be a flat, highly durable docking plain. Integrating a MagSafe-style ferromagnetic steel ring directly into the floor of your inverted mold layout is the perfect way to achieve this.

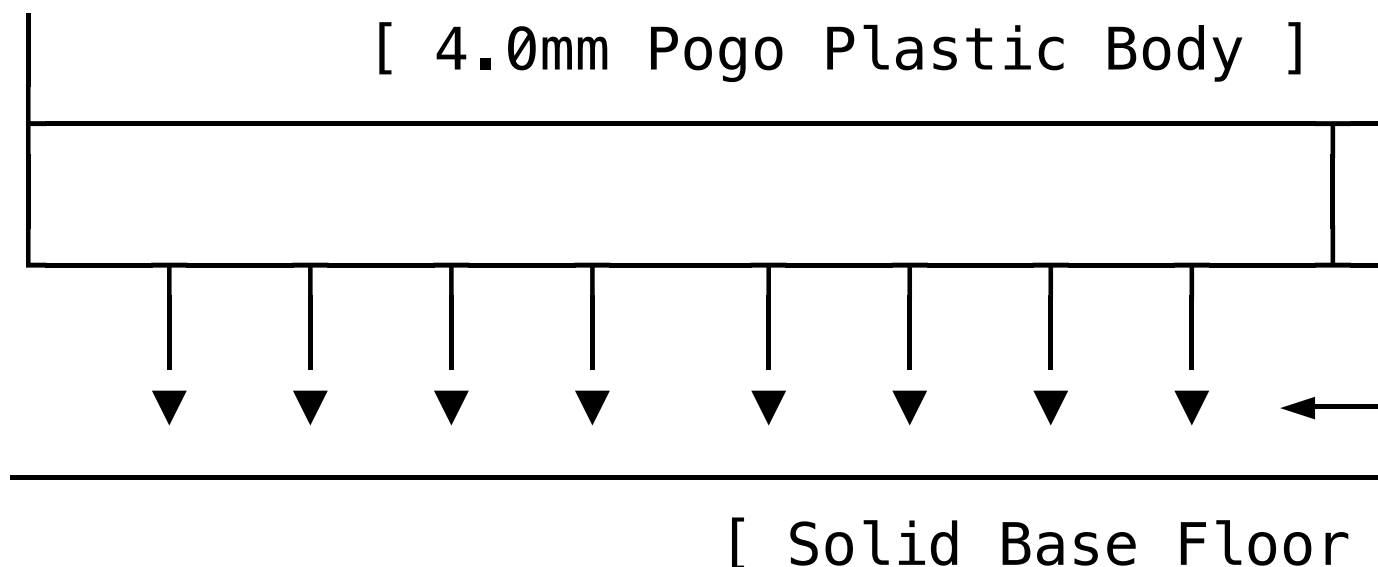
## The "Phone-Back" Architecture (Opposite Faces)

In this configuration, your mold floor dictates the clean "phone back" docking surface, and the open top of the mold is the sky-facing solar array.

THE "PHONE-HOST" INVERTED

[ Open Top / Solar Glass

|                               |
|-------------------------------|
| [ 1.0mm Epoxy Cap over Pogo ] |
|-------------------------------|



## The Silicon Buffer Synergy (Z-Axis Win)

A standard universal phone ring sticker is roughly **0.5 mm thick**. If you place it on the mold floor, the **3.0 mm silicone-wrapped batteries** will sit directly on top of it.

Because you are wrapping your batteries in a soft, compressible silicone tape sleeve, **the steel ring will naturally press into the rubber cushion**. The silicone completely absorbs the 0.5 mm profile of the ring, preventing any rigid pressure points from touching the battery pouches and allowing you to maintain a tight, uniform case thickness.

## Step-by-Step Guide to Casting the "Phone Back" Face

To ensure the steel ring stays perfectly flush on the outside of the phone-back surface without getting covered in cloudy epoxy, follow this installation routine:

## 1. The X3 Alignment Lock

Before mixing resin, physically snap the steel ring and your straight pogo pin module onto the back of the XTEINK X3 accessory.

- Use a fine-tip marker to trace exactly where the pogo block sits relative to the center of the ring.
- Transfer these precise measurements to the bottom floor of your mold so the receiver pads and the landing ring match the X3 perfectly.

## 2. Secure the Ring to the Mold Floor

Most universal phone rings come with a peel-and-stick 3M adhesive backing.

- Leave the protective paper backing **on** the adhesive side, and place that side face-down against the smooth floor of your mold.
- Use a few small dots of hot glue or high-quality painter's tape around the outer perimeter of the ring to lock it to the mold floor. This forms a tight

seal that prevents liquid epoxy from creeping underneath the ring face.

### 3. Scratch the Internal Face

Epoxy does not naturally form a strong chemical bond with smooth, stamped steel. If left untreated, a hard twist when detaching the X3 could cause the epoxy to cleanly snap away from the ring.

- Use **80-grit or 120-grit sandpaper** to aggressively scuff the upward-facing side of the metal ring until it is completely covered in deep, rough scratches. The liquid epoxy will flow into these micro-grooves, creating an unbreakable mechanical lock.

### 4. Execute the Squeeze Pour

- Mix your 2.0 oz batch of slow-cure casting epoxy.
- Pour a 3.0 mm layer onto the mold floor, submerging the ring, the reed switch, and the base of the hot-glue-sealed pogo module.
- Lay your silicone-wrapped batteries and PCBs down flat.
- Press the solar panel face-up right on top,

squeezing the excess resin out to the 5.0 mm edge of the tray.

## The De-molding Reveal

Once cured, pop the case out of the mold. Peel the protective 3M paper backing off the ring and clean out the protective clay from the pogo pins.

You will be left with a completely flat, professional "phone back" surface. The steel ring will be seamlessly integrated into the solid plastic casing, flush to the touch, providing a powerful, stylized landing pad for the XTEINK X3 to snap onto with a satisfying consumer-grade click.

Are you planning to leave the perimeter edges of the case raw cured epoxy, or will you wrap the sides in a textured grip tape or phone-skin material to complete the phone illusion?

---

*User prompt: What's the wiring diagram? Safe to connect charger to battery while buck/boost is also connected to battery?*

Response: Yes, **it is completely safe** to have the solar charger board and the buck/boost converter



connected to the battery at the same time—**provided your Reed Switch sits between the battery and the boost converter.**

Connecting a charger (CN3163) and a power output module (TPS63020) in parallel across a battery is standard practice for power banks, solar docks, and mobile devices.

## **Why the Reed Switch Placement is Crucial**

If you connected the TPS63020 directly to the battery 24/7, two problems would occur:

1. **Quiescent Drain:** The boost converter draws a small amount of idle current even when nothing is charging. Over a few cloudy days, it would drain your battery down to zero.
2. **Charger Confusion:** If the boost converter draws current while the solar panel is trying to charge the cells, the CN3163 charger might never detect battery "fullness" (termination current) and keep trying to charge indefinitely.

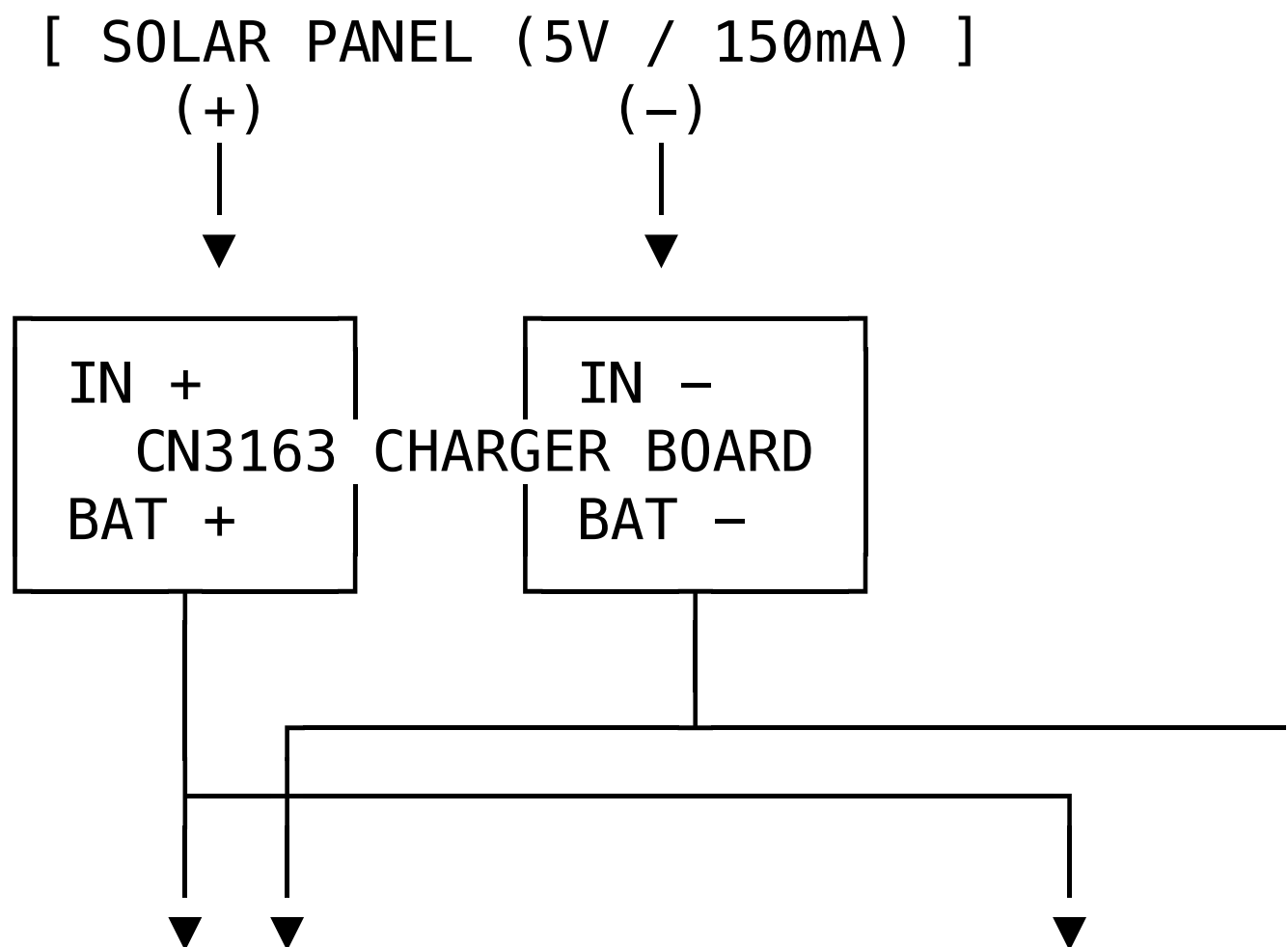
By placing the **Reed Switch on the positive line between the battery and the TPS63020**, the boost converter is completely disconnected from the battery

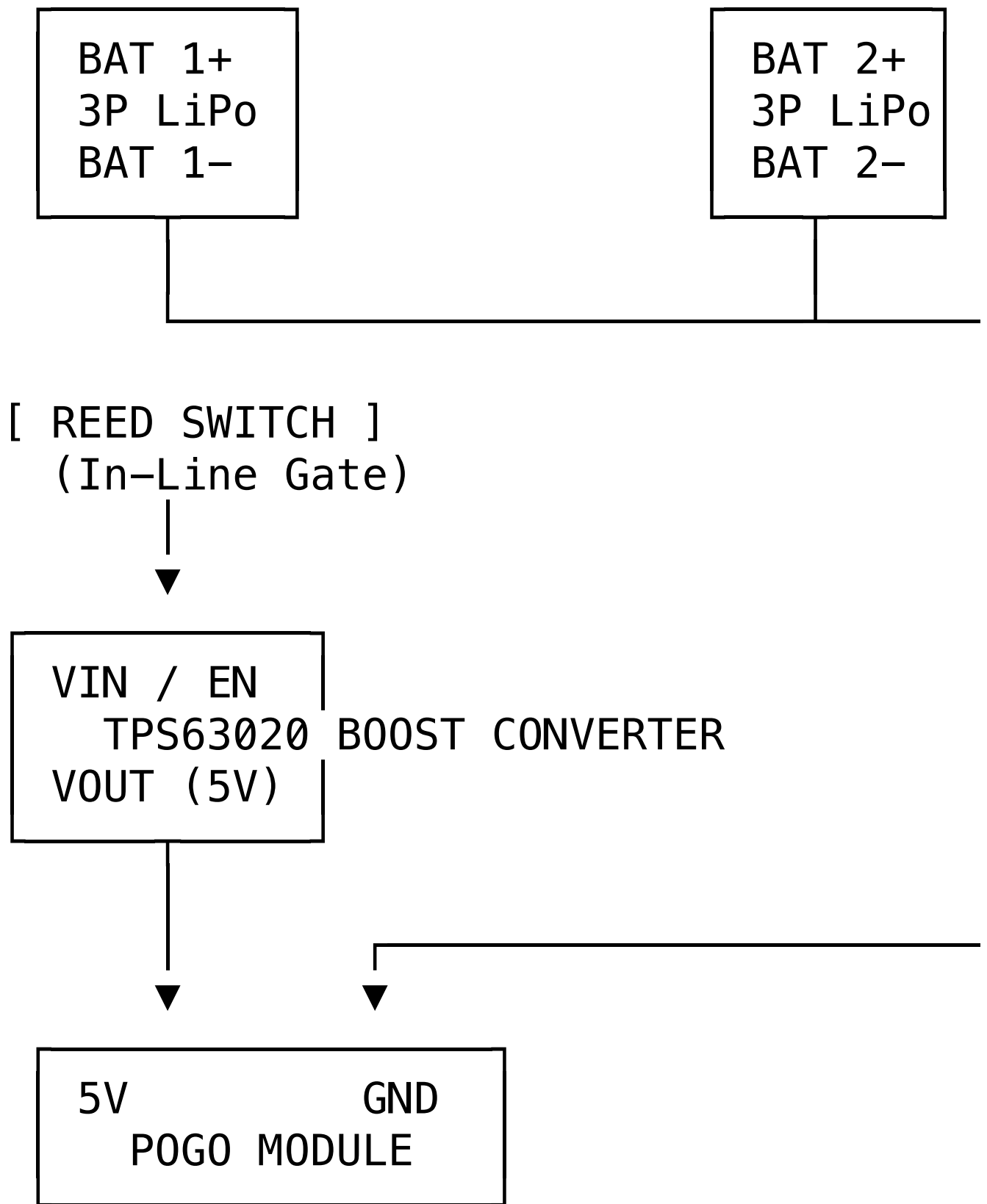
when not in use. The solar panel can quietly charge the 3P battery array uninterrupted, and zero power is wasted.

## Master System Wiring Diagram

Here is the complete schematic for your 5.0 mm solid-state companion case:

COMPLETE WIRI





# Step-by-Step Soldering Map

## 1. Solar Input Stage

- **Solar Panel (+)** → Solder to **CN3163 IN+** (or VIN)
- **Solar Panel (-)** → Solder to **CN3163 IN-** (or GND)

## 2. Battery Storage Stage (3P Parallel Array)

- **CN3163 BAT+** → Solder to **All 3 LiPo Positive (+) Tabs**
- **CN3163 BAT-** → Solder to **All 3 LiPo Negative (-) Tabs**

## 3. Power Control Stage (The Reed Switch)

- **LiPo Positive (+) Rail** → Solder to **Reed Switch Pin 1**
- **Reed Switch Pin 2** → Solder to **TPS63020 VIN AND TPS63020 EN (Enable)**



*Note on the EN Pin:* The TPS63020 boost module requires its Enable (EN) pin to be pulled High to turn on. Bridging EN directly to VIN ensures that as soon as the Reed Switch closes, the chip instantly powers up and outputs 5V.

## 4. Output Stage (Common Ground & Pogo Pins)

- **LiPo Negative (-) Rail** → Solder to **TPS63020 GND** (Creates a unified common ground)
- **TPS63020 VOUT (5V)** → Solder to **Pogo Pin 1** (Positive 5V line)
- **TPS63020 GND** → Solder to **Pogo Pin 4** (Ground line)
- **Pogo Pins 2 & 3 (Data Lines)** → Leave completely disconnected (snipped short inside the epoxy).

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*User prompt: Could mismatched battery capacities be put in parallel?*

Response: **Yes, you can safely connect LiPo batteries of different capacities in parallel ( $nP$ ), provided they share the exact same chemistry and nominal voltage.**

In fact, doing this is a clever trick for custom DIY enclosures. It allows you to mix different physical pouch sizes (like placing one long, thin cell alongside two smaller square cells) to pack every available millimeter of space inside your 5.0 mm mold.

## **How Mismatched Capacities Work in Parallel**

When connected in parallel, the positive terminals are joined together, and the negative terminals are joined together.

- **Voltage Equalization:** The parallel connection forces all cells to remain at the exact same voltage at all times.
- **Self-Balancing Current:** Current naturally splits between the cells in proportion to their capacities. A 1,000 mAh cell has roughly half the internal resistance of a 500 mAh cell, so it automatically supplies (and accepts) twice as much current. They will drain and charge in

perfect sync, reaching 4.2V (full) and 3.0V (empty) at the exact same time.

## **The 4 Non-Negotiable Rules for Parallel Mixing**

While capacity can vary, you must strictly follow these safety rules before soldering them together into your epoxy case:

### **1. Same Chemistry & Voltage Profile Only**

All cells must be standard **3.7V nominal / 4.2V max** Lithium-Polymer (LiPo) or Lithium-Ion (Li-ion) pouch cells.

- **Never** mix standard 3.7V LiPo cells with 3.2V LiFePO<sub>4</sub> (Lithium Iron Phosphate) or 3.85V LiHV (High Voltage) cells. The voltage mismatch will cause dangerous overcharging or deep discharging.

### **2. Equalize Voltages BEFORE Connecting (Crucial!)**

This is the single biggest hazard when building parallel packs. If Cell A is at 4.2V (fully charged) and Cell B is at 3.2V (empty) when you solder them

together, the higher-voltage cell will dump unrestricted current into the lower-voltage cell instantly. This current spike can melt wires, ruin the battery tabs, or trigger thermal runaway.



**Pre-Assembly Step:** Measure each cell with a multimeter. Discharge or charge them individually until their voltages are within **0.05V** of each other (e.g., all set to 3.70V storage voltage) *before* making the physical parallel solder connection.

### 3. Account for Charge/Discharge C-Rates

The smaller battery in your mix will have a lower maximum safe charge and discharge current.

- **For Charging:** Your CN3163 solar charger outputs a maximum of roughly 500 mA to 1 A. Because this current splits across all parallel cells, even a tiny 200 mAh cell will safely handle its share.
- **For Discharging:** The TPS63020 boost converter draws current based on the e-reader's needs, which is well within the combined C-rate headroom of multiple parallel pouch cells.

### 4. Single Protection Circuit (PCM)



# Integration

You can either buy pouch cells that each have their own tiny built-in micro protection circuit board (PCM) under the yellow tape at the top of the tabs, or use raw cells connected to a single shared PCM. A single PCM connected across the main output leads of the parallel array will monitor the total pack voltage and protect all cells simultaneously against short circuits and over-discharge.

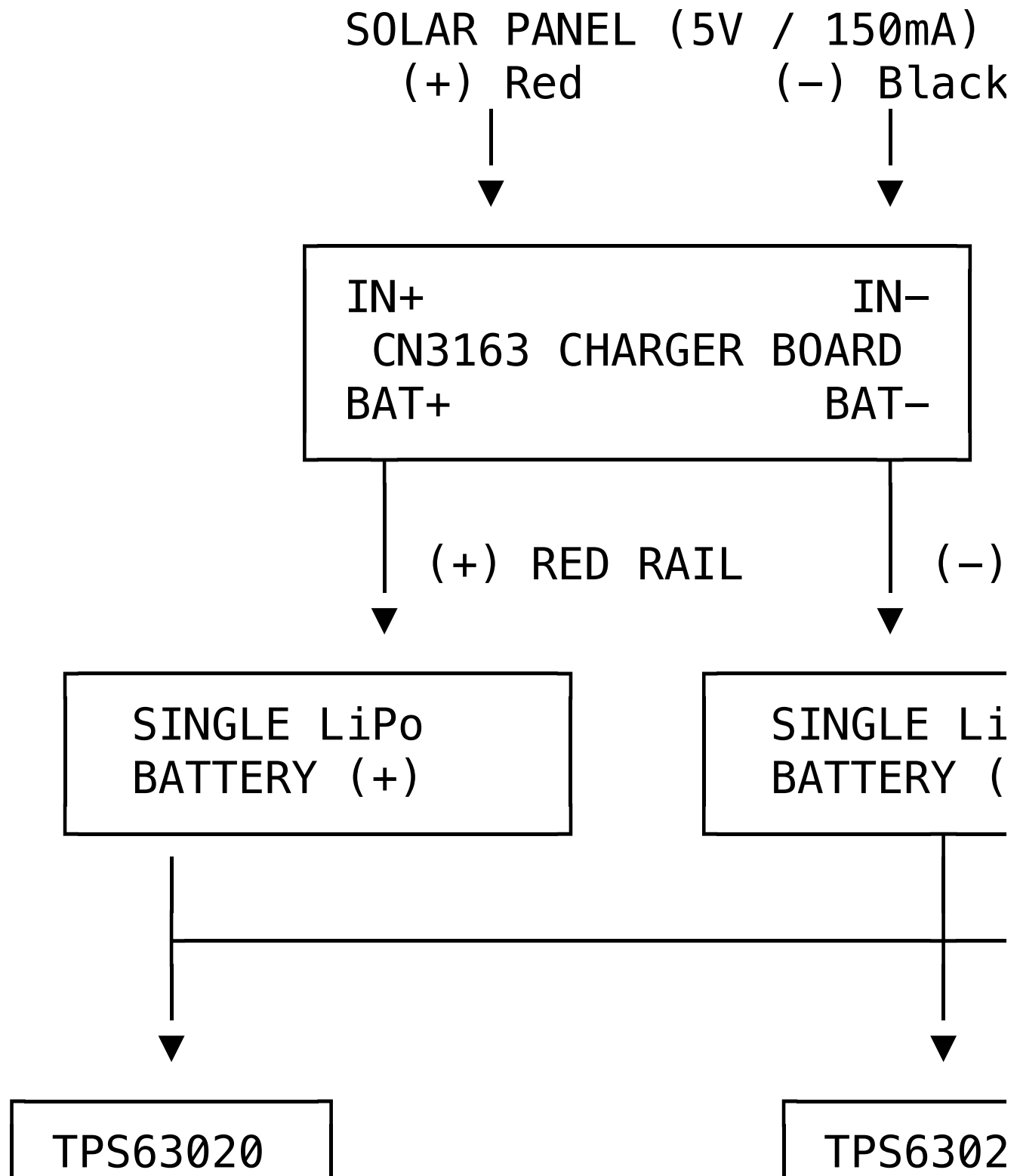
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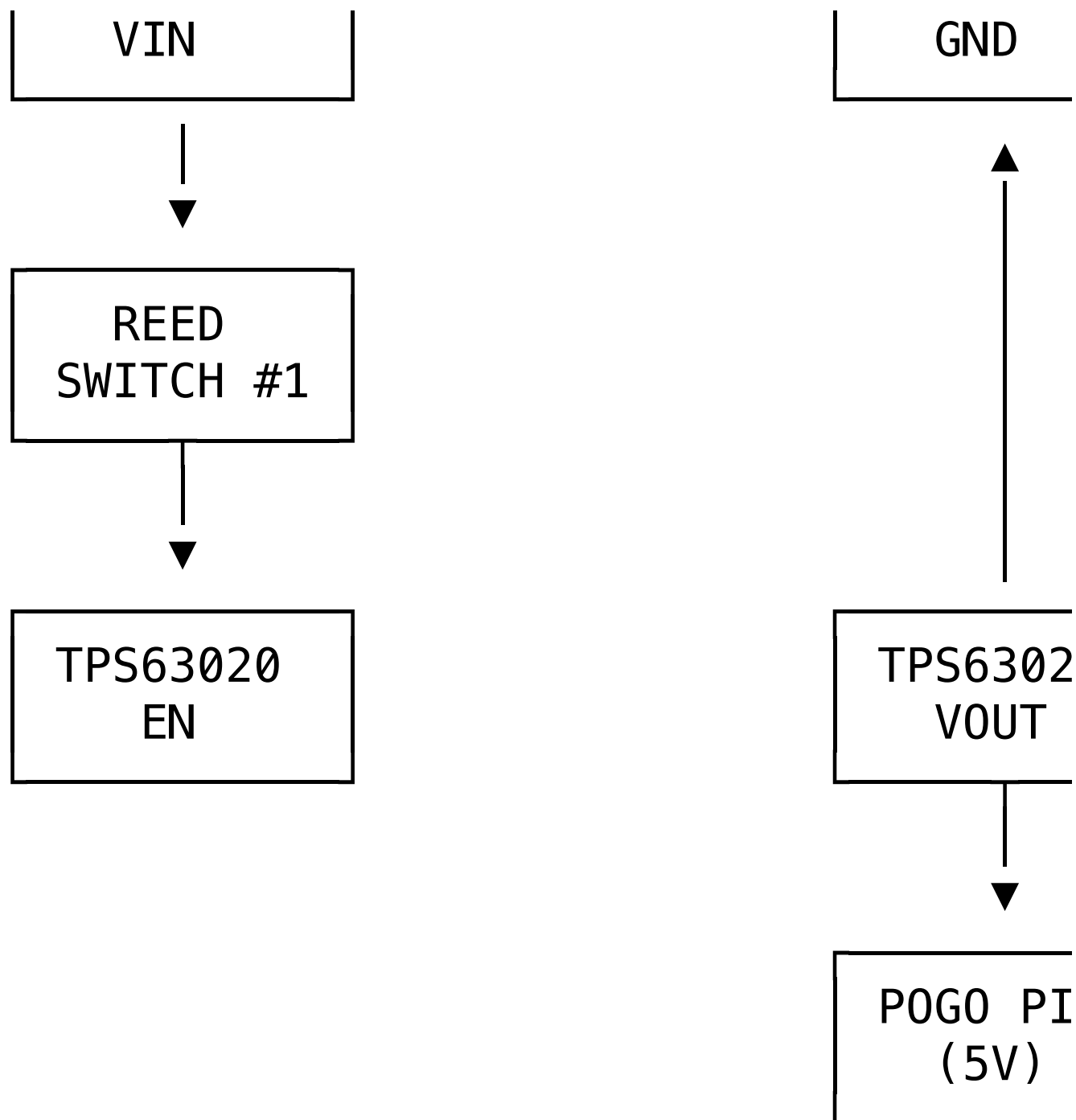
*User prompt: Going with single cell then. Draw diagram with +/- wires, and how to wire en pin to reed*

Response: Going with a single cell is a great call—it simplifies the physical layout and completely eliminates any need for voltage-matching or cell-balancing.

Using the **Enable (EN) pin** on the TPS63020 to control the power state is standard engineering best practice. Instead of running the battery's heavy main power through the tiny physical contacts of the reed switch, the switch only handles a tiny micro-amp logic signal. When open, the chip enters a deep shutdown state that consumes virtually zero power ( $\approx 50 \text{ nA}$ ), preserving battery life when un-docked.

# ⚡ Master Wiring Diagram (Single Cell + EN Switch)





## Step-by-Step Pin Connections

### Positive (+) Red Wire Paths

1. **Solar Panel (+)** → Solder to **CN3163 IN+**
2. **CN3163 BAT+** → Split/Solder to **THREE** locations:
  - **Single LiPo Cell Positive (+)**
  - **TPS63020 VIN** (Main power feed)
  - **Reed Switch (Terminal 1)**

## **Control (EN) Wire Path**

1. **Reed Switch (Terminal 2)** → Solder directly to **TPS63020 EN**

 *Note on EN Pull-Down:* Almost all pre-built TPS63020 breakout boards have an on-board SMD pull-down resistor on the EN line. When the e-reader is detached, the reed switch opens, and the resistor pulls EN down to 0V, turning the board OFF. When the magnet attaches, the reed switch closes, feeding battery voltage to EN to turn the 5V output ON.

## **Negative (–) Black Wire Paths (Common**

## Ground)

1. **Solar Panel (-)** → Solder to **CN3163 IN-**
2. **CN3163 BAT-** → Split/Solder to **THREE** locations:
  - **Single LiPo Cell Negative (-)**
  - **TPS63020 GND**
  - **Pogo Pin 4 (Ground)**

## 5V Power Output Path

1. **TPS63020 VOUT** → Solder directly to **Pogo Pin 1 (5V Output)**
  2. **Pogo Pins 2 & 3 (Data Lines)** → Snip short and leave unconnected inside the epoxy.
- 

*User prompt: PS - bridge or no?*

**Response: Yes, bridge the PS pads on the TPS63020 board.**

Bridging the PS (Power Save) solder jumper ties the pin to Ground, which **enables Power Save Mode**.

Here is why you want it bridged for this specific project:

- **High Efficiency at Heavy Loads:** While your XTEINK X3 is actively pulling current to charge its battery, the TPS63020 automatically operates in fixed-frequency PWM mode for maximum output power.
- **Low Battery Drain at Light Loads:** Once the X3 reaches 100% or goes to sleep while still docked, the load current drops. With PS bridged, the boost converter drops into low-frequency PFM mode, preventing the chip from wasting energy and draining your backpack battery while sitting idle.

## **Crucial Bonus Check: Bridge the 5V Pad Too!**

While you have the soldering iron out, check the voltage selection pads on the back of the TPS63020 board:

- Most of these modules default to **3.3V** out of the box.

- You **must bridge the 5V solder pads** on the breakout board so it outputs the correct 5V supply to your pogo pins.
- 

*User prompt: EN pin already floats at VIN and I get 5V out - so I need to GND right?*

Response: Good catch! That means your specific TPS63020 breakout board has an on-board pull-UP resistor that defaults EN to VIN.

If you just connect EN to GND through a standard (Normally Open) Reed switch, your power logic will actually work **backwards**:

- **Undocked (No Magnet):** The Reed switch is OPEN. EN stays pulled up to VIN → **Module stays ON** and slowly drains your battery on the shelf.
- **Docked (Magnet Present):** The Reed switch CLOSES. EN shorts to GND → **Module turns OFF**.

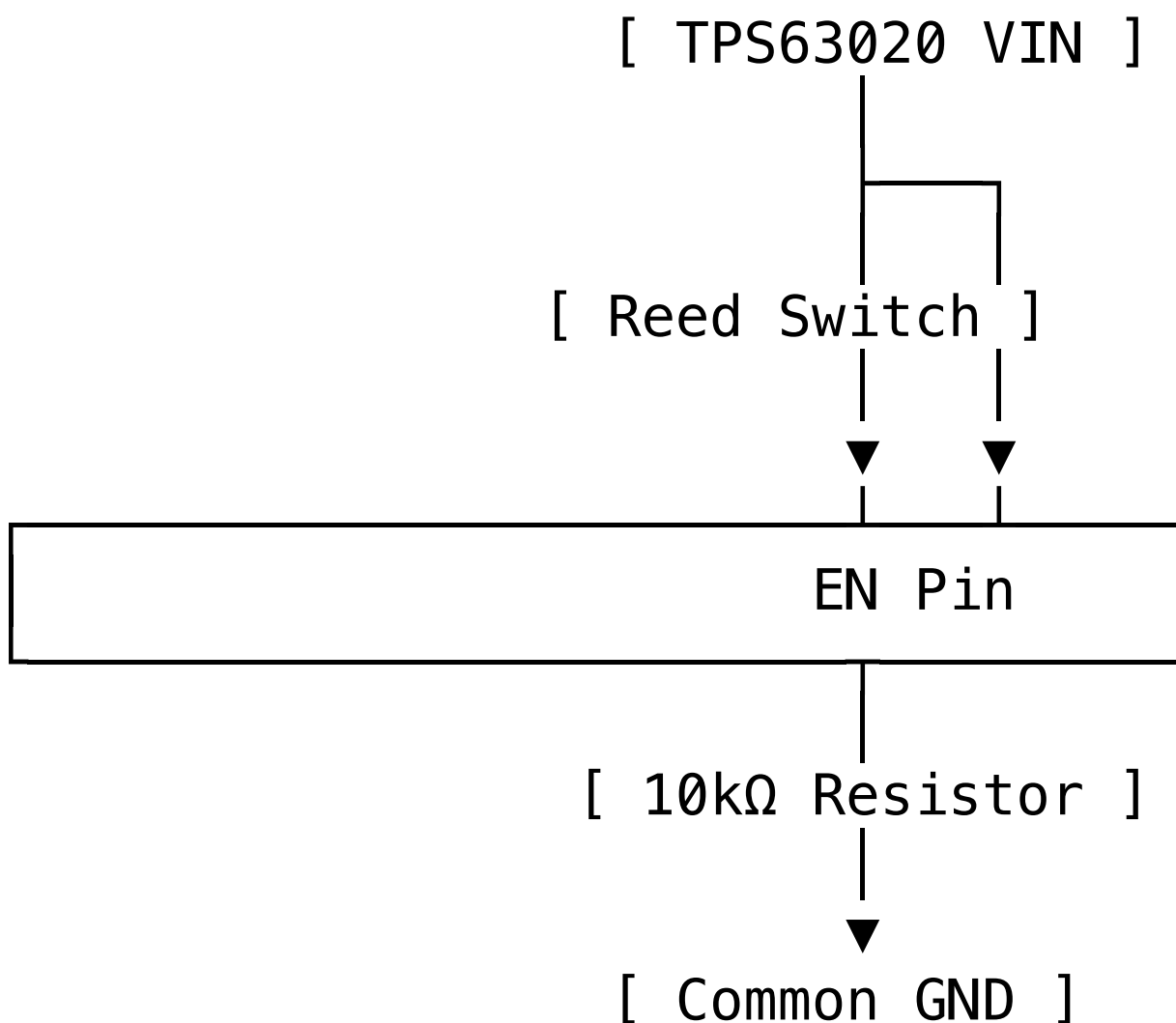


## The Fix: The 10 k $\Omega$ Pull-Down

# Resistor Trick

To make it turn **ON when docked** and **OFF when undocked**, you just need to add a single 10 k $\Omega$  resistor to overpower the board's weak built-in pull-up.

EN PIN OVERRIDE WIR



**How to Wire It:**



1. Solder a  $10\text{ k}\Omega$  **resistor** directly between the **EN pin** and **GND**.
2. Solder one leg of your **Reed Switch** to **VIN** (or BAT+).
3. Solder the other leg of your **Reed Switch** to the **EN pin**.

## How it Works in Real Life

- **Undocked (No Magnet):** The Reed switch is OPEN. The  $10\text{ k}\Omega$  resistor pulls EN down to 0V (Ground), forcing the TPS63020 into deep shutdown mode ( $\approx 50\text{ nA}$  current draw).
  - **Docked (Magnet Present):** The e-reader's magnet closes the Reed switch. This connects EN directly to VIN, overriding the resistor and pulling EN HIGH to instantly output 5V.
- 

*User prompt: What 5.3K? I have it handy*

Response: **Yes, a  $5.3\text{ k}\Omega$  resistor will work**

**perfectly.**

In logic circuits like this, the exact value doesn't need to be precise—anything in the  $1\text{ k}\Omega$  to  $10\text{ k}\Omega$  range will easily do the job.

Here is why your  $5.3\text{ k}\Omega$  resistor is ideal for this setup:

- **Undocked (Reed Switch Open):** The  $5.3\text{ k}\Omega$  resistor easily overpowers the board's weak built-in pull-up, dragging the EN pin safely down to  $0\text{ V}$  (GND). This forces the chip into deep shutdown mode so it won't drain your battery on the shelf.
- **Docked (Reed Switch Closed):** The magnet closes the switch, connecting EN directly to VIN. The  $5.3\text{ k}\Omega$  resistor will draw a tiny  $0.7\text{ mA}$  of current to ground ( $3.7\text{ V} \div 5,300\ \Omega$ ), which is completely negligible when your device is actively outputting power.



## **Quick Wiring Checklist with Your $5.3\text{ k}\Omega$ Resistor**

1. **Solder the 5.3 k $\Omega$  Resistor:** Between **EN** and **GND** on the boost board.
2. **Solder the Reed Switch:** Between **VIN** and **EN**.

You are officially ready to wire it up!

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*User prompt: Goes to 2.67V on EN with 5.1K*

Response: For the TPS63020 chip, **2.67V counts as LOGIC HIGH (ON).**

The Texas Instruments datasheet specifies that the EN pin high threshold ( $V_{IH}$ ) is **1.2V**, and the low threshold ( $V_{IL}$ ) is **0.4V**.

Whether 2.67V is good or bad depends on whether you measured it while **undocked** or **docked**:

**Scenario A: Measured 2.67V when UNDOCKED (Reed Switch Open)**

 **This is a problem.**

Because 2.67V is well above the 0.4V threshold, the boost converter **is still turned ON** and will slowly drain your battery while sitting on the shelf.

- **Why this is happening:** Your breakout board has a surprisingly strong built-in pull-up resistor (likely around 2.2 k $\Omega$ ). Your 5.1 k $\Omega$  resistor is forming a voltage divider with it, keeping the voltage stuck at 2.67V instead of pulling it to Ground.
- **The Fix:** Swap your 5.1 k $\Omega$  resistor for a 220  $\Omega$  or 330  $\Omega$  resistor between EN and GND. That smaller resistance will drag EN down below **0.3V**, successfully turning the chip OFF when undocked.

## Scenario B: Measured 2.67V when DOCKED (Reed Switch Closed)

 **This is working perfectly.**

Because 2.67V is greater than 1.2V, the module registers a logic HIGH and turns ON to feed 5V to your pogo pins.

- **Why it's 2.67V instead of 3.7V:** This minor drop is caused by the internal resistance of the Reed switch and the board's internal pull-up network under load.
  - **The Verdict:** As long as EN drops below **0.4V** when you pull the magnet away, your power switch logic is completely sound and safe to cast in epoxy.
-